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WITHDRAWN

11 kV AND 33 kV SINGLE CIRCUIT OVERHEAD LINES OF HEAVY CONSTRUCTION ON WOOD POLES

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11 kV AND 33 kV SINGLE CIRCUIT OVERHEAD LINES OF HEAVY CONSTRUCTION ON WOOD POLES

FOREWORD

In the past, 11 kV and 33 kV heavy duty overhead lines have been constructed to Specifications prepared by individual Area Boards. The advent of metrication and the coming into force of the Electricity (Overhead Lines) Regulations 1970 created the need for the revision of these Specifications and this Standard has been prepared to fulfil the needs of the Electricity Supply Industry.

1. SCOPE

This Standard covers 3-phase 11 kV and 33 kV overhead lines of heavy construction on wood poles, employing the following ACSR:

100 mm² (6/4.72 mm A1 + 7/1.57 mm St)
150 mm² (18/3.35 mm A1 + 1/3.35 mm St)
175 mm² (18/3.61 mm A1 + 1/3.61 mm St)
175 mm² (30/2.79 mm A1 + 7/2.79 mm St)
200 mm² (18/3.86 mm A1 + 1/3.86 mm St)

Other conductors may be used in conjunction with this Standard subject to users supplying their own erection data and verifying that the strength of components is adequate.

Lines to this Standard are designed to the conditions specified in The Electricity (Overhead Lines) Regulations 1970. Where these conditions may be exceeded due to geographical or weather conditions a modified form of construction as specified in Appendix A may be adopted.

2. DEFINITIONS

For the purpose of this Standard the following definitions apply:

Intermediate Support — A support in a straight run of a line on which the conductors are supported on pin or line post insulators.

Section Support — A support in a straight run of a line on which the conductors are made off on either side of the crossarm on tension insulator sets.

Pin Angle Support — A support at which a line deviates and the conductors are supported on pin or line post insulators.

Section Angle Support — A support at which a line deviates and the conductors are made off on either side of the crossarm on tension insulator sets.

Span — The horizontal distance between adjacent supports.

Wind Loading Span — The wind loading span associated with any support is half the sum of the spans adjacent to the support.

Average Span — The arithmetic average of a number of spans in a line or section of line.

Basic Span — The span length adopted for sag/tension calculations.

Recommended Span — The average span length in any section to which the line should be planned.

Maximum Span — The maximum span length permitted using normal conductor spacing.

Occasional Long Span — A span of line contained in a run of normal line construction but having a length above the maximum specified for normal span construction.

Sag — The vertical distance, under any system of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured mid-span.

Conductor Downpull — The vertical loading imposed by the conductors corresponding to a gradient measured between adjacent pole tops.

Pre-tensioning — The tension treatment applied to a conductor before final erection tension is established.

Overtensioning — Excess tension applied above normal tension at time of erection.

Auxiliary Equipment — Equipment other than that forming part of this line design which may be placed on supports, such as transformers, fusegear, etc.

3. REFERENCES

This Standard makes reference to the following documents:

BS 137, 'Insulators of Ceramic Material or Glass for Overhead Lines with a Nominal Voltage greater than 1000 V'.

BS 215, 'Aluminium Conductors and Aluminium Conductors, Steel Reinforced, for Overhead Power Transmission'.

Part 2, 'Aluminium Conductors, Steel-reinforced'.

BS 913, 'Wood Preservation by means of Pressure Creosoting'.

BS 1990, 'Wood Poles for Overhead Lines (Power and Telecommunication Lines)'.

BS 3288, 'Insulator and Conductor Fittings for Overhead Power Lines'.

BS 3452, 'Copper/Chrome Water-borne Wood Preservatives and their Application'.

BS 4072, 'Wood Preservation by means of Water-borne Copper/Chrome/Arsenic Compositions'.

BS 6485, 'PVC-covered Conductors for Overhead Power Lines'.

The Electricity (Overhead Lines) Regulations 1970, SI 1355.

*ESI Standard 43—89, 'Auxiliary Equipment Erected on Overhead Line Supports'.

ESI Standard 43—90, 'Anti-climbing Devices for H.V. Lines up to and including 132 kV'.

ESI Standard 43—91, 'Stay Strands and Stay Fittings for Overhead Lines'.

ESI Standard 43—92, 'Conductor Terminations, Joints and Insulation Binds for Overhead Lines up to and including 132 kV'.

ESI Standard 43—93, 'Line Insulators'.

ESI Standard 43—95, 'Steelwork for Overhead Lines'.

ESI Standard 43—96, 'Fasteners and Washers for Wood Pole Overhead Lines'.

*ESI Standard 43—97, 'Notices and Marker Posts for H.V. Lines up to and including 132 kV'.

*In course of preparation.

4. DESIGN DATA

This Standard is designed in accordance with the requirements of The Electricity (Overhead Lines) Regulations 1970.

4.1 Conductors

Conductor design tensions take into account the following conditions:

The minimum factors of safety on conductors shall be 2.0 on their calculated breaking loads.

The maximum working tension of conductors shall not exceed 2379 kgf (i.e. a factor of safety of 3 on 70 kN tension insulator fittings). The design tensions of aluminium based conductors at 5°C in still air shall not exceed 20 per cent of their calculated breaking load to obviate the risk of vibration.

The actual maximum working tensions and factors of safety are tabled below, together with the limiting design feature:

ACSR			M.W.T. (kgf)	F.O.S.	Limiting Design Feature
Size (mm ²)	Stranding (mm)	Type			
100	6/4.72 + 7/1.57	Bare PVC	1459 1501	2.29 2.22	Vibration Limit
150	18 + 1/3.35	Bare PVC	1628 1632	2.24 2.23	Vibration Limit
175	18 + 1/3.61	Bare PVC	1794 1793	2.34 2.34	Vibration Limit
175	30 + 7/2.79	Bare PVC	2379 2379	3.42 3.42	Tension Insulator Fittings
200	18 + 1/3.86	Bare PVC	1959 1958	2.42 2.42	Vibration Limit

4.2 Crossarm Assemblies

The minimum factor of safety on all crossarm assemblies is 2.5 on the ultimate tensile strength of the steel.

Crossarm assemblies in general are designed to support conductor downpull on each side of the support corresponding to a gradient of 1 in 10 on the maximum span length.

Exceptions are:

- 4.2.1 As shown in Tables 1, 2 and 3 for intermediate and pin angle crossarm.
- 4.2.2 Limit of 1 in 13 for the section angle crossarm for angles of deviation above 60° with 175 mm² (30 + 7/2.79 mm) ACSR.

4.3 Supports

The stresses created in intermediate supports are bending stresses due to windage on conductors, the loading points being 340 mm and 420 mm above pole tops for outer and centre conductors respectively. The stresses created in stayed supports are crippling stresses due to stay tension and vertical conductor loads acting at the pole top. The distribution of stresses in the limbs of 'H' supports on which all stays are adjusted to equal tension is considered to be as follows:

Stays on outer limb of angle support	All vertical components of stay tension plus half vertical conductor load taken on outer limb.
One stay on outer limb and two stays on inner limb of angle support.	60 per cent of stress in inner limb. 40 per cent of stress in outer limb.
Stayed equally on each limb	50 per cent of stress in each limb.

The strength of supports is derived from information contained in BS 1990 with respect to red fir poles.

The factor of safety for supports in normal conditions is 2.5.

Permissible strut loadings are given in Table 10.

5. MATERIALS

All components used in the construction of lines to this Standard shall comply with British Standards and/or ESI Standards indicated.

5.1 Conductors

ACSR	— BS 215, Part 2
PVC-covered conductors (Type 16)	— BS 6485

During stranding all inner wires shall be coated with an approved grease in such manner that the interstices between wires are effectively filled with grease within the circle circumscribing the penultimate layer of the complete conductor.

For extra protection of bare conductors in corrosive areas and when specified by the purchaser all wires shall be greased and the conductor subsequently wiped to remove excess grease from the outer surface whilst maintaining grease filling in the inter-wire crevices.

5.2 Wood Poles

Poles shall be European redwood complying with the requirements of BS 1990. They shall be fabricated to Fig. 1 and 2, all holes being drilled to template, the centre line of the template being placed along the effective centre line of the pole. Pole markings shall be made on the same side of the pole as that on which the drilling templates have been used. Pole tops shall be evenly quartered, the angle at the apex being 120°. When fully seasoned and fabricated, poles shall be creosoted in accordance with BS 913 by Rueping Process. An alternative method of wood preservative may be applied in accordance with BS 3452 or BS 4072 if specified by the purchaser. Where 'H' poles are required, the limbs shall be suitably matched and marked.

5.3 Crossarm Steelwork

Crossarm steelwork shall comply with ESI Standard 43—95.

5.4 Insulators

String insulator units, pin and line post insulators shall comply with BS 137 and with ESI Standard 43—93.

5.5 Insulator Fittings

Insulator fittings shall comply with BS 3288.

5.6 Conductor Terminations, Joints and Binders

Conductor terminations, joints and binders shall comply with ESI Standard 43—92.

5.7 Stays

Stay strand and stay fittings shall comply with ESI Standard 43—91.

5.8 Fasteners and Washers

Fasteners and washers shall comply with ESI Standard 43—96.

6. TECHNICAL REQUIREMENTS FOR THE ERECTION OF LINES

6.1 Span Lengths

Recommended and maximum span lengths for normal construction shall be 130 m and 150 m respectively for all conductors.

The occasional long span shall be 150 m to 180 m in length and provide a minimum ratio of 0.4 between mid-span spacing and still air design sag at 50°C.

6.2 Supports

Supports shall be single or 'H' type poles as specified in Table 1 and erected in accordance with Fig. 3—10. Where an occasional long span is the first or last span of a line, an 'H' terminal may be used as one support subject to the ratio limitation detailed at 6.1.

6.3 Maximum Angles of Line Deviation

The maximum angles of line deviation on pin and line post insulators are given in Table 2 (ACSR bare) and Table 3 (ACSR PVC-covered). These tables also indicate the changeover point for the crossarm from single to double member.

The maximum angles of line deviation on tension insulator strings are given in Table 1.

6.4 Wind Loading Spans

Wind loading spans are given in the following tables:

Table 4	—	Single Poles	—	ACSR Bare
Table 5	—	Single Poles	—	ACSR PVC-covered
Table 6	—	'H' Poles	—	ACSR Bare
Table 7	—	'H' Poles	—	ACSR PVC-covered

The tables could also apply to pin angle and section angle poles having an angle of deviation below 5°. Poles are assumed to have minimum top dimensions in accordance with BS 1990.

6.5 Staying Arrangements and Strut Loading for Angle and Terminal Supports

Staying arrangements are given in Table 8 (ACSR Bare) and Table 9 (ACSR PVC-covered).

Stay assemblies shall be to ESI Standard 43—91.

The strut loading imposed on angle and terminal supports shall be obtained from Nomograms 1—10. These loadings shall then be related to the safe working loads of the poles by reference to Table 10. When out-of-balance tension exists on a section pole, balancing stays should be fitted as required.

6.6 Conductor Design and Erection Sags and Tensions

Conductor design sags and tensions are given in Tables 13—18.

Ordinates for plotting uplift curves and ground clearance curves are given in Table 11 (ACSR Bare) and Table 12 (ACSR PVC-covered).

Ordinates for plotting uplift curves are based on conductor design tensions.

Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long-term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR	—	20 per cent reduction
6 + 7 ACSR	—	10 per cent reduction
30 + 7 ACSR	—	5 per cent reduction

The conductors shall be pre-tensioned for a minimum time of 60 minutes. At 15-minute intervals during this period the tension or sag shall be adjusted in order to maintain the correct value.

The conductors shall be then finally erected with an overtension equivalent to 10 per cent at 15°C.

Conductor pre-tension and erection sags and tensions are given in Tables 19—24. Erection figures include the appropriate value of overtension.

6.7 Use of ACSR PVC-covered

Technical data is provided in this Standard covering the erection of full section(s) of line conductored with ACSR PVC-covered. When a single span of ACSR PVC-covered is erected in a section of line the span shall not exceed 60 per cent of the average span.

6.8 Provision of Danger Plates

Each support on a line to this Standard shall be provided with a danger plate to ESI Standard 43—97. The plate shall be mounted approximately 3 m above ground level.

6.9 Precautions Against Access

The types of support requiring precautions against access and the methods to be employed are detailed in ESI Standard 43—90.

6.10 Minimum Height of Line Conductors and Clearance

The minimum height of line conductors shall comply with the Electricity (Overhead Lines) Regulation, 1970. Data are given in this Standard for conductors operating at maximum temperatures of 50°C and 75°C.

6.11 Erection of Auxiliary Equipment

Where a support carries auxiliary equipment the construction of the support shall be in accordance with ESI Standard 43—89, which also contains details of the earthing of these supports.

No allowance has been made in the basic support consideration for the additional mechanical loadings which would be imposed by auxiliary equipment and additional conductors.

6.12 Relationships with other Bodies

Where lines to this Standard are erected over or alongside the plant of the Post Office or British Rail the provisions agreed shall be complied with. Additionally, the requirements of the British Waterways Board should be noted.

Table 1 — Support Types and Duties

Duty	Assembly Fig. No.	Conductor Size mm ²	Deviation Angle		Reference for Downpull, Angles and Stays Table No.
			Maximum	Limit on H Poles stayed on one limb	
Straight line. Small deviation angles. Limited downpull.	3 Single Pole	100	0 to 6°	—	2, 3, 8 & 9
		150	0 to 5°	—	
		175(18/1)	0 to 4°	—	
		175(30/7)	0 to 3°	—	
		200	0 to 4°	—	
Straight line. Small deviation angles. 1:10 downpull	4 Single Pole	100	5° to 6°	—	2, 3, 8 & 9
		150	4° to 5°	—	
		175(18/1)	3° to 4°	—	
		175(30/7)	2° to 3°	—	
		200	3° to 4°	—	
Straight line section. Medium deviation angles 1:10 downpull	5 Single Pole	100	60°	—	8 & 9
		150	30°	—	
		175(18/1)	30°	—	
		175(30/7)	30°	—	
		200	30°	—	
Heavy deviation angles 1:10 downpull	6 ‘H’ Pole	100	60°	60°	8 & 9
		150	60°	60°	
		175(18/1)	60°	53°	
		175(30/7)	60°	38°	
		200	60°	48°	
Extra heavy deviation angles 1:10 downpull 1:13 downpull 175(30/7) above 60°	7 ‘H’ Pole	100	75°	68°	8 & 9
		150	75°	60°	
		175(18/1)	75°	53°	
		175(30/7)	75°	38°	
		200	75°	48°	
Occasional long span. Straight line on pin or line post insulators. Straight line section. Heavy deviation angles. 1:10 downpull	8 ‘H’ pole	100	60°	60°	8 & 9
		150	60°	58°	
		175(18/1)	60°	51°	
		175(30/7)	60°	36°	
		200	60°	46°	
Termination for all conductors 1:10 downpull Terminal for occasional long span if adjacent support is Fig. 8 construction and mid- span spacing complies with ratio specified in 6.1.	9 ‘H’ Pole	100	—	—	8 & 9
		150			
		175(18/1)			
		175(30/7)			
		200			
Termination for 100 mm ² ACSR 1:10 downpull	10 Single Pole	100	—	—	8 & 9

Downpull Limits and Maximum Angles of Line Deviation on Pin and Line Post Insulators

Table 2 — Bare Conductor

ACSR				Maximum Angle of Line Deviation (°)	
Size (mm ²)	Stranding (mm)	Wind Span (m)	Gradient	Single Member Crossarm	Double Member Crossarm
100	6/4.72 + 7/1.57	130	0 — 1:30	6	6
			1:30 — 1:20	6	6
			1:20 — 1:10	3	6
		150	0 — 1:30	5	5
			1:30 — 1:20	5	5
			1:20 — 1:10	*	5
150	18 + 1/3.35	130	0 — 1:30	5	5
			1:30 — 1:20	5	5
			1:20 — 1:10	*	5
		150	0 — 1:30	4	4
			1:30 — 1:20	4	4
			1:20 — 1:10	*	4
175	18 + 1/3.61	130	0 — 1:30	4	4
			1:30 — 1:20	4	4
			1:20 — 1:10	*	4
		150	0 — 1:30	3	3
			1:30 — 1:20	3	3
			1:20 — 1:10	*	3
175	30 + 7/2.79	130	0 — 1:30	3	3
			1:30 — 1:20	*	3
			1:20 — 1:10	*	3
		150	0 — 1:30	2	2
			1:30 — 1:20	*	2
			1:20 — 1:10	*	2
200	18 + 1/3.86	130	0 — 1:30	4	4
			1:30 — 1:20	4	4
			1:20 — 1:10	*	4
		150	0 — 1:30	3	3
			1:30 — 1:20	2	3
			1:20 — 1:10	*	3

* Use double member crossarm (Fig. 4) for intermediate or pin angle conditions.

Downpull Limits and Maximum Angles of Line Deviation on Pin and Line Post Insulators.

Table 3 – PVC-covered Conductor

ACSR				Maximum Angle of Line Deviation (°)	
Size (mm ²)	Stranding (mm)	Wind Span (m)	Gradient	Single Member Crossarm	Double Member Crossarm
100	6/4.72 + 7/1.57	130	0 – 1:30	5	5
			1:30 – 1:20	5	5
			1:20 – 1:10	*	5
		150	0 – 1:30	4	4
			1:30 – 1:20	4	4
			1:20 – 1:10	*	4
150	18 + 1/3.35	130	0 – 1:30	4	4
			1:30 – 1:20	4	4
			1:20 – 1:10	*	4
		150	0 – 1:30	3	3
			1:30 – 1:20	3	3
			1:20 – 1:10	*	3
175	18 + 1/3.61	130	0 – 1:30	4	4
			1:30 – 1:20	4	4
			1:20 – 1:10	*	4
		150	0 – 1:30	3	3
			1:30 – 1:20	0	3
			1:20 – 1:10	*	3
175	30 + 7/2.79	130	0 – 1:30	2	3
			1:30 – 1:20	*	3
			1:20 – 1:10	*	3
		150	0 – 1:30	*	2
			1:30 – 1:20	*	2
			1:20 – 1:10	*	2
200	18 + 1/3.86	130	0 – 1:30	3	3
			1:30 – 1:20	1	3
			1:20 – 1:10	*	3
		150	0 – 1:30	2	2
			1:30 – 1:20	*	2
			1:20 – 1:10	*	2

* Use double member crossarm (Fig. 4) for intermediate or pin angle conditions.

Wind Loading Spans for Single Poles

Table 4 — Bare Conductor

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100 mm ² ACSR 6/4.72 + 7/1.57 mm	3 x 150 mm ² ACSR 18/3.35 + 1/3.35 mm	3 x 175 mm ² ACSR 18/3.61 + 1/3.61 mm	3 x 175 mm ² ACSR 30/2.79 + 7/2.79 mm	3 x 200 mm ² ACSR 18/3.86 + 1/3.86 mm
8.5	1.8	215 (M)	69	64	61	59	59
		265 (S)	133	123	119	114	115
		275	149	139	134	129	129
		285	150	150	150	144	145
		295	—	—	—	150	150
9.0	1.8	220 (M)	69	64	61	59	59
		275 (S)	139	129	124	120	120
		285	150	144	139	134	135
		295	—	150	150	149	150
		305	—	—	—	150	—
9.5	1.8	225 (M)	69	64	61	59	59
		280 (S)	137	127	123	118	119
		290	150	142	137	132	133
		300	—	150	150	146	147
		310	—	—	—	150	150
10.0	1.8	230 (M)	69	64	61	59	59
		285 (S)	136	126	122	117	118
		295	150	141	136	130	131
		305	—	150	149	144	145
		315	—	—	150	150	150
10.5	1.8	235 (M)	69	64	62	59	60
		290 (S)	135	125	121	116	117
		300	150	139	134	129	130
		310	—	150	147	141	142
		320	—	—	150	150	150
11.0	1.8	240 (M)	69	64	62	60	60
		295 (S)	134	125	120	116	116
		305	148	138	133	128	128
		310	150	150	145	140	140
		325	—	—	150	150	150
11.5	1.8	245 (M)	69	64	62	60	60
		300 (S)	134	124	120	115	116
		310	147	136	132	126	127
		320	150	148	143	138	139
		330	—	150	150	149	150
		340	—	—	—	150	—

Table 4 (Cont'd.)

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100mm ² ACSR 6/4.72 + 7/1.57mm	3 x 150mm ² ACSR 18/3.35 + 1/3.35mm	3 x 175mm ² ACSR 18/3.61 + 1/3.61mm	3 x 175mm ² ACSR 30/2.79 + 7/2.79mm	3 x 200mm ² ACSR 18/3.86 + 1/3.86mm
12.0	1.8	250 (M)	70	65	62	60	60
		305 (S)	133	123	119	114	115
		315	146	135	130	125	126
		325	150	147	142	136	137
		335	—	150	150	147	148
		345	—	—	—	150	150
13.0	1.8	260 (M)	71	66	64	61	62
		320 (S)	139	129	124	120	120
		330	150	140	135	130	131
		340	—	150	146	141	142
		350	—	—	150	150	150
14.0	1.8	275 (M)	76	71	68	66	66
		335 (S)	144	133	129	124	124
		345	150	144	139	133	134
		355	—	150	149	143	144
		365	—	—	150	150	150
15.0	2.1	290 (M)	82	76	73	70	71
		350 (S)	148	137	132	127	128
		360	150	147	141	136	137
		370	—	150	150	145	146
		380	—	—	—	150	150
16.0	2.1	305 (M)	88	81	78	75	76
		365 (S)	150	143	138	133	133
		375	—	150	147	141	142
		385	—	—	150	150	150
17.0	2.1	320 (M)	95	88	85	82	82
		375 (S)	150	142	137	131	132
		385	—	150	145	140	141
		395	—	—	150	148	149
		405	—	—	—	150	150
18.0	2.1	330 (M)	96	89	86	82	83
		390 (S)	150	145	140	134	135
		400	—	150	148	142	143
		410	—	—	150	150	150
20.0	2.1	360 (M)	104	96	93	89	90
		415 (S)	150	146	141	136	136
		425	—	150	148	143	143
		435	—	—	150	149	150
		445	—	—	—	150	—

NOTES:

1. (M) or (S) against a pole diameter indicates that the pole is standard medium or stout pole in accordance with BS 1990.
2. Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Wind Loading Spans for Single Poles

Table 5 — PVC-covered Conductor

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100mm ² ACSR 6/4.72 + 7/1.57mm	3 x 150mm ² ACSR 18/3.35 + 1/3.35mm	3 x 175mm ² ACSR 18/3.61 + 1/3.61mm	3 x 175mm ² ACSR 30/2.79 + 7/2.79mm	3 x 200mm ² ACSR 18/3.86 + 1/3.86mm
8.5	1.8	215 (M)	61	57	55	53	54
		265 (S)	118	111	107	103	104
		275	133	125	121	116	117
		285	149	140	135	130	131
		295	150	150	150	144	146
		305				150	150
9.0	1.8	220 (M)	61	57	55	53	54
		275 (S)	124	116	112	108	109
		285	139	130	126	121	122
		295	150	144	140	135	136
		305		150	150	148	149
		315				150	150
9.5	1.8	225 (M)	61	57	55	53	54
		280 (S)	122	114	111	107	108
		290	137	128	124	120	120
		300	150	141	137	132	133
		310		150	150	144	146
		320				150	150
10.0	1.8	230 (M)	61	57	55	53	54
		285 (S)	121	113	110	106	107
		295	135	126	122	118	119
		305	149	139	135	130	131
		315	150	150	147	142	143
		325			150	150	150
10.5	1.8	235 (M)	61	57	56	54	54
		290 (S)	120	113	109	105	106
		300	134	125	121	117	117
		310	147	137	133	128	129
		320	150	149	144	139	140
		330		150	150	150	150
11.0	1.8	240 (M)	62	58	56	54	54
		295 (S)	120	112	108	104	105
		305	132	124	120	115	116
		315	144	135	131	126	127
		325	150	146	142	137	138
		335		150	150	147	148
		345				150	150
11.5	1.8	245 (M)	62	58	56	54	54
		300 (S)	119	111	108	104	105
		310	131	122	119	114	115
		320	143	133	129	125	125
		330	150	144	140	135	136
		340		150	150	144	146
		350				150	150
12.0	1.8	250 (M)	62	58	56	54	55
		305 (S)	118	111	107	103	104
		315	130	121	118	113	114
		325	141	132	128	123	124
		335	150	142	138	133	134
		345		150	147	142	143
		355			150	150	150

(Cont'd)

Table 5 (Cont'd.)

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100mm ² ACSR 6/4.72 + 7/1.57mm	3 x 150mm ² ACSR 18/3.35 + 1/3.35mm	3 x 175mm ² ACSR 18/3.61 + 1/3.61mm	3 x 175mm ² ACSR 30/2.79 + 7/2.79mm	3 x 200mm ² ACSR 18/3.86 + 1/3.86mm
13.0	1.8	260 (M)	63	59	57	55	56
		320 (S)	124	116	112	108	109
		330	135	126	122	118	119
		340	146	136	132	127	128
		350	150	146	141	137	137
		360	—	150	150	146	147
		370	—	—	—	150	150
14.0	1.8	275 (M)	68	64	62	59	60
		335 (S)	128	120	116	112	113
		345	138	129	125	121	121
		355	148	138	134	129	130
		365	150	147	143	138	139
		375	—	150	150	146	147
		385	—	—	—	150	150
15.0	2.1	290 (M)	73	68	66	64	64
		350 (S)	132	123	119	115	116
		360	141	132	128	123	124
		370	150	140	136	131	132
		380	—	149	144	139	140
		390	—	150	150	147	148
		400	—	—	—	150	150
16.0	2.1	305 (M)	78	73	71	68	69
		365 (S)	137	128	124	120	121
		375	146	137	133	128	129
		385	150	145	141	136	137
		395	—	150	149	143	144
		405	—	—	150	150	150
17.0	2.1	320 (M)	85	79	77	74	74
		375 (S)	136	127	123	119	120
		385	145	135	131	126	127
		395	150	143	139	134	135
		405	—	150	146	141	142
		415	—	—	150	148	149
		425	—	—	—	150	150
18.0	2.1	330 (M)	85	80	77	74	75
		390 (S)	139	130	126	121	122
		400	147	137	133	128	129
		410	150	145	140	135	136
		420	—	150	147	142	143
		430	—	—	150	149	150
		440	—	—	—	150	—
20.0	2.1	360 (M)	93	87	84	81	81
		415 (S)	141	131	127	123	124
		425	148	138	134	129	130
		435	150	145	140	135	136
		445	—	150	146	141	142
		455	—	—	150	147	148
		465	—	—	—	150	150

NOTES:

1. (M) or (S) against a pole diameter indicates that the pole is the standard medium or stout pole in accordance with BS 1990.
2. Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Wind Loading Spans for 'H' Poles

Table 6 – Bare Conductor

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100mm ² ACSR 6/4.72 + 7/1.57mm	3 x 150mm ² ACSR 18/3.35 + 1/3.35mm	3 x 175mm ² ACSR 18/3.61 + 1/3.61mm	3 x 175mm ² ACSR 30/2.79 + 7/2.79mm	3 x 200mm ² ACSR 18/3.86 + 1/3.86mm
8.5	1.8	215 (M) 265 (S)	150 180	139 180	135 180	129 180	130 180
9.0	1.8	220 (M) 275 (S)	150 180	139 180	134 180	129 180	130 180
9.5	1.8	225 (M) 280 (S)	149 180	139 180	134 180	129 180	129 180
10.0	1.8	230 (M) 285 (S)	150 180	139 180	134 180	129 180	129 180
10.5	1.8	235 (M) 290 (S)	150 180	139 180	134 180	129 180	130 180
11.0	1.8	240 (M) 295 (S)	151 180	140 180	135 180	130 180	130 180
11.5	1.8	245 (M) 300 (S)	151 180	140 180	135 180	130 180	131 180
12.0	1.8	250 (M) 305 (S)	151 180	140 180	136 180	130 180	131 180
13.0	1.8	260 (M) 320 (S)	155 180	143 180	138 180	133 180	134 180
14.0	1.8	275 (M) 335 (S)	165 180	153 180	148 180	142 180	143 180
15.0	2.1	290 (M) 350 (S)	177 180	164 180	159 180	152 180	153 180
16.0	2.1	305 (M) 365 (S)	180 —	176 180	169 180	163 180	164 180
17.0	2.1	320 (M) 375 (S)	180 —	180 —	180 —	176 180	177 180
18.0	2.1	330 (M) 390 (S)	180 —	180 —	180 —	178 180	179 180
20.0	2.1	360 (M)	180	180	180	180	180

NOTES:

1. (M) or (S) against a pole diameter indicates that the pole is the standard medium or stout pole in accordance with BS 1990.
2. Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Wind Loading Spans for 'H' Poles

Table 7 – PVC-covered Conductor

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100mm ² ACSR 6/4.72 + 7/1.57mm	3 x 150mm ² ACSR 18/3.35 + 1/3.35mm	3 x 175mm ² ACSR 18/3.61 + 1/3.61mm	3 x 175mm ² ACSR 30/2.79 + 7/2.79mm	3 x 200mm ² ACSR 18/3.86 + 1/3.86mm
8.5	1.8	215 (M) 265 (S)	134 180	125 180	121 180	117 180	118 180
9.0	1.8	220 (M) 275 (S)	133 180	125 180	121 180	116 180	117 180
9.5	1.8	225 (M) 280 (S)	133 180	124 180	121 180	116 180	117 180
10.0	1.8	230 (M) 285 (S)	133 180	125 180	121 180	116 180	117 180
10.5	1.8	235 (M) 290 (S)	134 180	125 180	121 180	117 180	118 180
11.0	1.8	240 (M) 295 (S)	134 180	125 180	121 180	117 180	118 180
11.5	1.8	245 (M) 300 (S)	135 180	126 180	122 180	118 180	118 180
12.0	1.8	250 (M) 305 (S)	135 180	126 180	122 180	118 180	119 180
13.0	1.8	260 (M) 320 (S)	138 180	129 180	125 180	120 180	121 180
14.0	1.8	275 (M) 335 (S)	147 180	138 180	133 180	129 180	130 180
15.0	2.1	290 (M) 350 (S)	158 180	148 180	143 180	138 180	139 180
16.0	2.1	305 (M) 365 (S)	169 180	158 180	153 180	147 180	148 180
17.0	2.1	320 (M) 375 (S)	180 —	171 180	165 180	159 180	160 180
18.0	2.1	330 (M) 390 (S)	180 —	172 180	166 180	161 180	162 180
20.0	2.1	360 (M) 415 (S)	180 —	180 —	180 —	174 180	175 180

NOTES:

- (M) or (S) against a pole diameter indicates that the pole is the standard medium or stout pole in accordance with BS 1990.
- Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Stay Details (Excluding strut loading considerations)

Table 8 — Bare Conductor

Conductors	Angle of Stay Slope	Maximum Angles of Line Deviation (°)															
		Grade 700 Stay Strand								Grade 1150 Stay Strand							
		7/4.00 mm Stays				19/3.55 mm Stays				7/3.25 mm Stays				7/4.00 mm Stays			
		1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
3 x 100mm ² ACSR 6/4.72 + 7/1.57 mm	30°	9	24	41	58	22	50	75	—	10	26	44	62	19	45	75	—
	35°	11	29	49	70	26	61	75	—	12	31	53	75	24	54	75	—
	40°	14	34	57	75	31	68	75	—	15	36	61	75	27	63	75	—
	45°	16	39	64	75	34	68	75	—	17	41	69	75	31	68	75	—
3 x 150mm ² ACSR 18/3.35 + 1/3.35 mm	30°	7	21	36	51	19	44	75	—	8	22	39	54	17	40	67	75
	35°	10	25	43	61	23	53	75	—	11	27	46	65	21	48	75	—
	40°	12	30	50	71	27	60	75	—	13	32	53	75	24	55	75	—
	45°	14	34	57	75	30	60	75	—	15	36	60	75	27	60	75	—
3 x 175mm ² ACSR 18/3.61 + 1/3.61 mm	30°	6	19	32	45	17	40	66	75	7	20	35	49	15	36	60	75
	35°	8	23	39	55	21	48	75	—	9	24	41	58	18	43	72	75
	40°	10	27	45	63	24	53	75	—	11	29	48	68	21	50	75	—
	45°	12	30	51	72	27	53	75	—	13	33	54	75	24	53	75	—
3 x 175mm ² ACSR 30/2.79 + 7/2.79 mm	30°	5	14	24	34	13	29	48	68	5	15	26	36	11	26	44	61
	35°	6	17	29	40	15	35	58	75	7	18	31	43	14	32	52	73
	40°	8	20	33	46	18	38	67	75	8	21	35	50	16	37	60	75
	45°	9	22	37	52	20	38	75	—	10	24	40	56	18	38	68	75
3 x 200mm ² ACSR 18/3.86 + 1/3.86 mm	30°	6	17	29	41	15	36	60	75	6	18	31	44	14	32	54	75
	35°	8	21	35	49	19	43	72	75	8	22	37	53	17	39	64	75
	40°	9	24	41	57	22	48	75	—	10	26	43	61	19	45	75	—
	45°	11	27	46	65	25	48	75	—	12	29	49	70	22	48	75	—

Table 9 — PVC-covered Conductor

Conductors	Angle of Stay Slope	Maximum Angles of Line Deviation (°)															
		Grade 700 Stay Strand								Grade 1150 Stay Strand							
		7/4.00 mm Stays				19/3.55 mm Stays				7/3.25 mm Stays				7/4.00 mm Stays			
		1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
3 x 100mm ² ACSR 6/4.72 + 7/1.57 mm	30°	7	22	39	54	20	48	75	—	8	24	42	59	18	43	73	75
	35°	10	27	47	67	24	58	75	—	11	29	50	71	21	52	75	—
	40°	12	32	54	75	28	67	75	—	13	34	58	75	25	60	75	—
	45°	14	37	61	75	32	68	75	—	15	39	66	75	29	68	75	—
3 x 150mm ² ACSR 18/3.35 + 1/3.35 mm	30°	6	20	35	50	18	43	73	75	7	21	38	53	16	39	66	75
	35°	8	24	42	60	22	52	75	—	9	26	45	64	19	47	75	—
	40°	10	29	49	70	25	60	75	—	11	31	52	75	23	54	75	—
	45°	12	33	55	75	29	60	75	—	13	35	59	75	26	60	75	—
3 x 175mm ² ACSR 18/3.61 + 1/3.61 mm	30°	5	18	31	45	16	39	66	75	6	19	34	48	14	35	59	75
	35°	7	22	38	54	19	47	75	—	8	24	41	58	17	42	71	75
	40°	9	26	44	63	23	53	75	—	10	28	47	67	20	49	75	—
	45°	11	30	50	71	26	53	75	—	12	32	53	75	23	53	75	—
3 x 175mm ² ACSR 30/2.79 + 7/2.79 mm	30°	4	13	23	33	12	29	48	67	4	14	25	35	10	26	43	60
	35°	5	16	28	39	14	34	57	75	6	17	30	42	13	31	51	71
	40°	6	19	32	46	17	38	66	75	7	21	35	49	15	36	59	75
	45°	8	22	37	52	19	38	75	—	9	23	39	55	17	38	67	75
3 x 200mm ² ACSR 18/3.86 + 1/3.86 mm	30°	5	16	28	40	14	35	59	75	5	17	31	43	12	32	53	75
	35°	6	20	34	49	17	42	71	75	7	21	37	52	15	38	64	75
	40°	8	23	40	57	21	48	75	—	9	25	43	61	18	44	74	75
	45°	10	27	45	64	23	48	75	—	11	29	48	69	21	48	75	—

Notes on Tables 8 & 9

1. Terminal stay requirements shall be matched to 60° angle of line deviation.
2. Maximum angles of line deviation shall be reduced by 2° for occasional long span requirements.
3. Reference must be made to Table 1 for limiting angles of deviation.

Table 10 — Permissible Strut Loading of Poles

Dia. at 1.5 m (mm)	Pole Class (BS 1990)	Pole Planting Depth (m)	Pole Length (m)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)
215	M	1.8	8.5	150	3388	165	4199	180	5148
220			9.0	150	3049	165	3774	180	4620
225			9.5	150	2769	165	3423	180	4185
230			10.0	150	2536	165	3129	185	4076
235			10.5	150	2338	170	3082	190	3990
240			11.0	150	2169	170	2853	190	3688
245			11.5	150	2024	170	2657	190	3428
250			12.0	150	1898	170	2487	190	3203
260			13.0	160	1936	180	2505	200	3191
275			14.0	160	1809	180	2327	205	3123
290		2.1	15.0	165	1872	185	2394	210	3193
305			16.0	170	1887	190	2392	215	3159
320			17.0	180	2026	200	2539	220	3143
330			18.0	180	1886	200	2356	220	2909
360			20.0	180	1765	205	2301	230	2952
265	S	1.8	8.5	190	8308	215	11029	240	14370
275			9.0	190	7664	215	10138	240	13169
280			9.5	190	6903	215	9121	245	12442
285			10.0	190	6270	215	8273	245	11269
290			10.5	190	5736	220	7969	250	10797
295			11.0	190	5282	220	7325	250	9910
300			11.5	190	4891	220	6771	250	9147
305			12.0	190	4553	220	6292	250	8487
320			13.0	195	4351	225	5953	255	7961
335			14.0	195	3988	225	5426	255	7221
350		2.1	15.0	195	3801	225	5167	260	7193
365			16.0	200	3736	230	5032	265	6939
375			17.0	200	3424	230	4595	265	6314
390			18.0	200	3247	230	4334	265	5925
415			20.0	200	2901	230	3840	265	5206
				Poles with top dia. equal to min. specified in BS 1990		Poles with mean value of top dia.		Poles with top dia. equal to max. specified in BS 1990	

For notes on Table 10 see Page 25

NOTES ON TABLE 10 — PERMISSIBLE STRUT LOADING OF POLES

1. The values of strut loading are safe working loads.
2. The table shall be used in conjunction with Nomograms 1—10.
3. The weight of any pole mounted equipment shall be added to the value of strut loading taken from the appropriate nomogram before reference is made to the table.
4. Pole diameters to match wind loading spans shall be adopted for stayed poles not having additional pole mounted equipment where the angle of deviation is below 5°.
5. The maximum allowable downpull for 175 mm² (30 + 7/2.79 mm) ACSR shall be 1 in 13 for angles of deviation above 60°.
6. The limbs of 'H' poles shall be selected as follows:
 - (a) The total strut loading shall be derived from the appropriate nomogram.
 - (b) The total strut loading shall be apportioned as follows:

(i) Stays on outer limb only	—	100 per cent
(ii) Stayed equally on each limb	—	50 per cent
(iii) Two stays on inner limb and one stay on outer limb	—	60 per cent
 - (c) Both limbs of 'H' pole shall be of the same size, matched and designed to withstand apportioned strut loading.

Ordinates for Plotting Uplift and Ground Clearance Curves – ACSR Basic Span 130 m

Table 11 – Bare Conductor

Span (m) (2X)	Sag (m)								
	100mm ² 6/4.72 + 7/1.57mm			175mm ² 30 + 7/2.79mm			150mm ² , 175mm ² , 200mm ² 18 + 1/3.35mm 18 + 1/3.61mm 18 + 1/3.86mm		
	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C
40	0.10	0.25	0.28	0.11	0.22	0.26	0.12	0.28	0.33
80	0.40	0.93	1.12	0.43	0.87	1.05	0.48	1.13	1.31
120	0.91	2.09	2.52	0.98	1.97	2.36	1.09	2.54	2.95
160	1.62	3.71	4.48	1.74	3.49	4.20	1.94	4.51	5.24
200	2.53	5.80	7.00	2.72	5.46	6.56	3.02	7.05	8.18
240	3.64	8.35	10.08	3.91	7.86	9.45	4.35	10.15	11.78
280	4.95	11.37	13.72	5.33	10.70	12.86	5.93	13.82	16.03
320	6.47	14.85	17.92	6.96	13.97	16.79	7.74	18.05	20.94
360	8.10	18.80	22.68	8.80	17.69	21.25	9.80	22.84	26.50
400	10.11	23.21	28.00	10.87	21.84	26.23	12.10	28.20	32.72
440	12.23	28.08	33.87	13.15	26.42	31.74	14.64	34.12	39.59
480	14.56	33.42	40.31	15.65	31.44	37.78	17.42	40.61	47.12
520	17.08	39.22	47.31	18.37	36.90	44.34	20.44	47.66	55.30
560	19.81	45.48	54.87	21.30	42.80	51.42	23.71	55.27	64.13
600	22.74	52.21	62.99	24.45	49.13	59.03	27.21	63.45	73.62
640	25.88	59.40	71.66	27.82	55.89	67.16	30.96	72.19	83.76
680	29.21	67.06	80.90	31.41	63.10	75.82	34.95	81.50	94.56

NOTE: Ordinates for plotting uplift curves are based on conductor design tensions. Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR – 20% reduction
6 + 7 ACSR – 10% reduction
30 + 7 ACSR – 5% reduction

Ordinates for Plotting Uplift and Ground Clearance Curves —ACSR Basic Span 130m

Table 12 — PVC-covered Conductor

Span (m) (2X)	Sag (m)								
	100mm ² 6/4.72 + 7/1.57mm			175mm ² 30 + 7/2.79mm			150mm ² , 175mm ² , 200mm ² 18 + 1/3.35mm 18 + 1/3.61mm 18 + 1/3.86mm		
	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C
40	0.15	0.27	0.32	0.14	0.25	0.29	0.17	0.33	0.37
80	0.59	1.09	1.26	0.57	1.00	1.16	0.67	1.31	1.47
120	1.32	2.46	2.84	1.29	2.25	2.61	1.51	2.95	3.31
160	2.35	4.38	5.05	2.29	4.01	4.65	2.68	5.24	5.88
200	3.67	6.84	7.89	3.58	6.26	7.26	4.18	8.19	9.19
240	5.28	9.84	11.36	5.16	9.01	10.45	6.03	11.79	13.23
280	7.19	13.40	15.46	7.02	12.26	14.22	8.20	16.05	18.01
320	9.39	17.50	20.20	9.17	16.02	18.58	10.71	20.96	23.52
360	11.88	22.15	25.56	11.61	20.27	23.51	13.56	26.53	29.77
400	14.67	27.34	31.56	14.33	25.03	29.03	16.74	32.75	36.75
440	17.75	33.08	38.18	17.34	30.28	35.12	20.25	39.63	44.47
480	21.12	39.37	45.44	20.64	36.04	41.80	24.10	47.16	52.92
520	24.79	46.21	53.33	24.22	42.30	49.05	28.29	55.35	62.11
560	28.75	53.59	61.85	28.09	49.05	56.89	32.81	64.19	72.04
600	33.00	61.52	71.00	32.25	56.31	65.31	37.66	73.69	82.69
640	37.55	69.99	80.78	36.69	64.07	74.30	42.85	83.84	94.09
680	42.39	79.02	91.19	41.42	72.33	83.88	48.37	94.65	106.22

NOTE: Ordinates for plotting uplift curves are based on conductor design tensions. Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR — 20% reduction
6 + 7 ACSR — 10% reduction
30 + 7 ACSR — 5% reduction

Design Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR — Basic Span 130 m

Table 13 — Bare Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	722.4	0.11	0.17	0.25	0.34	0.45	0.56	0.70	0.84	1.00	1.18	1.36	1.57	1.78	2.01	2.25
5.0	666.9	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.91	1.09	1.27	1.48	1.70	1.93	2.18	2.44
10.0	616.6	0.13	0.20	0.29	0.40	0.52	0.66	0.82	0.99	1.17	1.38	1.60	1.83	2.09	2.36	2.64
15.0	571.4	0.14	0.22	0.32	0.43	0.56	0.71	0.88	1.06	1.27	1.49	1.72	1.98	2.25	2.54	2.85
20.0	531.2	0.15	0.24	0.34	0.46	0.61	0.77	0.95	1.14	1.36	1.60	1.85	2.13	2.42	2.73	3.06
25.0	495.7	0.16	0.25	0.37	0.50	0.65	0.82	1.01	1.23	1.46	1.71	1.99	2.28	2.59	2.93	3.28
30.0	464.5	0.17	0.27	0.39	0.53	0.69	0.88	1.08	1.31	1.56	1.83	2.12	2.43	2.77	3.13	3.51
35.0	436.9	0.18	0.29	0.42	0.56	0.74	0.93	1.15	1.39	1.66	1.94	2.25	2.59	2.94	3.32	3.73
40.0	412.6	0.20	0.31	0.44	0.60	0.78	0.99	1.22	1.47	1.75	2.06	2.39	2.74	3.12	3.52	3.95
45.0	391.2	0.21	0.32	0.46	0.63	0.82	1.04	1.28	1.55	1.85	2.17	2.52	2.89	3.29	3.71	4.16
50.0	372.2	0.22	0.34	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.65	3.04	3.46	3.90	4.37
55.0	355.2	0.23	0.36	0.51	0.69	0.91	1.15	1.41	1.71	2.04	2.39	2.77	3.18	3.62	4.09	4.58
60.0	340.0	0.24	0.37	0.53	0.72	0.95	1.20	1.48	1.79	2.13	2.50	2.90	3.33	3.78	4.27	4.79
65.0	326.4	0.25	0.39	0.56	0.75	0.99	1.25	1.54	1.86	2.22	2.60	3.02	3.46	3.94	4.45	4.99
70.0	314.1	0.26	0.40	0.58	0.78	1.02	1.30	1.60	1.94	2.30	2.70	3.14	3.60	4.10	4.62	5.18
75.0	302.9	0.27	0.42	0.60	0.81	1.06	1.34	1.66	2.01	2.39	2.80	3.25	3.73	4.25	4.79	5.37

Design Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR — Basic Span 130 m

Table 14 — PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	707.2	0.16	0.25	0.35	0.48	0.63	0.80	0.99	1.19	1.42	1.67	1.93	2.22	2.53	2.85	3.20
5.0	666.9	0.17	0.26	0.38	0.51	0.67	0.85	1.05	1.27	1.51	1.77	2.05	2.35	2.68	3.02	3.39
10.0	630.4	0.18	0.28	0.40	0.54	0.71	0.90	1.11	1.34	1.59	1.87	2.17	2.49	2.83	3.20	3.58
15.0	597.4	0.19	0.29	0.42	0.57	0.75	0.95	1.17	1.41	1.68	1.97	2.29	2.63	2.99	3.37	3.78
20.0	567.6	0.20	0.31	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	2.41	2.76	3.15	3.55	3.98
25.0	540.7	0.21	0.32	0.47	0.63	0.83	1.04	1.29	1.56	1.86	2.18	2.53	2.90	3.30	3.73	4.18
30.0	516.5	0.22	0.34	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.65	3.04	3.46	3.90	4.38
35.0	494.5	0.23	0.35	0.51	0.69	0.90	1.14	1.41	1.71	2.03	2.38	2.76	3.17	3.61	4.08	4.57
40.0	474.5	0.24	0.37	0.53	0.72	0.94	1.19	1.47	1.78	2.12	2.48	2.88	3.31	3.76	4.25	4.76
45.0	456.4	0.25	0.38	0.55	0.75	0.98	1.24	1.53	1.85	2.20	2.58	3.00	3.44	3.91	4.42	4.95
50.0	439.8	0.25	0.40	0.57	0.78	1.01	1.28	1.59	1.92	2.28	2.68	3.11	3.57	4.06	4.58	5.14
55.0	424.7	0.26	0.41	0.59	0.80	1.05	1.33	1.64	1.99	2.37	2.78	3.22	3.70	4.20	4.75	5.32
60.0	410.8	0.27	0.43	0.61	0.83	1.09	1.38	1.70	2.05	2.45	2.87	3.33	3.82	4.35	4.91	5.50
65.0	398.0	0.28	0.44	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.96	3.44	3.94	4.49	5.07	5.68
70.0	386.2	0.29	0.45	0.65	0.89	1.16	1.46	1.81	2.19	2.60	3.05	3.54	4.06	4.62	5.22	5.85
75.0	375.2	0.30	0.47	0.67	0.91	1.19	1.51	1.86	2.25	2.68	3.14	3.64	4.18	4.76	5.37	6.02

Design Sag/Tension Tables for 175 mm² (30 + 7/2.79) ACSR — Basic Span 130 m

Table 15 — Bare Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)															
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0	
0.0	1480.5	0.12	0.18	0.27	0.36	0.47	0.60	0.74	0.89	1.06	1.25	1.45	1.66	1.89	2.13	2.39	
5.0	1382.6	0.13	0.20	0.28	0.39	0.51	0.64	0.79	0.96	1.14	1.34	1.55	1.78	2.02	2.28	2.56	
10.0	1292.9	0.14	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.22	1.43	1.66	1.90	2.16	2.44	2.74	
15.0	1211.4	0.14	0.23	0.32	0.44	0.58	0.73	0.90	1.09	1.30	1.52	1.77	2.03	2.31	2.61	2.92	
20.0	1137.6	0.15	0.24	0.35	0.47	0.61	0.78	0.96	1.16	1.38	1.62	1.88	2.16	2.46	2.78	3.11	
25.0	1071.3	0.16	0.25	0.37	0.50	0.65	0.83	1.02	1.23	1.47	1.72	2.00	2.29	2.61	2.95	3.30	
30.0	1011.8	0.17	0.27	0.39	0.53	0.69	0.87	1.08	1.31	1.55	1.82	2.12	2.43	2.76	3.12	3.50	
35.0	958.5	0.18	0.28	0.41	0.56	0.73	0.92	1.14	1.38	1.64	1.93	2.23	2.56	2.92	3.29	3.69	
40.0	910.7	0.19	0.30	0.43	0.59	0.77	0.97	1.20	1.45	1.73	2.03	2.35	2.70	3.07	3.47	3.89	
45.0	867.7	0.20	0.31	0.45	0.62	0.81	1.02	1.26	1.52	1.81	2.13	2.47	2.83	3.22	3.64	4.08	
50.0	829.1	0.21	0.33	0.47	0.65	0.84	1.07	1.32	1.59	1.90	2.23	2.58	2.96	3.37	3.81	4.27	
55.0	794.3	0.22	0.34	0.50	0.67	0.88	1.11	1.38	1.66	1.98	2.32	2.70	3.09	3.52	3.97	4.46	
60.0	762.8	0.23	0.36	0.52	0.70	0.92	1.16	1.43	1.73	2.06	2.42	2.81	3.22	3.67	4.14	4.64	
65.0	734.2	0.24	0.37	0.54	0.73	0.95	1.21	1.49	1.80	2.14	2.51	2.92	3.35	3.81	4.30	4.82	
70.0	708.1	0.25	0.39	0.56	0.76	0.99	1.25	1.54	1.87	2.22	2.61	3.02	3.47	3.95	4.46	5.00	
75.0	684.2	0.26	0.40	0.57	0.78	1.02	1.29	1.60	1.93	2.30	2.70	3.13	3.59	4.09	4.61	5.17	

Design Sag/Tension Tables for 175 mm² (30 + 7/2.79) ACSR — Basic Span 130 m

Table 16 — PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	1410.2	0.15	0.24	0.35	0.47	0.61	0.78	0.96	1.16	1.38	1.62	1.88	2.16	2.46	2.77	3.11
5.0	1334.8	0.16	0.25	0.36	0.50	0.65	0.82	1.01	1.23	1.46	1.71	1.99	2.28	2.59	2.93	3.28
10.0	1266.0	0.17	0.27	0.38	0.52	0.68	0.87	1.07	1.29	1.54	1.81	2.09	2.40	2.73	3.09	3.46
15.0	1203.3	0.18	0.28	0.40	0.55	0.72	0.91	1.12	1.36	1.62	1.90	2.20	2.53	2.88	3.25	3.64
20.0	1146.2	0.19	0.29	0.42	0.58	0.76	0.96	1.18	1.43	1.70	1.99	2.31	2.65	3.02	3.41	3.82
25.0	1094.4	0.20	0.31	0.44	0.61	0.79	1.00	1.24	1.50	1.78	2.09	2.42	2.78	3.16	3.57	4.00
30.0	1047.2	0.21	0.32	0.46	0.63	0.83	1.05	1.29	1.56	1.86	2.18	2.53	2.91	3.31	3.73	4.18
35.0	1004.2	0.22	0.34	0.48	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.64	3.03	3.45	3.89	4.36
40.0	965.0	0.22	0.35	0.50	0.69	0.90	1.14	1.40	1.70	2.02	2.37	2.75	3.15	3.59	4.05	4.54
45.0	929.2	0.23	0.36	0.52	0.71	0.93	1.18	1.46	1.76	2.10	2.46	2.85	3.28	3.73	4.21	4.72
50.0	896.4	0.24	0.38	0.54	0.74	0.97	1.22	1.51	1.83	2.17	2.55	2.96	3.39	3.86	4.36	4.89
55.0	866.2	0.25	0.39	0.56	0.77	1.00	1.26	1.56	1.89	2.25	2.64	3.06	3.51	4.00	4.51	5.06
60.0	838.5	0.26	0.40	0.58	0.79	1.03	1.31	1.61	1.95	2.32	2.73	3.16	3.63	4.13	4.66	5.23
65.0	812.8	0.27	0.42	0.60	0.82	1.06	1.35	1.66	2.01	2.40	2.81	3.26	3.74	4.26	4.81	5.39
70.0	789.1	0.27	0.43	0.62	0.84	1.10	1.39	1.71	2.07	2.47	2.90	3.36	3.86	4.39	4.95	5.55
75.0	767.1	0.28	0.44	0.63	0.86	1.13	1.43	1.76	2.13	2.54	2.98	3.46	3.97	4.51	5.10	5.71

Design Sag/Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 130 m

Table 17 — Bare Conductor

Temp (°C)	Tension (kgf)			Sag (m) for span lengths (m)														
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	787.9	906.9	1026.7	0.13	0.21	0.30	0.41	0.53	0.68	0.83	1.01	1.20	1.41	1.63	1.88	2.13	2.41	2.70
5.0	728.1	838.2	949.3	0.14	0.23	0.33	0.44	0.58	0.73	0.90	1.09	1.30	1.52	1.77	2.03	2.31	2.61	2.92
10.0	675.1	777.4	881.0	0.16	0.24	0.35	0.48	0.62	0.79	0.97	1.18	1.40	1.64	1.91	2.19	2.49	2.81	3.15
15.0	628.5	724.0	820.9	0.17	0.26	0.38	0.51	0.67	0.85	1.04	1.26	1.50	1.77	2.05	2.35	2.67	3.02	3.38
20.0	587.7	677.3	768.3	0.18	0.28	0.40	0.55	0.71	0.90	1.12	1.35	1.61	1.89	2.19	2.51	2.86	3.23	3.62
25.0	552.0	636.3	722.3	0.19	0.30	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	2.33	2.67	3.04	3.43	3.85
30.0	520.6	600.3	681.8	0.20	0.32	0.45	0.62	0.81	1.02	1.26	1.52	1.81	2.13	2.47	2.83	3.22	3.64	4.08
35.0	493.0	568.7	646.2	0.21	0.33	0.48	0.65	0.85	1.08	1.33	1.61	1.91	2.25	2.61	2.99	3.40	3.84	4.31
40.0	468.6	540.7	614.7	0.22	0.35	0.50	0.69	0.90	1.13	1.40	1.69	2.01	2.36	2.74	3.15	3.58	4.04	4.53
45.0	446.9	515.8	586.7	0.23	0.37	0.53	0.72	0.94	1.19	1.47	1.77	2.11	2.48	2.87	3.30	3.75	4.24	4.75
50.0	427.6	493.7	561.6	0.25	0.38	0.55	0.75	0.98	1.24	1.53	1.85	2.21	2.59	3.00	3.45	3.92	4.43	4.96
55.0	410.2	473.7	539.2	0.26	0.40	0.58	0.78	1.02	1.29	1.60	1.93	2.30	2.70	3.13	3.59	4.09	4.61	5.17
60.0	394.6	455.8	518.9	0.27	0.42	0.60	0.81	1.06	1.34	1.66	2.01	2.39	2.80	3.25	3.73	4.25	4.80	5.38
65.0	380.5	439.5	500.5	0.28	0.43	0.62	0.84	1.10	1.39	1.72	2.08	2.48	2.91	3.37	3.87	4.40	4.97	5.57
70.0	367.6	424.7	483.7	0.29	0.45	0.64	0.87	1.14	1.44	1.78	2.15	2.56	3.01	3.49	4.01	4.56	5.15	5.77
75.0	355.8	411.2	468.4	0.29	0.46	0.66	0.90	1.18	1.49	1.84	2.23	2.65	3.11	3.61	4.14	4.71	5.32	5.96

Design/Sag Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 130 m

Table 18 — PVC-covered Conductor

Temp (°C)	Tension (kgf)			Sag (m) for span lengths (m)														
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	770.3	887.2	1006.1	0.18	0.28	0.41	0.55	0.72	0.91	1.12	1.36	1.62	1.90	2.20	2.53	2.88	3.25	3.64
5.0	728.1	838.2	949.3	0.19	0.30	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	2.33	2.68	3.05	3.44	3.86
10.0	690.2	794.3	898.6	0.20	0.31	0.45	0.62	0.80	1.02	1.26	1.52	1.81	2.12	2.46	2.83	3.21	3.63	4.07
15.0	656.3	755.0	853.2	0.21	0.33	0.48	0.65	0.85	1.07	1.32	1.60	1.90	2.23	2.59	2.97	3.38	3.82	4.28
20.0	625.9	719.7	812.5	0.22	0.35	0.50	0.68	0.89	1.12	1.39	1.68	2.00	2.34	2.72	3.12	3.55	4.01	4.49
25.0	598.4	687.9	776.0	0.23	0.36	0.52	0.71	0.93	1.17	1.45	1.75	2.09	2.45	2.84	3.26	3.71	4.19	4.70
30.0	573.7	659.3	743.1	0.24	0.38	0.55	0.74	0.97	1.23	1.51	1.83	2.18	2.56	2.97	3.40	3.87	4.37	4.90
35.0	551.2	633.4	713.4	0.25	0.39	0.57	0.77	1.01	1.28	1.57	1.91	2.27	2.66	3.09	3.54	4.03	4.55	5.10
40.0	530.8	609.8	686.4	0.26	0.41	0.59	0.80	1.05	1.32	1.64	1.98	2.36	2.76	3.21	3.68	4.19	4.73	5.30
45.0	512.2	588.4	661.9	0.27	0.42	0.61	0.83	1.09	1.37	1.70	2.05	2.44	2.87	3.32	3.81	4.34	4.90	5.49
50.0	495.2	568.7	639.5	0.28	0.44	0.63	0.86	1.12	1.42	1.75	2.12	2.53	2.96	3.44	3.95	4.49	5.07	5.68
55.0	479.6	550.7	618.9	0.29	0.45	0.65	0.89	1.16	1.47	1.81	2.19	2.61	3.06	3.55	4.08	4.64	5.23	5.87
60.0	465.2	534.1	600.0	0.30	0.47	0.67	0.92	1.20	1.51	1.87	2.26	2.69	3.16	3.66	4.20	4.78	5.40	6.05
65.0	451.9	518.8	582.6	0.31	0.48	0.69	0.94	1.23	1.56	1.92	2.33	2.77	3.25	3.77	4.33	4.92	5.56	6.23
70.0	439.5	504.6	566.4	0.32	0.49	0.71	0.97	1.27	1.60	1.98	2.39	2.85	3.34	3.87	4.45	5.06	5.71	6.41
75.0	428.1	491.4	551.4	0.33	0.51	0.73	0.99	1.30	1.64	2.03	2.46	2.92	3.43	3.98	4.57	5.20	5.87	6.58

Pre-tension and Erection Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR – Basic Span 130 m

Table 19 – Bare Conductor

Pre-tension kgf (All Temps)	Sag (m) for span lengths (m) at all temperatures														
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
750	0.11	0.17	0.25	0.33	0.44	0.55	0.68	0.83	0.98	1.15	1.34	1.53	1.75	1.97	2.21

Erection Temp. (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	798.4	0.10	0.16	0.23	0.31	0.40	0.51	0.63	0.76	0.91	1.06	1.23	1.42	1.61	1.82	2.04
5.0	736.9	0.11	0.17	0.25	0.33	0.44	0.55	0.68	0.83	0.98	1.15	1.34	1.53	1.75	1.97	2.21
10.0	680.2	0.12	0.18	0.27	0.36	0.47	0.60	0.74	0.89	1.06	1.25	1.45	1.66	1.89	2.14	2.39
15.0	628.5	0.13	0.20	0.29	0.39	0.51	0.65	0.80	0.97	1.15	1.35	1.57	1.80	2.05	2.31	2.59
20.0	582.1	0.14	0.22	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.46	1.69	1.94	2.21	2.49	2.80
25.0	540.8	0.15	0.23	0.34	0.46	0.59	0.75	0.93	1.12	1.34	1.57	1.82	2.09	2.38	2.69	3.01
30.0	504.1	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.68	1.95	2.24	2.55	2.88	3.23
35.0	471.8	0.17	0.27	0.39	0.52	0.68	0.86	1.06	1.29	1.53	1.80	2.09	2.40	2.73	3.08	3.45
40.0	443.4	0.18	0.29	0.41	0.56	0.73	0.92	1.13	1.37	1.63	1.92	2.22	2.55	2.90	3.28	3.67
45.0	418.4	0.19	0.30	0.44	0.59	0.77	0.97	1.20	1.45	1.73	2.03	2.35	2.70	3.07	3.47	3.89
50.0	396.3	0.20	0.32	0.46	0.62	0.81	1.03	1.27	1.53	1.83	2.14	2.49	2.85	3.25	3.66	4.11
55.0	376.7	0.21	0.34	0.48	0.65	0.85	1.08	1.33	1.61	1.92	2.25	2.61	3.00	3.42	3.86	4.32
60.0	359.2	0.23	0.35	0.51	0.69	0.90	1.13	1.40	1.69	2.01	2.36	2.74	3.15	3.58	4.04	4.53
65.0	343.6	0.24	0.37	0.53	0.72	0.94	1.18	1.46	1.77	2.11	2.47	2.87	3.29	3.74	4.23	4.74
70.0	329.6	0.25	0.38	0.55	0.75	0.98	1.23	1.52	1.84	2.20	2.58	2.99	3.43	3.90	4.41	4.94
75.0	317.0	0.26	0.40	0.57	0.78	1.01	1.28	1.59	1.92	2.28	2.68	3.11	3.57	4.06	4.58	5.14

Pre-tension and Erection Tables for 100 mm² (6/4.72 + 7/1.57) ACSR — Basic Span 130 m

Table 20 — PVC-covered Conductor

Pre-tension kgf (All Temps)	Sag (m) for span lengths (m) at all temperatures														
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
750	0.15	0.24	0.34	0.46	0.60	0.76	0.94	1.14	1.36	1.59	1.85	2.12	2.41	2.73	3.06

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	787.0	0.14	0.22	0.32	0.43	0.57	0.72	0.89	1.07	1.28	1.50	1.74	1.99	2.27	2.56	2.87
5.0	739.7	0.15	0.24	0.34	0.46	0.60	0.76	0.94	1.14	1.36	1.59	1.85	2.12	2.41	2.73	3.06
10.0	696.4	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.69	1.96	2.25	2.56	2.89	3.24
15.0	657.1	0.17	0.27	0.38	0.52	0.68	0.86	1.06	1.28	1.53	1.79	2.08	2.39	2.72	3.07	3.44
20.0	621.6	0.18	0.28	0.40	0.55	0.72	0.91	1.12	1.36	1.62	1.90	2.20	2.52	2.87	3.24	3.64
25.0	589.5	0.19	0.30	0.43	0.58	0.76	0.96	1.18	1.43	1.70	2.00	2.32	2.66	3.03	3.42	3.83
30.0	560.5	0.20	0.31	0.45	0.61	0.80	1.01	1.24	1.51	1.79	2.10	2.44	2.80	3.19	3.60	4.03
35.0	534.3	0.21	0.33	0.47	0.64	0.84	1.06	1.31	1.58	1.88	2.21	2.56	2.94	3.34	3.77	4.23
40.0	510.6	0.22	0.34	0.49	0.67	0.87	1.11	1.37	1.65	1.97	2.31	2.68	3.07	3.50	3.95	4.43
45.0	489.2	0.23	0.36	0.52	0.70	0.91	1.16	1.43	1.73	2.05	2.41	2.79	3.21	3.65	4.12	4.62
50.0	469.7	0.24	0.37	0.54	0.73	0.95	1.20	1.49	1.80	2.14	2.51	2.91	3.34	3.80	4.29	4.81
55.0	452.0	0.25	0.39	0.56	0.76	0.99	1.25	1.54	1.87	2.22	2.61	3.02	3.47	3.95	4.46	5.00
60.0	435.8	0.26	0.40	0.58	0.78	1.02	1.30	1.60	1.94	2.30	2.70	3.14	3.60	4.10	4.63	5.19
65.0	421.0	0.27	0.42	0.60	0.81	1.06	1.34	1.66	2.00	2.39	2.80	3.25	3.73	4.24	4.79	5.37
70.0	407.4	0.27	0.43	0.62	0.84	1.10	1.39	1.71	2.07	2.47	2.89	3.36	3.85	4.38	4.95	5.55
75.0	394.9	0.28	0.44	0.64	0.87	1.13	1.43	1.77	2.14	2.54	2.99	3.46	3.97	4.52	5.11	5.72

Pre-tension and Erection Tables for 175 mm² (30 + 7/2.79) ACSR — Basic Span 130 m

Table 21 — Bare Conductor

Pre-tension kgf (All Temps)	Sag (m) for span lengths (m) at all temperatures															
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0	
1500	0.11	0.18	0.26	0.35	0.46	0.58	0.72	0.87	1.03	1.21	1.40	1.61	1.83	2.07	2.32	

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	1636.7	0.11	0.17	0.24	0.33	0.43	0.54	0.67	0.81	0.96	1.13	1.31	1.50	1.71	1.93	2.16
5.0	1527.4	0.11	0.18	0.26	0.35	0.46	0.58	0.72	0.87	1.03	1.21	1.40	1.61	1.83	2.07	2.32
10.0	1425.9	0.12	0.19	0.28	0.38	0.49	0.62	0.77	0.93	1.10	1.29	1.50	1.72	1.96	2.21	2.48
15.0	1332.5	0.13	0.20	0.30	0.40	0.52	0.66	0.82	0.99	1.18	1.39	1.61	1.84	2.10	2.37	2.66
20.0	1247.3	0.14	0.22	0.32	0.43	0.56	0.71	0.88	1.06	1.26	1.48	1.72	1.97	2.24	2.53	2.84
25.0	1170.0	0.15	0.23	0.34	0.46	0.60	0.76	0.93	1.13	1.34	1.58	1.83	2.10	2.39	2.70	3.03
30.0	1100.4	0.16	0.25	0.36	0.49	0.64	0.80	0.99	1.20	1.43	1.68	1.95	2.23	2.54	2.87	3.22
35.0	1037.9	0.17	0.26	0.38	0.52	0.67	0.85	1.05	1.27	1.52	1.78	2.06	2.37	2.69	3.04	3.41
40.0	981.9	0.18	0.28	0.40	0.55	0.71	0.90	1.11	1.35	1.60	1.88	2.18	2.50	2.85	3.22	3.61
45.0	931.6	0.19	0.29	0.42	0.57	0.75	0.95	1.17	1.42	1.69	1.98	2.30	2.64	3.00	3.39	3.80
50.0	886.6	0.20	0.31	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	2.42	2.77	3.15	3.56	3.99
55.0	846.1	0.21	0.32	0.46	0.63	0.83	1.05	1.29	1.56	1.86	2.18	2.53	2.91	3.31	3.73	4.18
60.0	809.6	0.22	0.34	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.64	3.04	3.45	3.90	4.37
65.0	776.7	0.23	0.35	0.51	0.69	0.90	1.14	1.41	1.70	2.03	2.38	2.76	3.16	3.60	4.07	4.56
70.0	746.8	0.23	0.37	0.53	0.72	0.94	1.18	1.46	1.77	2.11	2.47	2.87	3.29	3.75	4.23	4.74
75.0	719.6	0.24	0.38	0.55	0.74	0.97	1.23	1.52	1.84	2.19	2.57	2.98	3.42	3.89	4.39	4.92

Pre-tension and Erection Tables for 175 mm² (30 + 7/2.79) ACSR — Basic Span 130 m

Table 22 — PVC-covered Conductor

Pre-tension kgf (All Temps)	Sag (m) for span lengths (m) at all temperatures														
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
	0.15	0.23	0.33	0.45	0.59	0.74	0.91	1.11	1.32	1.55	1.79	2.06	2.34	2.64	2.96

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	1567.3	0.14	0.22	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.46	1.69	1.94	2.21	2.49	2.80
5.0	1479.1	0.15	0.23	0.33	0.45	0.59	0.74	0.91	1.11	1.32	1.55	1.79	2.06	2.34	2.64	2.96
10.0	1397.9	0.15	0.24	0.35	0.47	0.62	0.78	0.97	1.17	1.39	1.64	1.90	2.18	2.48	2.80	3.13
15.0	1323.6	0.16	0.26	0.37	0.50	0.65	0.83	1.02	1.24	1.47	1.73	2.00	2.30	2.62	2.95	3.31
20.0	1255.8	0.17	0.27	0.39	0.53	0.69	0.87	1.08	1.30	1.55	1.82	2.11	2.42	2.76	3.11	3.49
25.0	1194.0	0.18	0.28	0.41	0.56	0.72	0.92	1.13	1.37	1.63	1.91	2.22	2.55	2.90	3.27	3.67
30.0	1137.8	0.19	0.30	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	2.33	2.67	3.04	3.44	3.85
35.0	1086.7	0.20	0.31	0.45	0.61	0.80	1.01	1.24	1.51	1.79	2.10	2.44	2.80	3.19	3.60	4.03
40.0	1040.2	0.21	0.33	0.47	0.64	0.83	1.05	1.30	1.57	1.87	2.20	2.55	2.93	3.33	3.76	4.21
45.0	997.8	0.22	0.34	0.49	0.66	0.87	1.10	1.36	1.64	1.95	2.29	2.66	3.05	3.47	3.92	4.39
50.0	959.2	0.23	0.35	0.51	0.69	0.90	1.14	1.41	1.71	2.03	2.38	2.76	3.17	3.61	4.08	4.57
55.0	923.8	0.23	0.37	0.53	0.72	0.94	1.19	1.46	1.77	2.11	2.47	2.87	3.29	3.75	4.23	4.74
60.0	891.5	0.24	0.38	0.55	0.74	0.97	1.23	1.52	1.84	2.18	2.56	2.97	3.41	3.88	4.38	4.92
65.0	861.7	0.25	0.39	0.56	0.77	1.00	1.27	1.57	1.90	2.26	2.65	3.08	3.53	4.02	4.54	5.09
70.0	834.3	0.26	0.41	0.58	0.79	1.04	1.31	1.62	1.96	2.33	2.74	3.18	3.65	4.15	4.69	5.25
75.0	809.0	0.27	0.42	0.60	0.82	1.07	1.35	1.67	2.02	2.41	2.83	3.28	3.76	4.28	4.83	5.42

Pre-tension and Erection Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 130 m

Table 23 — Bare Conductor

Pre-tension kgf (All Temps)		Sag (m) for span lengths (m) at all temperatures															
150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
800	950	1050	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	1.59	1.83	2.08	2.35	2.63

Erection Temp. (°C)	Erection Tension (kgf)		Sag (m) for span lengths (m)															
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	878.0	1010.9	1145.0	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.91	1.08	1.26	1.47	1.68	1.92	2.16	2.42
5.0	808.7	931.2	1054.9	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	1.59	1.83	2.08	2.35	2.63
10.0	746.5	859.8	974.4	0.14	0.22	0.32	0.43	0.56	0.71	0.88	1.06	1.27	1.49	1.72	1.98	2.25	2.54	2.85
15.0	691.4	796.4	903.0	0.15	0.24	0.34	0.47	0.61	0.77	0.95	1.15	1.37	1.60	1.86	2.14	2.43	2.74	3.08
20.0	642.8	740.7	840.3	0.16	0.26	0.37	0.50	0.65	0.83	1.02	1.24	1.47	1.73	2.00	2.30	2.61	2.95	3.31
25.0	600.2	691.9	785.3	0.18	0.27	0.39	0.54	0.70	0.89	1.09	1.32	1.57	1.85	2.14	2.46	2.80	3.16	3.54
30.0	562.9	649.1	737.1	0.19	0.29	0.42	0.57	0.75	0.94	1.17	1.41	1.68	1.97	2.28	2.62	2.98	3.37	3.77
35.0	530.2	611.6	694.9	0.20	0.31	0.45	0.61	0.79	1.00	1.24	1.50	1.78	2.09	2.42	2.78	3.17	3.57	4.01
40.0	501.5	578.6	657.7	0.21	0.33	0.47	0.64	0.84	1.06	1.31	1.58	1.88	2.21	2.56	2.94	3.35	3.78	4.23
45.0	476.1	549.5	624.9	0.22	0.34	0.50	0.67	0.88	1.11	1.38	1.67	1.98	2.33	2.70	3.10	3.52	3.98	4.46
50.0	453.6	523.7	595.7	0.23	0.36	0.52	0.71	0.92	1.17	1.44	1.75	2.08	2.44	2.83	3.25	3.70	4.17	4.68
55.0	433.6	500.6	569.8	0.24	0.38	0.54	0.74	0.97	1.22	1.51	1.83	2.18	2.55	2.96	3.40	3.87	4.37	4.89
60.0	415.6	480.0	546.5	0.25	0.39	0.57	0.77	1.01	1.28	1.58	1.91	2.27	2.66	3.09	3.54	4.03	4.55	5.10
65.0	399.5	461.5	525.5	0.26	0.41	0.59	0.80	1.05	1.33	1.64	1.98	2.36	2.77	3.21	3.69	4.20	4.74	5.31
70.0	384.9	444.7	506.5	0.27	0.43	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.87	3.33	3.83	4.35	4.92	5.51
75.0	371.6	429.4	489.2	0.28	0.44	0.63	0.86	1.13	1.43	1.76	2.13	2.54	2.98	3.45	3.96	4.51	5.09	5.71

Pre-tension and Erection Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR – Basic Span 130 m

Table 24 – PVC-covered Conductor

Pre-tension kgf (All Temps)			Sag (m) for span lengths (m) at all temperatures														
150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
800	950	1050	0.17	0.27	0.39	0.52	0.68	0.87	1.07	1.29	1.54	1.81	2.10	2.41	2.74	3.09	3.47

Erection Temp (°C)	Erection Tension (kgf)			Sag (m) for span lengths (m)														
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	861.2	992.1	1125.2	0.16	0.25	0.36	0.49	0.64	0.81	1.01	1.22	1.45	1.70	1.97	2.26	2.57	2.91	3.26
5.0	809.7	932.3	1056.0	0.17	0.27	0.39	0.52	0.68	0.87	1.07	1.29	1.54	1.81	2.10	2.41	2.74	3.09	3.47
10.0	763.4	878.6	994.0	0.18	0.28	0.41	0.56	0.73	0.92	1.14	1.37	1.63	1.92	2.23	2.55	2.91	3.28	3.68
15.0	722.0	830.5	938.5	0.19	0.30	0.43	0.59	0.77	0.97	1.20	1.45	1.73	2.03	2.35	2.70	3.07	3.47	3.89
20.0	684.8	787.4	888.9	0.20	0.32	0.46	0.62	0.81	1.03	1.27	1.53	1.82	2.14	2.48	2.85	3.24	3.66	4.10
25.0	651.4	748.8	844.5	0.21	0.33	0.48	0.65	0.85	1.08	1.33	1.61	1.92	2.25	2.61	3.00	3.41	3.85	4.32
30.0	621.4	714.1	804.7	0.22	0.35	0.50	0.68	0.89	1.13	1.40	1.69	2.01	2.36	2.74	3.14	3.58	4.04	4.53
35.0	594.4	682.9	769.0	0.23	0.37	0.53	0.72	0.93	1.18	1.46	1.77	2.10	2.47	2.86	3.29	3.74	4.22	4.73
40.0	570.1	654.8	736.8	0.24	0.38	0.55	0.75	0.97	1.23	1.52	1.84	2.19	2.57	2.99	3.43	3.90	4.40	4.94
45.0	548.0	629.3	707.6	0.25	0.40	0.57	0.78	1.01	1.28	1.59	1.92	2.28	2.68	3.11	3.57	4.06	4.58	5.14
50.0	527.9	606.1	681.2	0.26	0.41	0.59	0.81	1.05	1.33	1.65	1.99	2.37	2.78	3.23	3.70	4.21	4.76	5.33
55.0	509.5	584.9	657.1	0.27	0.43	0.61	0.84	1.09	1.38	1.71	2.06	2.46	2.88	3.34	3.84	4.37	4.93	5.53
60.0	492.7	565.6	635.1	0.28	0.44	0.64	0.86	1.13	1.43	1.76	2.13	2.54	2.98	3.46	3.97	4.51	5.10	5.71
65.0	477.3	547.8	614.9	0.29	0.46	0.66	0.89	1.17	1.47	1.82	2.20	2.62	3.08	3.57	4.10	4.66	5.26	5.90
70.0	463.1	531.4	596.3	0.30	0.47	0.68	0.92	1.20	1.52	1.88	2.27	2.70	3.17	3.68	4.22	4.81	5.42	6.08
75.0	449.9	516.3	579.2	0.31	0.48	0.70	0.95	1.24	1.56	1.93	2.34	2.78	3.27	3.79	4.35	4.95	5.58	6.26

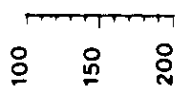
NOMOGRAM I

100mm'(6/4.72+7/1.57)ACSR.

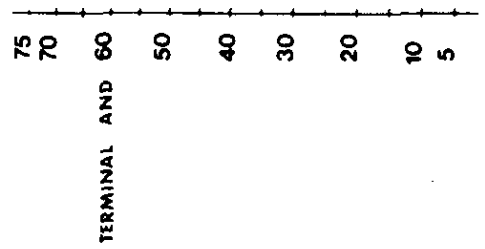
BARE

LEVEL GROUND

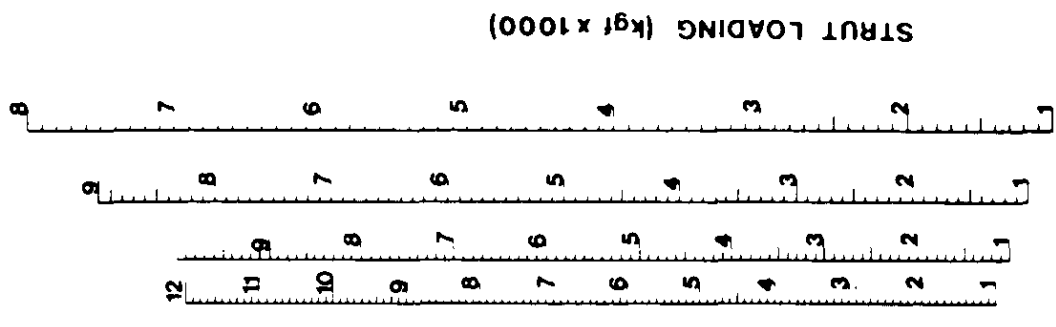
FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf
1 : 30	290
1 : 20	440
1 : 10	870



WIND
SPAN
(m)



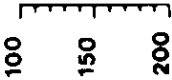
ANGLE OF
DEVIATION
(°)



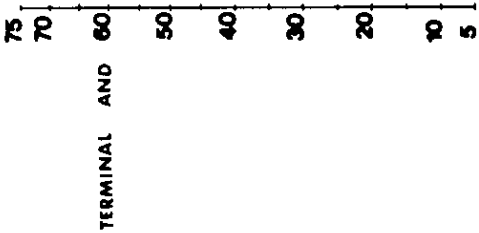
30° 35° 40° 45°
STAY SLOPE

NOMOGRAM 2
150mm' (18/3.35 + 1/3.35) ACSR
BARE
LEVEL GROUND

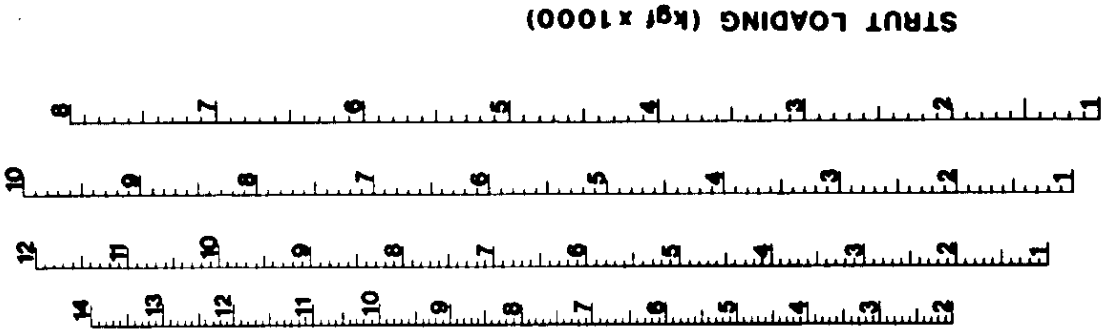
FOR DOWNPULL CONDITIONS		
DECLINATION	ADDITIONAL kgf	
1 : 30	330	
1 : 20	490	
1 : 10	970	



**WIND
SPAN
(m)**



**ANGLE OF
DEVIATION
(°)**



30° 35' 40' 45'
STAY SLOPE

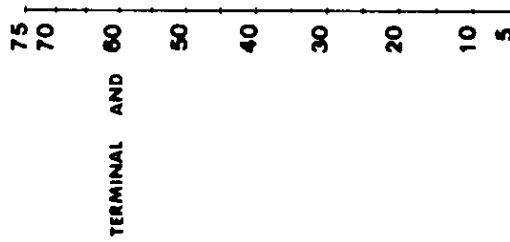
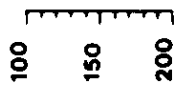
STRUT LOADING (kgf x 1000)

NOMOGRAM 3

175 mm² (18/3.61 + 1/3.61) A CSR.

BARE
 LEVEL GROUND

FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf
1 : 30	360
1 : 20	540
1 : 10	1070



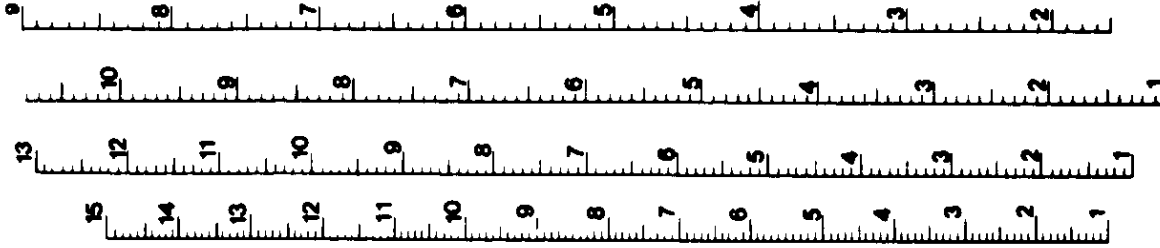
WIND
 SPAN
 (m)

ANGLE OF
 DEVIATION
 (°)

30° 35° 40° 45°
 STAY SLOPE

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STRUT LOADING (kgf x 1000)



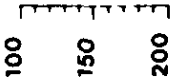
NOMOGRAM 4

175 mm' (30/2.79 + 7/2.79) ACSR

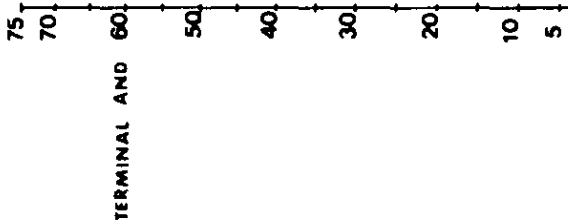
BARE

LEVEL GROUND

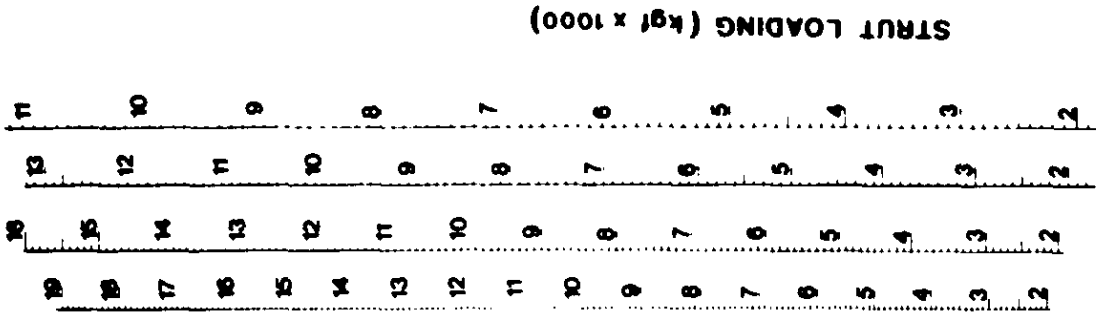
FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf
1 : 30	480
1 : 20	710
1 : 13	1100
1 : 10	1420



**WIND
SPAN
(m)**



**ANGLE OF
DEVIATION
(°)**



**30° 35° 40° 45°
STAY SLOPE**

STRUT LOADING (kgf x 1000)

NOMOGRAM 5
200mm' (18/3.86 + 1/3.86) ACSR
BARE
LEVEL GROUND

FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf
1 : 30	390
1 : 20	590
1 : 10	1170

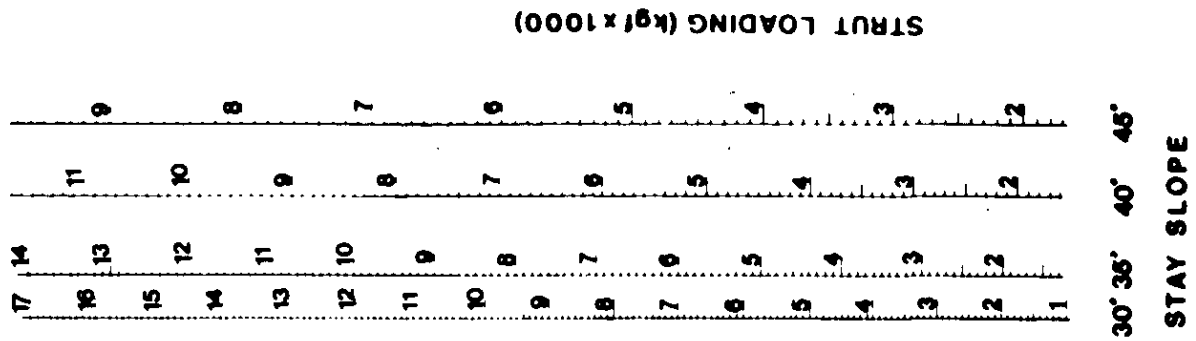
100
150
200

75
70
60
50
40
30
20
10
5

TERMINAL AND

WIND
SPAN
(m)

ANGLE OF
DEVIATION
(°)



NOMOGRAM 6

100mm' (6/4.72 + 7/1.57) ACSR

PVC COVERED

LEVEL GROUND

FOR DOWNPULL CONDITIONS		
DECLINATION	ADDITIONAL kgf.	
1 : 30	300	
1 : 20	450	
1 : 10	900	

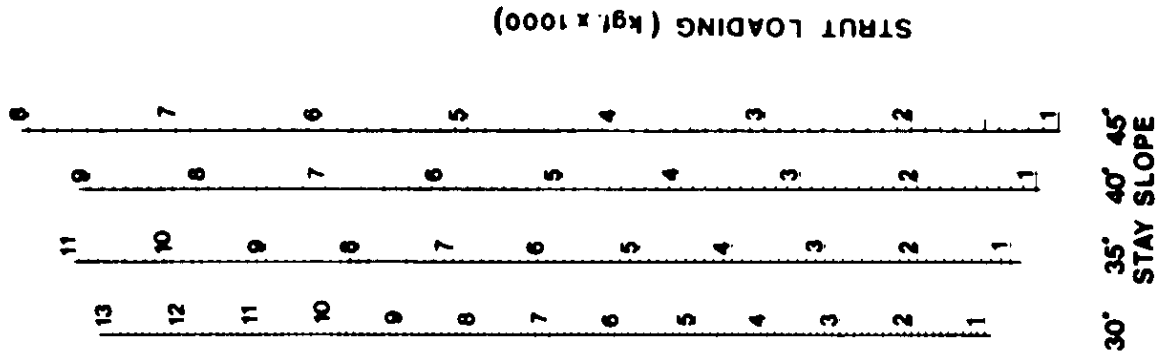
100
150
200

75
70
60
50
40
30
20
10
5

TERMINAL AND

WIND
SPAN
(m)

ANGLE OF
DEVIATION
(°)



NOMOGRAM 7
 150 mm' (18/ 3.35+ 1/3.35) ACSR
 PVC COVERED
 LEVEL GROUND

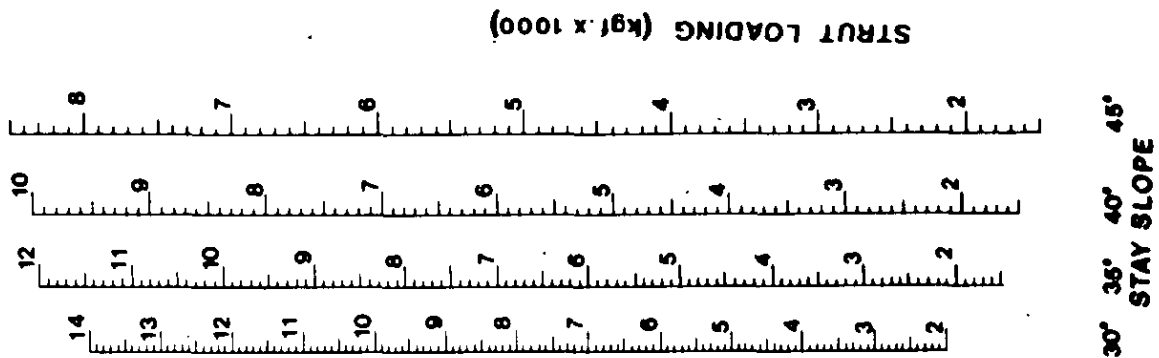
FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf.
1 : 30	330
1 : 20	490
1 : 10	980

100
 150
 200

75
 70
 60
 50
 40
 30
 20
 10
 5
 TERMINAL AND 60

WIND
 SPAN
 (m)

ANGLE OF
 DEVIATION
 (°)



NOMOGRAM 8
 175mm² (18/3.61/ +1/3.61) ACSR
 PVC COVERED
 LEVEL GROUND

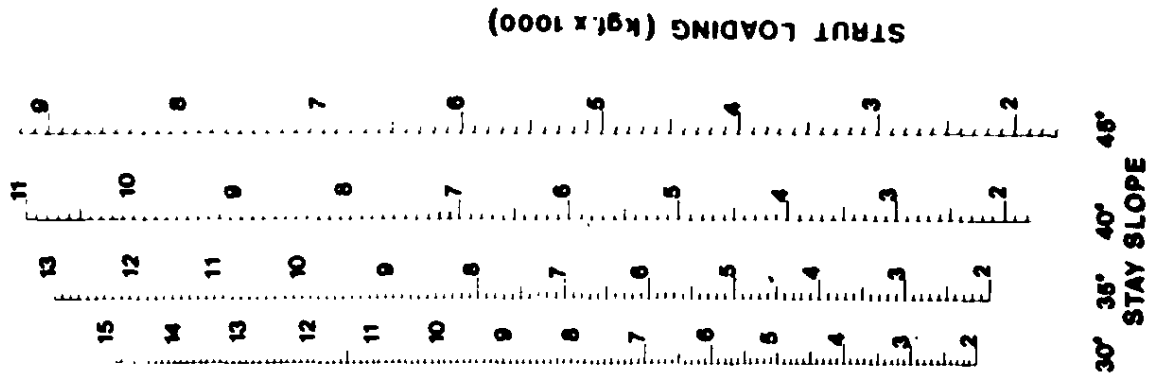
FOR DOWNPULL CONDITIONS		
DECLINATION	ADDITIONAL kgf.	
1 : 30	360	
1 : 20	540	
1 : 10	1070	

100
 150
 200

WIND
 SPAN
 (m)

75
 70
 60
 50
 40
 30
 20
 10
 5
 0
 TERMINAL AND 60

ANGLE OF
 DEVIATION
 (°)



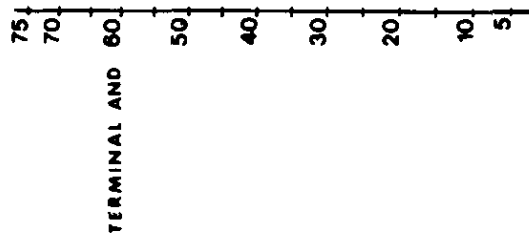
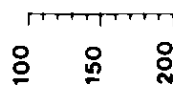
NOMOGRAM 9

175mm² (30/2.79 + 7/2.79) ACSR

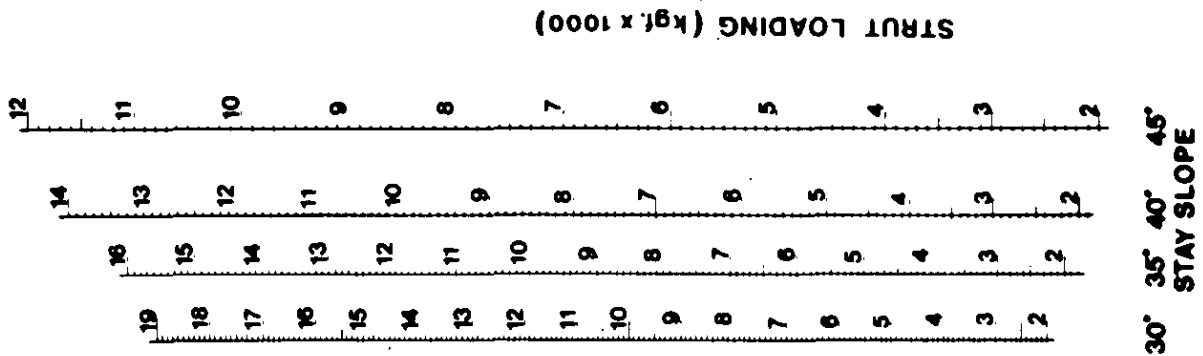
PVC COVERED

LEVEL GROUND

FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf.
1 : 30	480
1 : 20	710
1 : 13	1100
1 : 10	1420



TERMINAL AND



NOMOGRAM 10

200mm' (18/3.86 + 1/3.86) ACSR
PVC COVERED
LEVEL GROUND

FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf.
1 : 30	390
1 : 20	590
1 : 10	1170

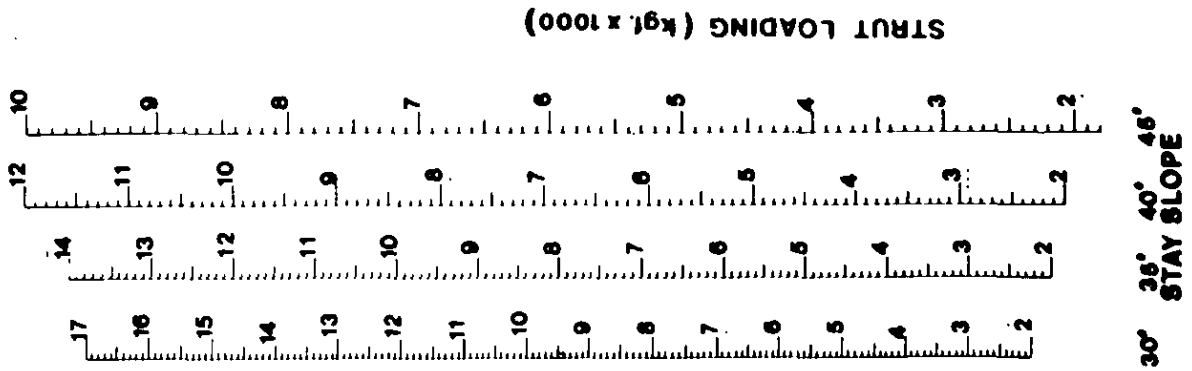
100
150
200

75
70
60
50
40
30
20
10
5

TERMINAL AND

WIND
SPAN
(m)

ANGLE OF
DEVIATION
(°)



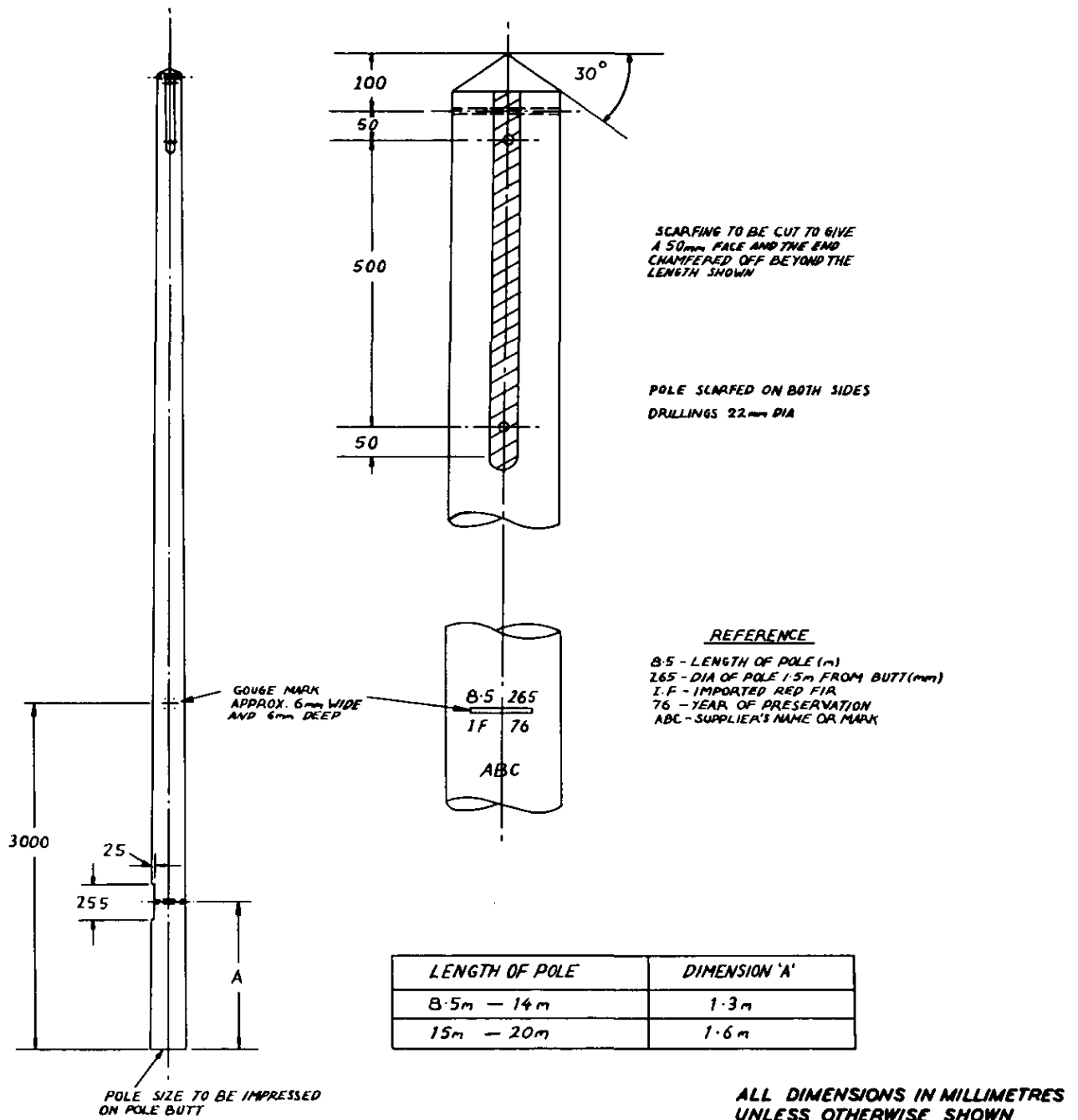


Fig. 1 Single pole drilling, scarfing and marking
(ESI Drawing 432001)

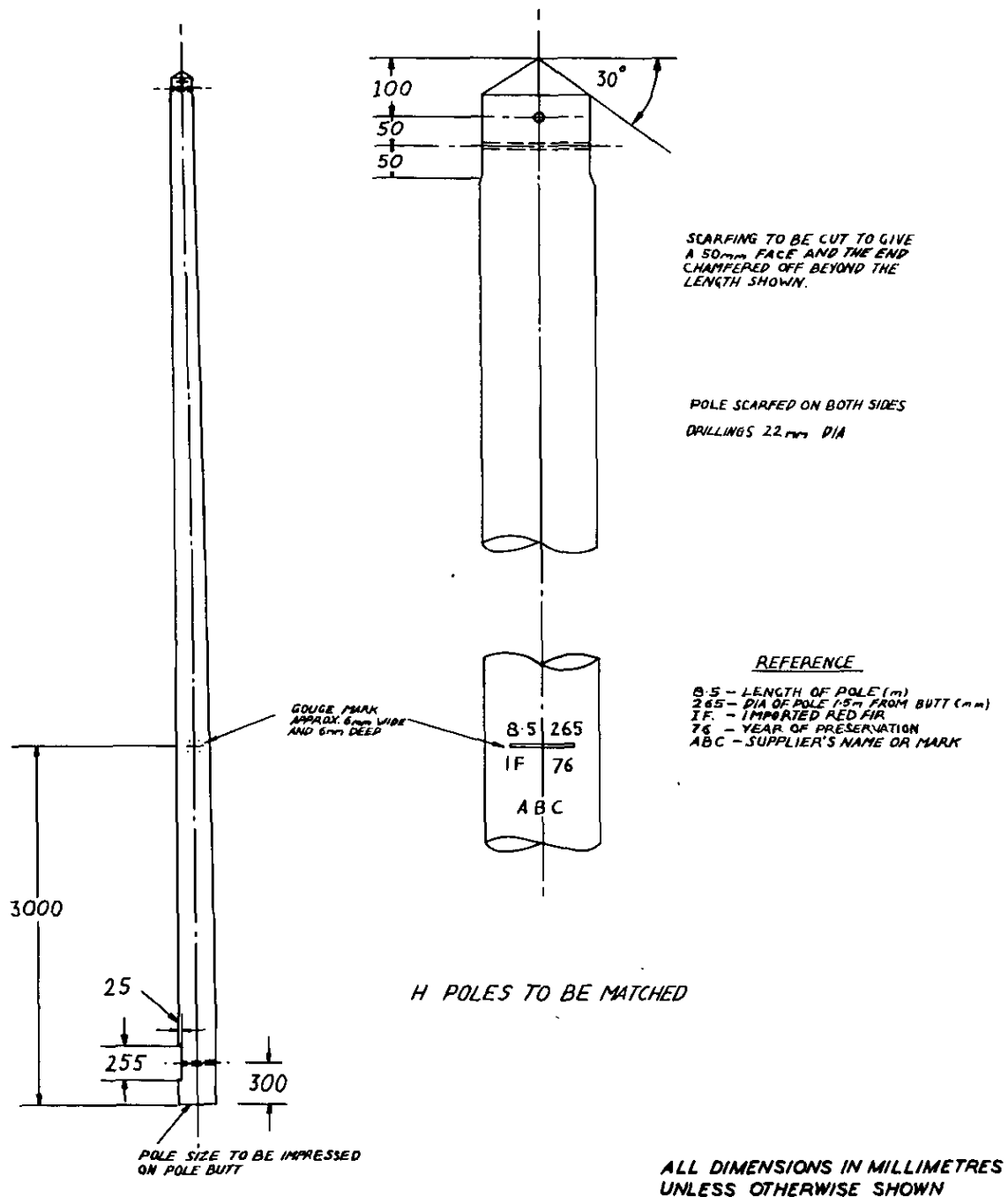
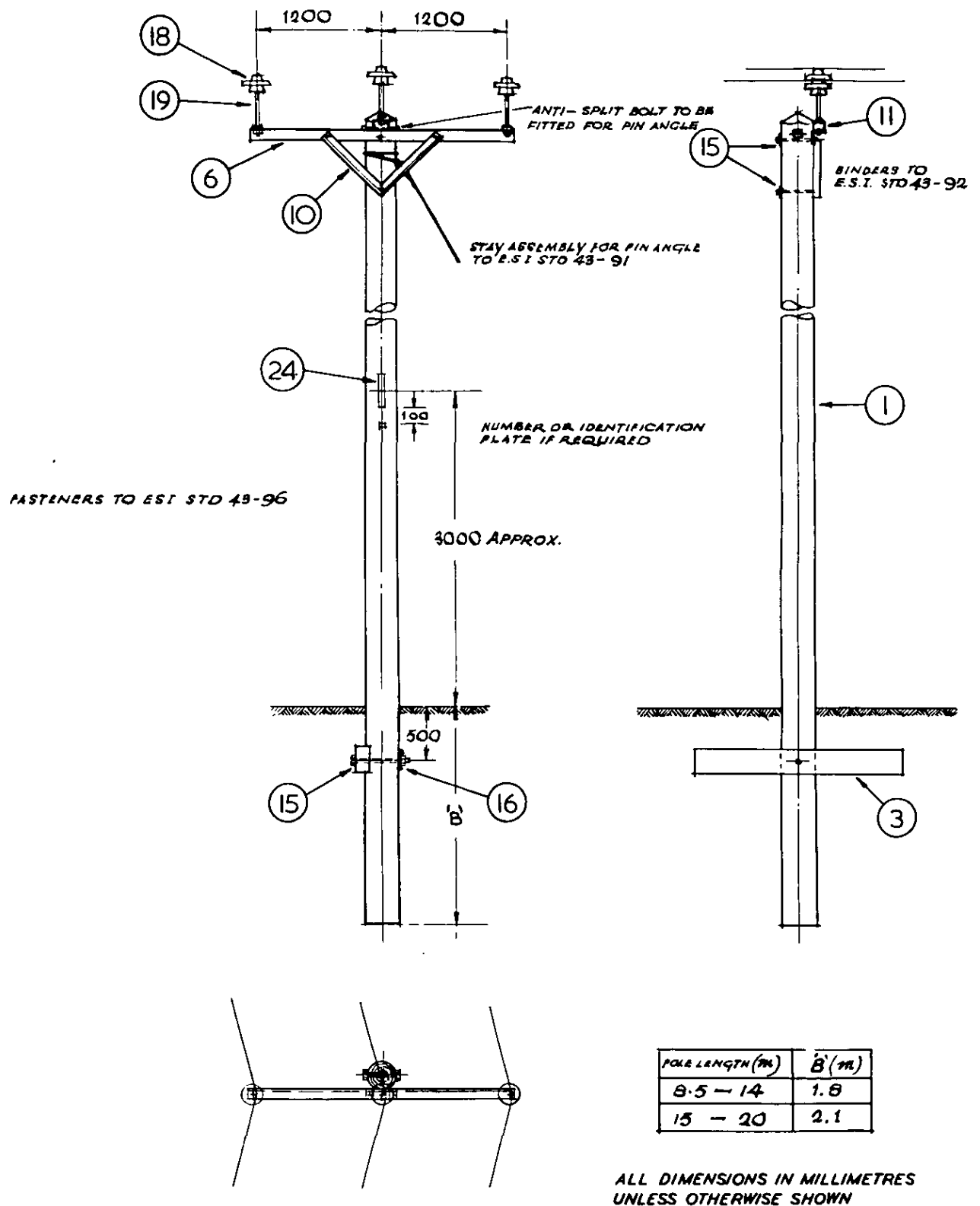


Fig. 2 'H' pole member drilling, scarfing and marking
(ESI Drawing 432002)



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Fig. 3 General assembly – intermediate or pin angle pole
(Single member crossarm)
(ESI Drawing 432003)

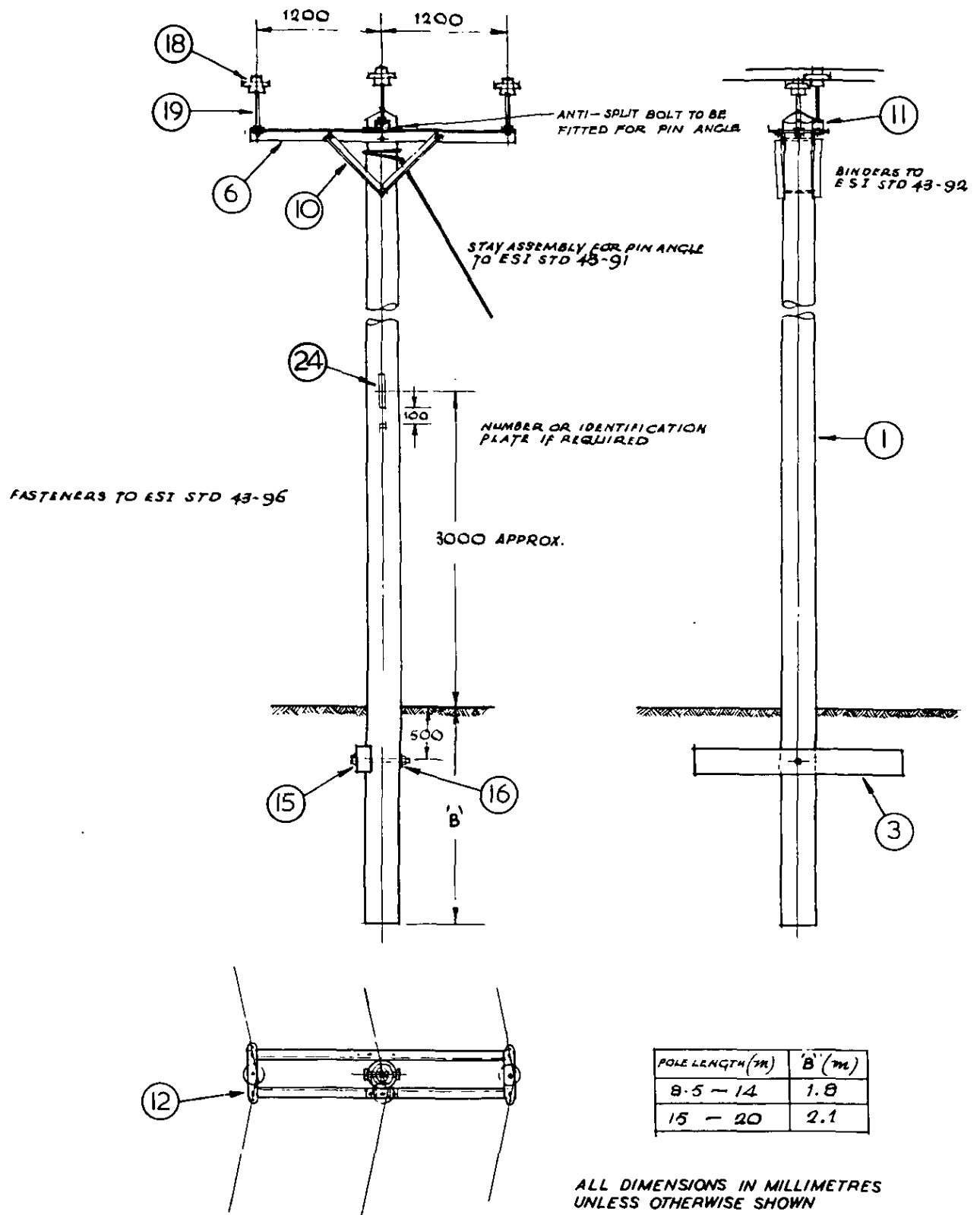
Material List for Fig. 3

Intermediate or Pin Angle Pole
(Single Member Crossarm)

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref.	M 20 Bolts
1	1	Wood Pole	432001	2 or 3 *
3	1	Kicking Block	439103 Type 2	1
6	1	Crossarm Member	439519	—
10	2	Crossarm Strut	439526	2
11	1	Insulator Bracket	439518	2
15	3	Square Flat Washer	439604	—
16	1	Square Curved Washer	439605	—
18	3	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —
19	3	11 kV Pin (50 mm shank) 33 kV Pin (50 mm shank)	BS 3288 Ref. 29 BS 3288 Ref. 31	— —
24	1	Danger Plate	439701	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Pin Angle — (i) *Anti-split bolt with two square curved washers to be fitted.
(ii) Additional round washers and locknuts required for bonding stay strand to pole top bolts.



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Fig. 4 General assembly – intermediate or pin angle pole
(Double member crossarm)
(ESI Drawing 432004)

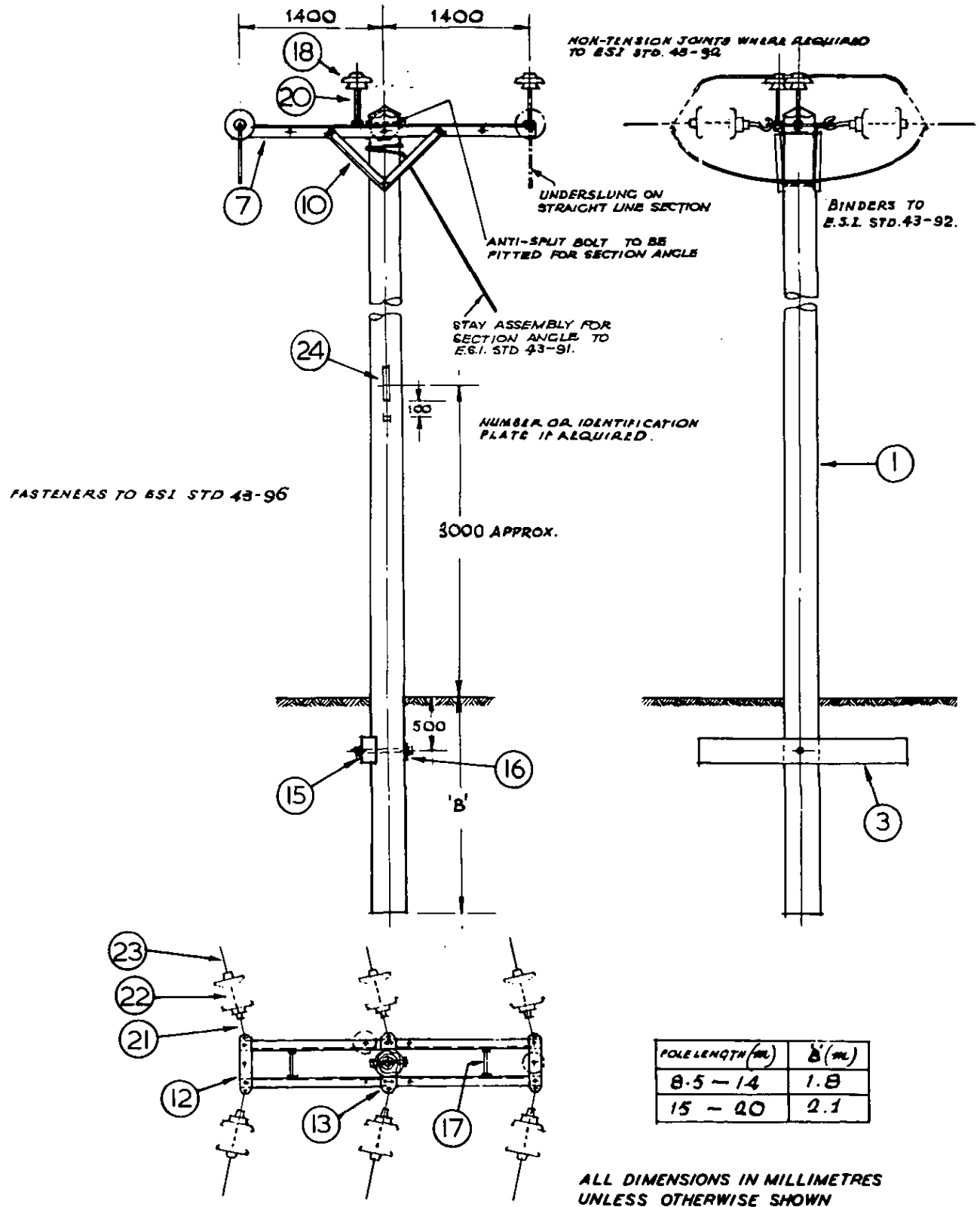
Material List for Fig. 4

Intermediate or Pin Angle Pole (Double Member Crossarm)

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
1	1	Wood Pole	432001	2 or 3*	—
3	1	Kicking Block	439103 Type 2	1	—
6	2	Crossarm Member	439519	—	—
10	4	Crossarm Strut	439526	4	—
11	1	Insulator Bracket	439518	2	—
12	2	Section Strap	439525	4	8
15	1	Square Flat Washer	439604	—	—
16	1	Square Curved Washer	439605	—	—
18	3	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —	— — —
19	3	11 kV Pin (50 mm shank) 33 kV Pin (50 mm shank)	BS 3288 Ref. 29 BS 3288 Ref. 31	— —	— —
24	1	Danger Plate	439701	—	—

NOTES:

- All bolts 60 mm long with the exception of pole bolts.
- Pin Angle — (i) *Anti-split bolt with two square curved washers to be fitted.
(ii) Additional round washers and locknuts required for bonding stay strand to pole top bolts.



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Fig. 5 General assembly — single pole straight line section or section angle (see Table 1)
(ESI Drawing 432005)

Material List for Fig. 5

Single Pole Straight Line Section or Section Angle

Item No.	Quantity	Description	ESI Std., Dwg. No or BS Ref	M 20 Bolts	Round Washers
1	1	Wood Pole	432001	2 or 3*	—
3	1	Kicking Block	439103 Type 2	1	—
7	2	Crossarm Member	439520	—	—
10	4	Crossarm Strut	439526	4	—
12	2	Section Strap	439525	4	4
13	2	Terminal Strap	439528	4	—
15	1	Square Flat Washer	439604	—	—
16	1	Square Curved Washer	439605	—	—
17	2	Tie Rod	439608 Type 3	—	—
18	1 or 2	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —	— — —
20	1 or 2	Pilot Pin (50 mm shank)	BS 3288 Ref. 50	—	—
21	6	Ball-ended Hook	BS 3288 Ref. 15/32	—	—
22	6	Insulator String	43-93	—	—
23	6	Conductor Termination	43-92	—	—
24	1	Danger Plate	439701	—	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Section Angle — (i) *Anti-split bolt with two square curved washers to be fitted.
(ii) Additional round washers and locknuts required for bonding stay strand to pole top bolts.

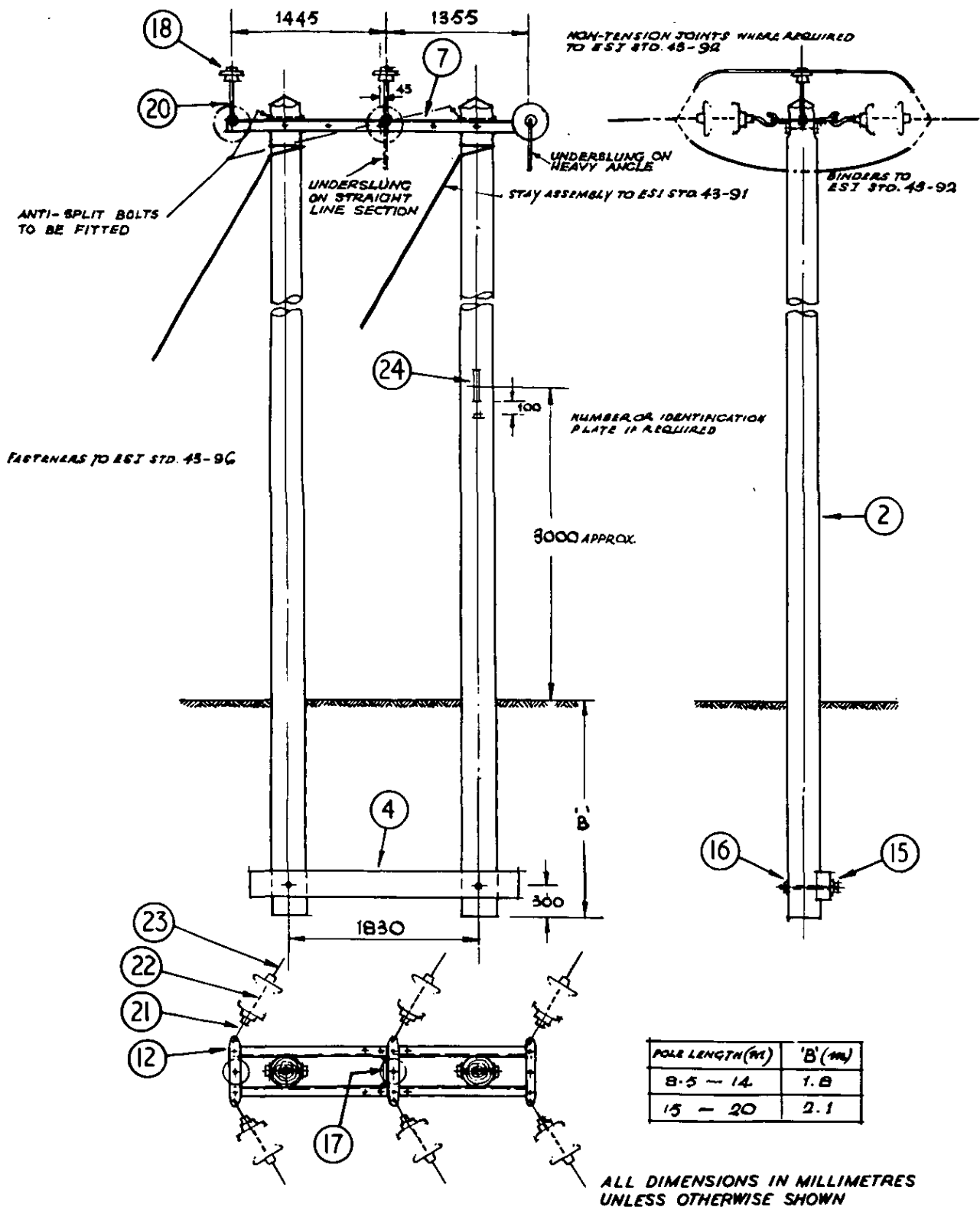


Fig. 6 General assembly - 'H' pole section angle up to 60°
(ESI Drawing 432006)

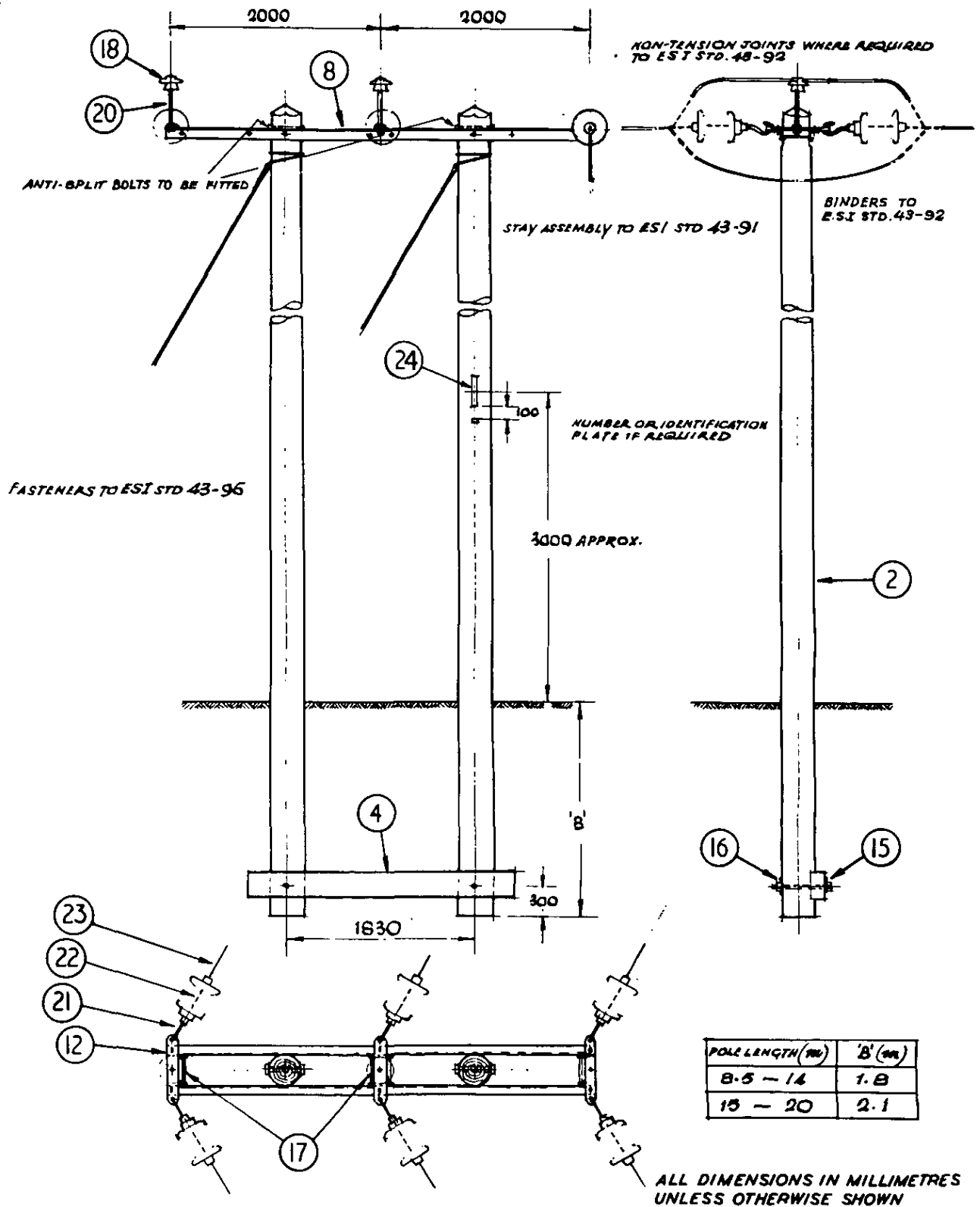
Material List for Fig. 6

'H' Pole Section Angle up to 60°

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
2	2	Wood Pole	432002	4 *	—
4	1	Brace Block	439112 Type 1	2	—
7	2	Crossarm Member	439520	—	—
12	3	Section Strap	439525	6	6
15	2	Square Flat Washer	439604	—	—
16	6	Square Curved Washer	439605	—	—
17	1	Tie Rod	439608 Type 3	—	—
18	2 or 3	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —	— — —
20	2 or 3	Pilot Pin (50 mm shank)	BS 3288 Ref. 50	—	—
21	6	Ball-ended Hook	BS 3288 Ref. 15/32	—	—
22	6	Insulator String	43-93	—	—
23	6	Conductor Termination	43-92	—	—
24	1	Danger Plate	439701	—	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Additional round washers and locknuts required for bonding stay strand to pole top bolts.
3. *Pole top bolts include two anti-split bolts each fitted with two square curved washers.



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Fig. 7 General assembly - 'H' pole section angle between 60° and 75°
(ESI Drawing 432007)

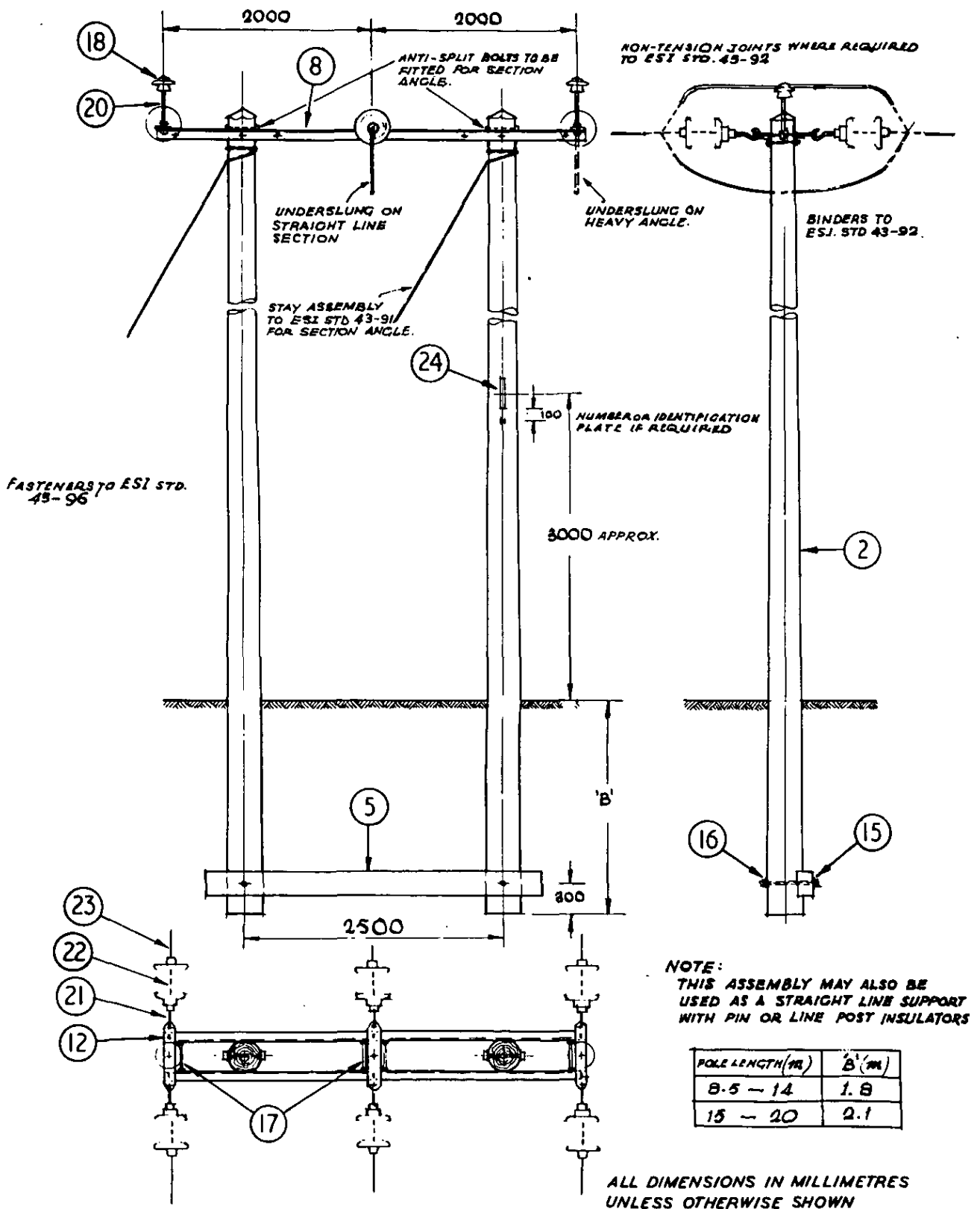
Material List for Fig. 7

'H' Pole Section Angle between 60° and 75°

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref	M 20 Bolts	Round Washers
2	2	Wood Pole	432002	4 *	—
4	1	Brace Block	439112 Type 1	2	—
8	2	Crossarm Member	439521	—	—
12	3	Section Strap	439525	6	6
15	2	Square Flat Washer	439604	—	—
16	6	Square Curved Washer	439605	—	—
17	4	Tie Rod	439608 Type 3	—	—
18	2	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —	— — —
20	2	Pilot Pin (55 mm shank)	BS 3288 Ref. 50	—	—
21	6	Ball-ended Hook	BS 3288 Ref. 15/32	—	—
22	6	Insulator String	43-93	—	—
23	6	Conductor Termination	43-92	—	—
24	1	Danger Plate	439701	—	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Additional round washers and locknuts required for bonding stay strand to pole top bolts.
3. *Pole top bolts include two anti-split bolts each fitted with two square curved washers.



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Fig. 8 General assembly — 'H' pole straight line section or section angle up to 60° for occasional long span (ESI Drawing 432008)

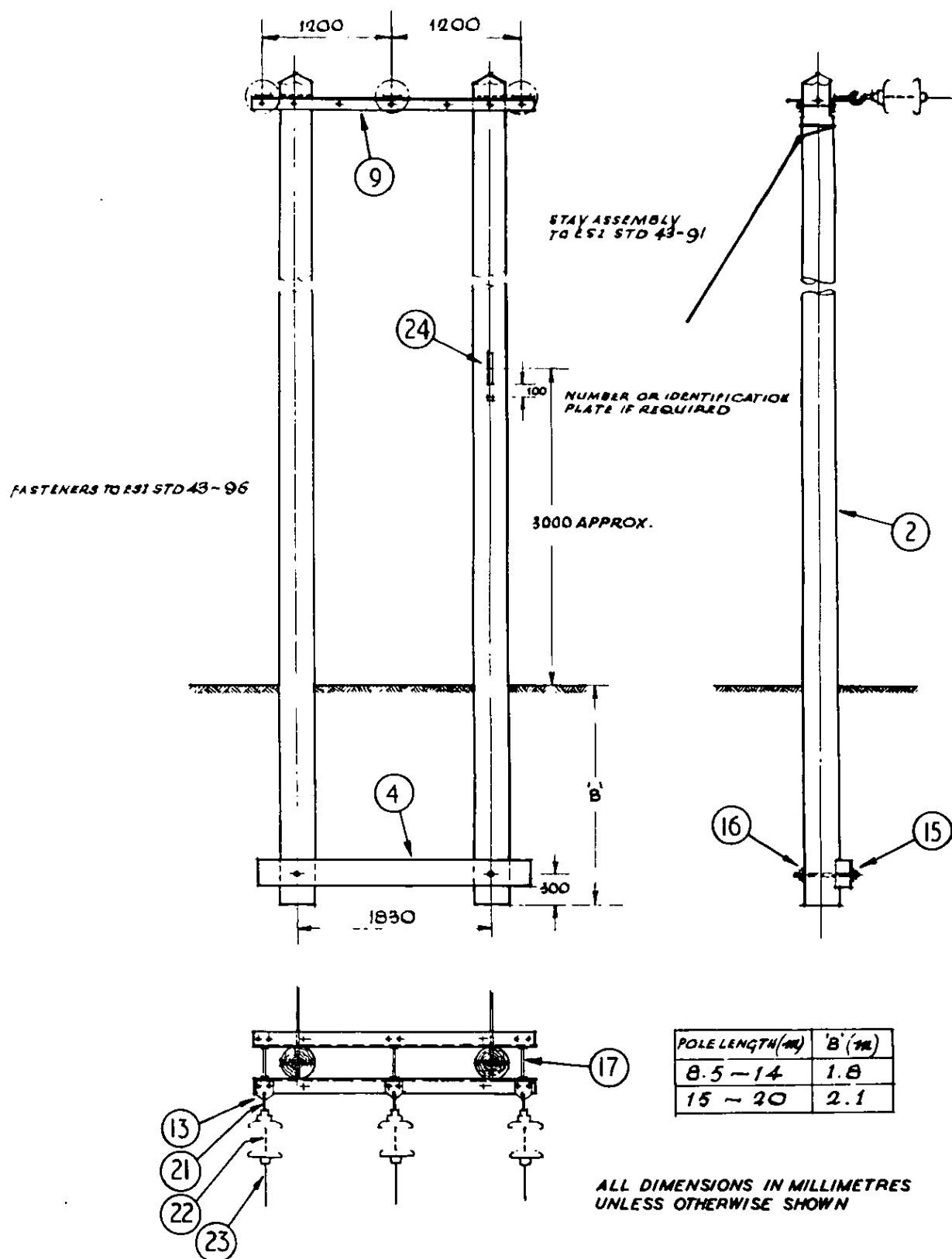
Material List for Fig. 8

'H' Pole Straight Line Section or
Section Angle up to 60° for Occasional Long Span

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref	M 20 Bolts	Round Washers
2	2	Wood Pole	432002	2 or 4*	--
5	1	Brace Block	439112 Type 2	2	—
8	2	Crossarm Member	439521	—	—
12	3	Section Strap	439525	6	6
15	2	Square Flat Washer	439604	—	—
16	2	Square Curved Washer	439605	—	—
17	4	Tie Rod	439608 Type 3	—	—
18	2 or 3	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —	-- — —
20	2 or 3	Pilot Pin (50 mm shank)	BS 3288 Ref. 50	--	—
21	6	Ball-ended Hook	BS 3288 Ref. 15/32	—	—
22	6	Insulator String	43-93	—	—
23	6	Conductor Termination	43-92	—	—
24	1	Danger Plate	439701	--	--

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Section Angle — (i) *Two anti-split bolts, each fitted with two square curved washers.
(ii) Additional round washers and locknuts required for bonding stay strand to pole top bolts.
When used as a straight line support with pin or line post insulators, items 20-23 are replaced by item 19(3 off).



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Fig. 9 General assembly — 'H' pole terminal
(ESI Drawing 432009)

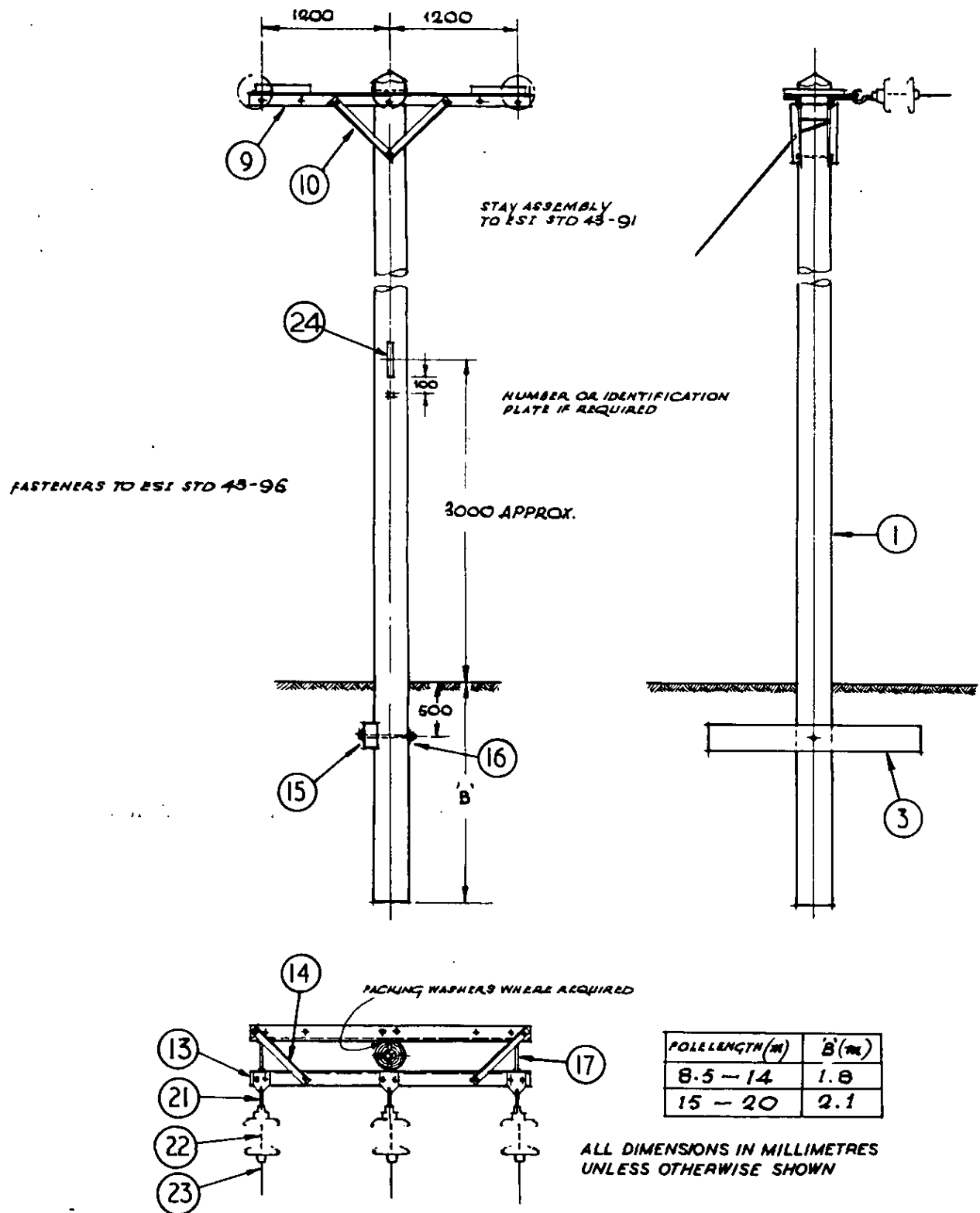
Material List for Fig. 9

'H' Pole Terminal

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref.	M 20 Bolts
2	2	Wood Pole	432002	2
4	1	Brace Block	439112 Type 1	2
9	2	Crossarm Member	439522	—
13	3	Terminal Strap	439528	6
15	2	Square Flat Washer	439604	—
16	2	Square Curved Washer	439605	—
17	3	Tie Rod	439608 Type 3	—
21	3	Ball-ended Hook	BS 3288 Ref. 15/32	—
22	3	Insulator String	43-93	—
23	3	Conductor Termination	43-92	—
24	1	Danger Plate	439701	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Additional round washers and locknuts required for bonding stay strand to pole top bolts.



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Fig. 10 General assembly — single pole terminal
(ESI Drawing 432010)

Material List for Fig. 10

Single Pole Terminal

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref.	M 20 Bolts
1	1	Wood Pole	432001	2
3	1	Kicking Block	439103 Type 2	1
9	2	Crossarm Member	439522	—
10	4	Crossarm Strut	439526	4
13	3	Terminal Strap	439528	6
14	2	Crossarm Brace	439515	4
15	1	Square Flat Washer	439604	—
16	1	Square Curved Washer	439605	—
17	2	Tie Rod	439608 Type 3	—
21	3	Ball-ended Hook	BS 3288 Ref. 15/32	—
22	3	Insulator String	43-93	—
23	3	Conductor Termination	43-92	—
24	1	Danger plate	439701	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Additional round washers and locknuts required for bonding stay strand to pole top bolts.

APPENDIX A

LINES IN SEVERE ENVIRONMENTS

The purpose of this Appendix is to specify modifications to this Standard to cater for lines in situations where the design conditions specified in The Electricity (Overhead Lines) Regulations, 1970 may be exceeded due to geographical or weather conditions. A definition of such situations is not given because it is best determined by individual Electricity Boards based upon local knowledge and operating experience.

A1. SPAN LENGTHS

Recommended and maximum span lengths for normal construction shall be 90 m and 105 m respectively for all conductors.

The occasional long span shall be 130 m to 150 m in length and provide a minimum ratio of 0.5 between mid-span spacing and still air design sag at 50°C.

A2. SUPPORTS

A2.1 Section Angle and Terminal Supports

Requirements shall be determined by the method laid down in 6.5 on page 12, but, to provide for increased factor of safety of 3.5 on poles, the loadings so determined shall be related to Table A1 of this Appendix with necessary reference to notes on Table 10 of this Standard.

A2.2 Intermediate and Section Supports

When determining the diameters of intermediate and section supports, the wind loading span shall be that shown in Tables 4—7, Pages 17—22, multiplied by 0.70.

A3. STAYS

It is recommended that the largest angle of stay slope possible up to a maximum of 45° be adopted.

A4. CONDUCTOR DESIGN AND ERECTION SAGS AND TENSIONS

Ordinates for plotting uplift curves and ground clearance curves are given in Table A2 (bare conductor) and Table A3 (PVC-covered conductor). These are based on a basic span of 90 m.

Conductor design sags and tensions are given in Tables A4—A9.

Conductor pre-tension and erection sags and tensions are given in Tables A10—A15.

Table A1 — Permissible Strut Loading of Poles — Factor of Safety 3.5

Dia. at 1.5 m (mm)	Pole Class (BS 1990)	Pole Planting Depth (m)	Pole Length (m)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)
215	M	1.8	8.5	150	2420	160	2999	180	3677
220			9.0	150	2178	165	2695	180	3300
225			9.5	150	1978	165	2445	180	2989
230			10.0	150	1811	165	2235	185	2911
235			10.5	150	1670	170	2201	190	2850
240			11.0	150	1550	170	2039	190	2635
245			11.5	150	1446	170	1898	190	2449
250			12.0	150	1356	170	1777	190	2288
260			13.0	160	1383	180	1790	200	280
275			14.0	160	1292	180	1663	205	2231
290		2.1	15.0	165	1337	185	1710	210	2281
305			16.0	170	1348	190	1709	215	2256
320			17.0	180	1447	200	1813	220	2245
330			18.0	180	1347	200	1683	220	2078
360			20.0	180	1261	205	1644	230	2108
265	S	1.8	8.5	190	5934	215	7878	240	10264
275			9.0	190	5474	215	7242	240	9407
280			9.5	190	4931	215	6515	245	8887
285			10.0	190	4479	215	5909	245	8049
290			10.5	190	4097	220	5692	250	7712
295			11.0	190	3773	220	5232	250	7079
300			11.5	190	3494	220	4837	250	6534
305			12.0	190	3252	220	4494	250	6062
320			13.0	195	3108	225	4252	255	5686
335			14.0	195	2849	225	3876	255	5158
350		2.1	15.0	195	2716	225	3691	260	5138
365			16.0	200	2669	230	3594	265	4956
375			17.0	200	2446	230	3282	265	4510
390			18.0	200	2319	230	3096	265	4232
415			20.0	200	2072	230	2743	265	3719
				Poles with top dia. equal to min. specified in BS 1990		Poles with mean value of top dia.		Poles with top dia. equal to max. specified in BS 1990	

For notes on Table A1 see page 25

Ordinates for Plotting Uplift and Ground Clearance Curves – ACSR Basic Span 90 m

Table A2 – Bare Conductor

Span (m) (2X)	Sag (m)								
	100mm ² 6/4.72 + 7/1.57mm			175mm ² 30 + 7/2.79mm			150mm ² , 175mm ² , 200mm ² 18 + 1.35mm 18 + 1/3.61mm 18 + 1/3.86mm		
	-6° C (Still Air)	50° C	75° C	-6° C (Still Air)	50° C	75° C	-6° C (Still Air)	50° C	75° C
40	0.10	0.29	0.37	0.09	0.24	0.31	0.11	0.35	0.42
80	0.39	1.15	1.47	0.36	0.95	1.25	0.46	1.39	1.69
120	0.88	2.60	3.30	0.81	2.12	2.80	1.03	3.12	3.79
160	1.56	4.61	5.86	1.44	3.78	4.98	1.83	5.55	6.74
200	2.43	7.21	9.16	2.26	5.90	7.78	2.86	8.68	10.53
240	3.51	10.38	13.19	3.25	8.50	11.21	4.12	12.48	15.16
280	4.77	14.12	17.95	4.42	11.56	15.25	5.61	16.98	20.63
320	6.23	18.45	23.45	5.78	15.10	19.92	7.32	22.18	26.95
360	7.89	23.35	29.68	7.31	19.11	25.22	9.27	28.07	34.11
400	9.74	28.82	36.64	9.03	23.60	31.13	11.44	34.65	42.11
440	11.78	34.88	44.33	10.93	28.55	37.67	13.84	41.93	50.95
480	14.02	41.51	52.76	13.00	33.98	44.83	16.47	49.90	60.64
520	16.46	48.71	61.92	15.26	39.88	52.61	19.33	58.57	71.16
560	19.09	56.49	71.81	17.70	46.25	61.02	22.42	67.92	82.53
600	21.91	64.85	82.43	20.32	53.09	70.04	25.74	77.97	94.74
640	24.93	73.79	93.79	23.11	60.41	79.69	29.28	88.71	107.79
680	28.14	83.30	105.88	26.09	68.20	89.96	33.06	100.15	121.69

NOTE: Ordinates for plotting uplift curves are based on conductor design tensions. Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR – 20% reduction
6 + 7 ACSR – 10% reduction
30 + 7 ACSR – 5% reduction

Ordinates for Plotting Uplift and Ground Clearance Curves —ACSR Basic Span 90 m

Table A3 — PVC-covered Conductor

Span (m) (2X)	Span (m)								
	100mm ² 6/4.72 + 7/1.57mm			175mm ² 30 + 7/2.79mm			150mm ² , 175mm ² , 200mm ² 18 + 1/3.35mm 18 + 1/3.61mm 18 + 1/3.86mm		
	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C
40	0.14	0.33	0.40	0.12	0.27	0.34	0.16	0.39	0.46
80	0.56	1.32	1.60	0.47	1.07	1.35	0.63	1.56	1.83
120	1.26	2.96	3.60	1.05	2.41	3.04	1.41	3.51	4.12
160	2.23	5.27	6.39	1.87	4.29	5.40	2.51	6.24	7.32
200	3.49	8.23	9.99	2.92	6.70	8.43	3.93	9.75	11.44
240	5.03	11.85	14.38	4.20	9.64	12.14	5.66	14.04	16.47
280	6.84	16.13	19.57	5.72	13.12	16.53	7.70	19.11	22.42
320	8.94	21.07	25.56	7.47	17.14	21.59	10.06	24.96	29.29
360	11.31	26.67	32.35	9.46	21.70	27.32	12.73	31.58	37.06
400	13.96	32.92	39.94	11.68	26.78	33.73	15.71	38.99	45.76
440	16.90	39.84	48.33	14.13	32.41	40.82	19.01	47.18	55.37
480	20.11	47.41	57.51	16.81	38.57	48.57	22.63	56.15	65.89
520	23.60	55.64	67.50	19.73	45.26	57.01	26.56	65.90	77.33
560	27.37	64.53	78.28	22.89	52.49	66.11	30.80	76.42	89.68
600	31.42	74.08	89.86	26.27	60.26	75.90	35.36	87.73	102.95
640	35.75	84.28	102.24	29.89	68.56	86.35	40.23	99.82	117.14
680	40.36	95.15	115.42	33.75	77.40	97.48	45.41	112.69	132.24

NOTE: Ordinates for plotting uplift curves are based on conductor design tensions. Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR — 20% reduction
6 + 7 ACSR — 10% reduction
30 + 7 ACSR — 5% reduction

Design Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR — Basic Span 90 m

Table A4 — Bare Conductor

Temp.	Tension	Sag (m) for span lengths (m)											
(°C)	(kgf)	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	736.6	0.11	0.17	0.25	0.33	0.44	0.55	0.68	0.83	0.98	1.15	1.34	1.53
5.0	666.9	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.91	1.09	1.27	1.48	1.70
10.0	602.3	0.13	0.21	0.30	0.41	0.53	0.68	0.83	1.01	1.20	1.41	1.64	1.88
15.0	543.6	0.15	0.23	0.33	0.45	0.59	0.75	0.92	1.12	1.33	1.56	1.81	2.08
20.0	491.3	0.16	0.26	0.37	0.50	0.65	0.83	1.02	1.24	1.47	1.73	2.00	2.30
25.0	445.7	0.18	0.28	0.41	0.55	0.72	0.91	1.13	1.36	1.62	1.91	2.21	2.54
30.0	406.4	0.20	0.31	0.45	0.61	0.79	1.00	1.24	1.50	1.78	2.09	2.42	2.78
35.0	372.9	0.22	0.34	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.64	3.03
40.0	344.4	0.23	0.36	0.53	0.71	0.93	1.18	1.46	1.77	2.10	2.47	2.86	3.28
45.0	320.2	0.25	0.39	0.57	0.77	1.00	1.27	1.57	1.90	2.26	2.65	3.08	3.53
50.0	299.5	0.27	0.42	0.60	0.82	1.07	1.36	1.68	2.03	2.42	2.84	3.29	3.78
55.0	281.6	0.29	0.45	0.64	0.87	1.14	1.45	1.78	2.16	2.57	3.02	3.50	4.01
60.0	266.2	0.30	0.47	0.68	0.92	1.21	1.53	1.89	2.28	2.72	3.19	3.70	4.25
65.0	252.8	0.32	0.50	0.72	0.97	1.27	1.61	1.99	2.41	2.86	3.36	3.90	4.47
70.0	240.9	0.33	0.52	0.75	1.02	1.33	1.69	2.09	2.52	3.00	3.52	4.09	4.69
75.0	230.4	0.35	0.55	0.79	1.07	1.40	1.77	2.18	2.64	3.14	3.69	4.27	4.91

Table A5 — PVC-covered Conductor

Temp.	Tension	Sag (m) for span lengths (m)											
(°C)	(kgf)	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	724.1	0.15	0.24	0.35	0.47	0.62	0.78	0.96	1.17	1.39	1.63	1.89	2.17
5.0	666.9	0.17	0.26	0.38	0.51	0.67	0.85	1.05	1.27	1.51	1.77	2.05	2.35
10.0	614.9	0.18	0.28	0.41	0.56	0.73	0.92	1.13	1.37	1.63	1.92	2.22	2.55
15.0	568.2	0.20	0.31	0.44	0.60	0.79	0.99	1.23	1.49	1.77	2.07	2.41	2.76
20.0	526.8	0.21	0.33	0.48	0.65	0.85	1.07	1.32	1.60	1.91	2.24	2.60	2.98
25.0	490.2	0.23	0.36	0.51	0.70	0.91	1.15	1.42	1.72	2.05	2.40	2.79	3.20
30.0	458.1	0.24	0.38	0.55	0.75	0.97	1.23	1.52	1.84	2.19	2.57	2.98	3.43
35.0	429.8	0.26	0.41	0.58	0.80	1.04	1.31	1.62	1.96	2.34	2.74	3.18	3.65
40.0	405.1	0.28	0.43	0.62	0.84	1.10	1.39	1.72	2.08	2.48	2.91	3.37	3.87
45.0	383.3	0.29	0.45	0.66	0.89	1.16	1.47	1.82	2.20	2.62	3.08	3.57	4.09
50.0	364.0	0.31	0.48	0.69	0.94	1.23	1.55	1.92	2.32	2.76	3.24	3.76	4.31
55.0	346.9	0.32	0.50	0.72	0.99	1.29	1.63	2.01	2.43	2.90	3.40	3.94	4.52
60.0	331.7	0.34	0.53	0.76	1.03	1.35	1.70	2.10	2.54	3.03	3.55	4.12	4.73
65.0	318.0	0.35	0.55	0.79	1.07	1.40	1.78	2.19	2.65	3.16	3.71	4.30	4.94
70.0	305.7	0.37	0.57	0.82	1.12	1.46	1.85	2.28	2.76	3.29	3.86	4.47	5.13
75.0	294.5	0.38	0.59	0.85	1.16	1.52	1.92	2.37	2.87	3.41	4.00	4.64	5.33

Design Sag/Tension Tables for 175 mm² (30 + 7/2.79) ACSR – Basic Span 90 m

Table A6 – Bare Conductor

Temp. (°C)	Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	1764.8	0.10	0.15	0.22	0.30	0.40	0.50	0.62	0.75	0.89	1.05	1.21	1.39
5.0	1627.4	0.11	0.17	0.24	0.33	0.43	0.54	0.67	0.81	0.97	1.13	1.32	1.51
10.0	1496.1	0.12	0.18	0.26	0.36	0.47	0.59	0.73	0.88	1.05	1.23	1.43	1.64
15.0	1372.1	0.13	0.20	0.29	0.39	0.51	0.64	0.80	0.96	1.15	1.35	1.56	1.79
20.0	1256.5	0.14	0.22	0.31	0.43	0.56	0.70	0.87	1.05	1.25	1.47	1.70	1.96
25.0	1150.4	0.15	0.24	0.34	0.47	0.61	0.77	0.95	1.15	1.37	1.60	1.86	2.14
30.0	1054.5	0.17	0.26	0.37	0.51	0.66	0.84	1.04	1.25	1.49	1.75	2.03	2.33
35.0	969.1	0.18	0.28	0.41	0.55	0.72	0.91	1.13	1.36	1.62	1.91	2.21	2.54
40.0	893.8	0.20	0.31	0.44	0.60	0.78	0.99	1.22	1.48	1.76	2.07	2.40	2.75
45.0	828.0	0.21	0.33	0.48	0.65	0.84	1.07	1.32	1.60	1.90	2.23	2.59	2.97
50.0	770.7	0.23	0.35	0.51	0.69	0.91	1.15	1.42	1.72	2.04	2.40	2.78	3.19
55.0	720.9	0.24	0.38	0.55	0.74	0.97	1.23	1.52	1.83	2.18	2.56	2.97	3.41
60.0	677.6	0.26	0.40	0.58	0.79	1.03	1.31	1.61	1.95	2.32	2.72	3.16	3.63
65.0	639.7	0.27	0.43	0.61	0.84	1.09	1.38	1.71	2.07	2.46	2.89	3.35	3.84
70.0	606.4	0.29	0.45	0.65	0.88	1.15	1.46	1.80	2.18	2.59	3.04	3.53	4.05
75.0	577.1	0.30	0.47	0.68	0.93	1.21	1.53	1.89	2.29	2.73	3.20	3.71	4.26

Table A7 – PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	1697.0	0.13	0.20	0.29	0.39	0.51	0.65	0.80	0.96	1.15	1.35	1.56	1.79
5.0	1573.9	0.14	0.21	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.45	1.68	1.93
10.0	1458.2	0.15	0.23	0.33	0.45	0.59	0.75	0.93	1.12	1.34	1.57	1.82	2.09
15.0	1350.8	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.69	1.96	2.25
20.0	1252.0	0.17	0.27	0.39	0.53	0.69	0.87	1.08	1.31	1.56	1.83	2.12	2.43
25.0	1162.4	0.19	0.29	0.42	0.57	0.74	0.94	1.16	1.41	1.68	1.97	2.28	2.62
30.0	1081.7	0.20	0.31	0.45	0.61	0.80	1.01	1.25	1.51	1.80	2.11	2.45	2.81
35.0	1009.6	0.21	0.33	0.48	0.66	0.86	1.09	1.34	1.62	1.93	2.26	2.63	3.01
40.0	945.5	0.23	0.36	0.51	0.70	0.92	1.16	1.43	1.73	2.06	2.42	2.80	3.22
45.0	888.8	0.24	0.38	0.55	0.75	0.97	1.23	1.52	1.84	2.19	2.57	2.98	3.42
50.0	838.5	0.26	0.40	0.58	0.79	1.03	1.31	1.61	1.95	2.32	2.73	3.16	3.63
55.0	794.0	0.27	0.43	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.88	3.34	3.83
60.0	754.4	0.29	0.45	0.65	0.88	1.15	1.45	1.79	2.17	2.58	3.03	3.51	4.03
65.0	719.0	0.30	0.47	0.68	0.92	1.20	1.52	1.88	2.28	2.71	3.18	3.69	4.23
70.0	687.4	0.31	0.49	0.71	0.96	1.26	1.59	1.97	2.38	2.83	3.32	3.86	4.43
75.0	659.0	0.33	0.51	0.74	1.01	1.31	1.66	2.05	2.48	2.96	3.47	4.02	4.62

Design Sag/Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 90 m

Table A8 — Bare Conductor

Temp (°C)	Tension (kgf)			Sag (m) for span lengths (m)											
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	810.0	932.7	1056.3	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	1.59	1.82
5.0	728.1	838.2	949.3	0.14	0.23	0.32	0.44	0.58	0.73	0.90	1.09	1.30	1.52	1.77	2.03
10.0	654.8	753.8	854.0	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.70	1.97	2.26
15.0	590.7	680.1	770.8	0.18	0.28	0.40	0.54	0.71	0.90	1.11	1.35	1.60	1.88	2.18	2.50
20.0	535.5	616.8	699.5	0.20	0.31	0.44	0.60	0.78	0.99	1.23	1.48	1.77	2.07	2.40	2.76
25.0	488.7	563.1	639.0	0.21	0.34	0.48	0.66	0.86	1.09	1.34	1.63	1.93	2.27	2.63	3.02
30.0	449.1	517.7	587.8	0.23	0.37	0.53	0.72	0.93	1.18	1.46	1.77	2.10	2.47	2.86	3.29
35.0	415.6	479.3	544.6	0.25	0.39	0.57	0.77	1.01	1.28	1.58	1.91	2.27	2.67	3.09	3.55
40.0	387.3	446.8	508.0	0.27	0.42	0.61	0.83	1.08	1.37	1.69	2.05	2.44	2.86	3.32	3.81
45.0	363.1	419.0	476.6	0.29	0.45	0.65	0.88	1.16	1.46	1.80	2.18	2.60	3.05	3.54	4.06
50.0	342.2	395.1	449.7	0.31	0.48	0.69	0.94	1.22	1.55	1.91	2.32	2.76	3.23	3.75	4.31
55.0	324.2	374.4	426.2	0.32	0.51	0.73	0.99	1.29	1.64	2.02	2.44	2.91	3.41	3.96	4.55
60.0	308.4	356.2	405.7	0.34	0.53	0.76	1.04	1.36	1.72	2.12	2.57	3.06	3.59	4.16	4.78
65.0	294.4	340.2	387.6	0.36	0.56	0.80	1.09	1.42	1.80	2.22	2.69	3.20	3.76	4.36	5.00
70.0	282.1	326.0	371.5	0.37	0.58	0.84	1.14	1.48	1.88	2.32	2.81	3.34	3.92	4.55	5.22
75.0	271.0	313.3	357.1	0.39	0.60	0.87	1.18	1.55	1.96	2.41	2.92	3.48	4.08	4.73	5.43

Design Sag/Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 90 m

Table A9 — PVC-covered Conductor

Temp (°C)	Tension (kgf)			Sag (m) for span lengths (m)											
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	793.0	913.6	1036.4	0.17	0.27	0.39	0.54	0.70	0.88	1.09	1.32	1.57	1.85	2.14	2.46
5.0	728.1	838.2	949.3	0.19	0.30	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	2.33	2.68
10.0	670.7	771.6	872.5	0.21	0.32	0.47	0.63	0.83	1.05	1.29	1.56	1.86	2.18	2.53	2.91
15.0	620.3	713.3	805.4	0.22	0.35	0.50	0.69	0.90	1.13	1.40	1.69	2.01	2.36	2.74	3.15
20.0	576.5	662.5	747.1	0.24	0.38	0.54	0.74	0.96	1.22	1.51	1.82	2.17	2.54	2.95	3.39
25.0	538.3	618.4	696.5	0.26	0.40	0.58	0.79	1.03	1.31	1.61	1.95	2.32	2.73	3.16	3.63
30.0	505.2	580.1	652.7	0.28	0.43	0.62	0.84	1.10	1.39	1.72	2.08	2.48	2.91	3.37	3.87
35.0	476.2	546.7	614.5	0.29	0.46	0.66	0.89	1.17	1.48	1.82	2.21	2.63	3.08	3.58	4.11
40.0	450.8	517.4	581.2	0.31	0.48	0.69	0.94	1.23	1.56	1.93	2.33	2.78	3.26	3.78	4.34
45.0	428.4	491.6	551.9	0.32	0.51	0.73	0.99	1.30	1.64	2.03	2.45	2.92	3.43	3.98	4.57
50.0	408.6	468.8	526.0	0.34	0.53	0.77	1.04	1.36	1.72	2.13	2.57	3.06	3.60	4.17	4.79
55.0	391.0	448.5	503.0	0.36	0.56	0.80	1.09	1.42	1.80	2.22	2.69	3.20	3.76	4.36	5.00
60.0	375.2	430.4	482.4	0.37	0.58	0.83	1.14	1.48	1.88	2.32	2.80	3.34	3.92	4.54	5.21
65.0	361.0	414.0	463.9	0.39	0.60	0.87	1.18	1.54	1.95	2.41	2.92	3.47	4.07	4.72	5.42
70.0	348.1	399.2	447.2	0.40	0.62	0.90	1.22	1.60	2.02	2.50	3.02	3.60	4.22	4.90	5.62
75.0	336.4	385.8	432.0	0.41	0.65	0.93	1.27	1.65	2.09	2.59	3.13	3.72	4.37	5.07	5.82

Pre-tension and Erection Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR – Basic Span 90 m

Table A10 – Bare Conductor

Pre-tension kgf (All Temps.)	Sag (m) for span lengths (m) at all temperatures											
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
750	0.11	0.17	0.25	0.34	0.44	0.56	0.69	0.83	0.99	1.16	1.35	1.55

Erection Temp. (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	805.1	0.10	0.16	0.22	0.31	0.40	0.51	0.62	0.76	0.90	1.05	1.22	1.40
5.0	731.5	0.11	0.17	0.25	0.34	0.44	0.56	0.69	0.83	0.99	1.16	1.35	1.55
10.0	662.1	0.12	0.19	0.27	0.37	0.49	0.61	0.76	0.92	1.09	1.28	1.49	1.71
15.0	597.9	0.13	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.21	1.42	1.55	1.89
20.0	539.7	0.15	0.23	0.34	0.46	0.60	0.75	0.93	1.13	1.34	1.57	1.82	2.09
25.0	487.9	0.16	0.26	0.37	0.50	0.66	0.83	1.03	1.25	1.48	1.74	2.02	2.32
30.0	442.7	0.18	0.28	0.41	0.56	0.73	0.92	1.14	1.37	1.63	1.92	2.22	2.55
35.0	403.9	0.20	0.31	0.45	0.61	0.80	1.01	1.24	1.51	1.79	2.10	2.44	2.80
40.0	370.7	0.22	0.34	0.49	0.66	0.87	1.10	1.36	1.64	1.95	2.29	2.66	3.05
45.0	342.6	0.23	0.37	0.53	0.72	0.94	1.19	1.47	1.77	2.11	2.48	2.88	3.30
50.0	318.6	0.25	0.39	0.57	0.77	1.01	1.28	1.58	1.91	2.27	2.67	3.09	3.55
55.0	298.1	0.27	0.42	0.61	0.83	1.08	1.37	1.69	2.04	2.43	2.85	3.30	3.79
60.0	280.5	0.29	0.45	0.64	0.88	1.15	1.45	1.79	2.17	2.58	3.03	3.51	4.03
65.0	265.2	0.30	0.47	0.68	0.93	1.21	1.53	1.89	2.29	2.73	3.20	3.71	4.26
70.0	251.9	0.32	0.50	0.72	0.98	1.28	1.62	2.00	2.41	2.87	3.37	3.91	4.49
75.0	240.1	0.33	0.52	0.75	1.03	1.34	1.69	2.09	2.53	3.01	3.54	4.10	4.71

Pre-tension and Erection Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR – Basic Span 90 m

Table A11 – PVC-covered Conductor

Pre-tension kgf (All Temps.)	Sag (m) for span lengths (m) at all temperatures											
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
750	0.15	0.24	0.34	0.46	0.61	0.77	0.95	1.15	1.36	1.60	1.86	2.13

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	799.5	0.14	0.22	0.31	0.43	0.56	0.71	0.87	1.06	1.26	1.47	1.71	1.96
5.0	736.4	0.15	0.24	0.34	0.46	0.61	0.77	0.95	1.15	1.36	1.60	1.86	2.13
10.0	678.1	0.16	0.26	0.37	0.50	0.66	0.83	1.03	1.24	1.48	1.74	2.02	2.31
15.0	625.1	0.18	0.28	0.40	0.55	0.71	0.90	1.12	1.35	1.61	1.89	2.19	2.51
20.0	577.3	0.19	0.30	0.43	0.59	0.77	0.98	1.21	1.46	1.74	2.04	2.37	2.72
25.0	534.8	0.21	0.33	0.47	0.64	0.83	1.06	1.30	1.58	1.88	2.20	2.56	2.93
30.0	497.3	0.22	0.35	0.50	0.69	0.90	1.14	1.40	1.70	2.02	2.37	2.75	3.16
35.0	464.3	0.24	0.38	0.54	0.74	0.96	1.22	1.50	1.82	2.16	2.54	2.94	3.38
40.0	435.3	0.26	0.40	0.58	0.79	1.03	1.30	1.60	1.94	2.31	2.71	3.14	3.61
45.0	409.9	0.27	0.43	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.88	3.34	3.83
50.0	387.5	0.29	0.45	0.65	0.88	1.15	1.46	1.80	2.18	2.59	3.04	3.53	4.05
55.0	367.8	0.30	0.47	0.68	0.93	1.21	1.54	1.90	2.29	2.73	3.21	3.72	4.27
60.0	350.2	0.32	0.50	0.72	0.98	1.27	1.61	1.99	2.41	2.87	3.37	3.90	4.48
65.0	334.6	0.33	0.52	0.75	1.02	1.33	1.69	2.08	2.52	3.00	3.52	4.09	4.69
70.0	320.7	0.35	0.54	0.78	1.07	1.39	1.76	2.18	2.63	3.13	3.68	4.26	4.89
75.0	308.1	0.36	0.57	0.82	1.11	1.45	1.83	2.26	2.74	3.26	3.83	4.44	5.09

Pre-tension and Erection Sag/Tension Tables for 175 mm² (30 + 7/2.19) ACSR — Basic Span 90 m.

Table A12 — Bare Conductor.

Pre-tension kgf (All Temps.)	Sag (m) for span lengths (m) at all temperatures											
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
1800	0.10	0.15	0.22	0.30	0.39	0.50	0.61	0.74	0.88	1.04	1.20	1.38

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	1921.9	0.09	0.14	0.20	0.28	0.36	0.46	0.57	0.69	0.82	0.96	1.11	1.28
5.0	1779.2	0.10	0.15	0.22	0.30	0.39	0.50	0.61	0.74	0.88	1.04	1.20	1.38
10.0	1641.3	0.11	0.17	0.24	0.33	0.43	0.54	0.67	0.81	0.96	1.12	1.30	1.50
15.0	1509.3	0.12	0.18	0.26	0.35	0.46	0.59	0.72	0.88	1.04	1.22	1.42	1.63
20.0	1384.4	0.13	0.20	0.28	0.39	0.51	0.64	0.79	0.95	1.14	1.33	1.55	1.78
25.0	1267.9	0.14	0.22	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.46	1.69	1.94
30.0	1160.8	0.15	0.24	0.34	0.46	0.60	0.76	0.94	1.14	1.36	1.59	1.84	2.12
35.0	1063.9	0.16	0.26	0.37	0.50	0.66	0.83	1.03	1.24	1.48	1.74	2.01	2.31
40.0	977.3	0.18	0.28	0.40	0.55	0.72	0.91	1.12	1.35	1.61	1.89	2.19	2.52
45.0	901.0	0.19	0.30	0.44	0.59	0.78	0.98	1.21	1.47	1.75	2.05	2.38	2.73
50.0	834.3	0.21	0.33	0.47	0.64	0.84	1.06	1.31	1.58	1.89	2.21	2.57	2.95
55.0	776.2	0.23	0.35	0.51	0.69	0.90	1.14	1.41	1.70	2.03	2.38	2.76	3.17
60.0	725.7	0.24	0.38	0.54	0.74	0.96	1.22	1.51	1.82	2.17	2.54	2.95	3.39
65.0	681.7	0.26	0.40	0.58	0.79	1.03	1.30	1.60	1.94	2.31	2.71	3.14	3.61
70.0	643.3	0.27	0.42	0.61	0.83	1.09	1.38	1.70	2.05	2.45	2.87	3.33	3.82
75.0	609.7	0.29	0.45	0.65	0.88	1.15	1.45	1.79	2.17	2.58	3.03	3.51	4.03

Pre-tension and Erection Sag/Tension Tables for 175 mm² (30 + 7/2.19) ACSR – Basic Span 90 m

Table A13 – PVC-covered Conductor

Pre-tension kgf (All Temps)	Sag (m) for span lengths (m) at all temperatures											
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
1800	0.13	0.20	0.28	0.38	0.50	0.63	0.78	0.95	1.13	1.32	1.53	1.76

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	1859.2	0.12	0.18	0.26	0.36	0.47	0.59	0.73	0.88	1.05	1.23	1.43	1.64
5.0	1728.2	0.13	0.20	0.28	0.38	0.50	0.63	0.78	0.95	1.13	1.32	1.53	1.76
10.0	1603.4	0.13	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.21	1.43	1.65	1.90
15.0	1485.8	0.15	0.23	0.33	0.45	0.58	0.74	0.91	1.10	1.31	1.54	1.78	2.05
20.0	1376.3	0.16	0.25	0.35	0.48	0.63	0.80	0.98	1.19	1.42	1.66	1.93	2.21
25.0	1275.4	0.17	0.27	0.38	0.52	0.68	0.86	1.06	1.28	1.53	1.79	2.08	2.39
30.0	1183.5	0.18	0.29	0.41	0.56	0.73	0.93	1.14	1.38	1.65	1.93	2.24	2.57
35.0	1100.6	0.20	0.31	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	2.41	2.76
40.0	1026.5	0.21	0.33	0.47	0.65	0.84	1.07	1.32	1.59	1.90	2.23	2.58	2.96
45.0	960.5	0.23	0.35	0.51	0.69	0.90	1.14	1.41	1.70	2.03	2.38	2.76	3.17
50.0	902.0	0.24	0.37	0.54	0.73	0.96	1.21	1.50	1.81	2.16	2.53	2.94	3.37
55.0	850.3	0.25	0.40	0.57	0.78	1.02	1.29	1.59	1.92	2.29	2.69	3.12	3.58
60.0	804.4	0.27	0.42	0.61	0.82	1.08	1.36	1.68	2.03	2.42	2.84	3.30	3.78
65.0	763.6	0.28	0.44	0.64	0.87	1.13	1.43	1.77	2.14	2.55	2.99	3.47	3.99
70.0	727.3	0.30	0.46	0.67	0.91	1.19	1.51	1.86	2.25	2.68	3.14	3.64	4.18
75.0	694.9	0.31	0.49	0.70	0.95	1.25	1.58	1.95	2.36	2.80	3.29	3.81	4.38

Pre-tension and Erection Sag/Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 90 m

Table A14 — Bare Conductor

Pre-tension kgf (All Temps)			Sag (m) for span lengths (m) at all temperatures											
150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
800	950	1050	0.13	0.20	0.29	0.40	0.52	0.66	0.82	0.99	1.18	1.38	1.60	1.84

Erection Temp (°C)	Erection Tension (kgf)			Sag (m) for span lengths (m)											
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	892.7	1028.4	1165.4	0.12	0.18	0.26	0.36	0.47	0.60	0.74	0.89	1.06	1.24	1.44	1.65
5.0	803.6	925.5	1048.6	0.13	0.20	0.29	0.40	0.52	0.66	0.82	0.99	1.18	1.38	1.60	1.84
10.0	722.3	831.7	942.4	0.15	0.23	0.33	0.45	0.58	0.74	0.91	1.10	1.31	1.54	1.78	2.05
15.0	649.7	748.1	847.9	0.16	0.25	0.36	0.50	0.65	0.82	1.01	1.22	1.46	1.71	1.98	2.27
20.0	586.3	675.2	765.6	0.18	0.28	0.40	0.55	0.72	0.91	1.12	1.36	1.61	1.89	2.20	2.52
25.0	531.8	612.6	695.0	0.20	0.31	0.44	0.60	0.79	1.00	1.23	1.49	1.78	2.09	2.42	2.78
30.0	485.5	559.5	635.2	0.22	0.34	0.49	0.66	0.87	1.09	1.35	1.64	1.95	2.28	2.65	3.04
35.0	446.4	514.7	584.6	0.24	0.37	0.53	0.72	0.94	1.19	1.47	1.78	2.12	2.48	2.88	3.31
40.0	413.4	476.8	541.9	0.25	0.40	0.57	0.78	1.02	1.28	1.59	1.92	2.28	2.68	3.11	3.57
45.0	385.3	444.6	505.7	0.27	0.43	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.87	3.33	3.83
50.0	361.4	417.2	474.7	0.29	0.45	0.65	0.89	1.16	1.47	1.81	2.19	2.61	3.06	3.55	4.08
55.0	340.8	393.5	447.9	0.31	0.48	0.69	0.94	1.23	1.56	1.92	2.33	2.77	3.25	3.77	4.32
60.0	322.9	373.0	424.7	0.32	0.51	0.73	0.99	1.30	1.64	2.03	2.45	2.92	3.43	3.97	4.56
65.0	307.3	355.0	404.4	0.34	0.53	0.77	1.04	1.36	1.73	2.13	2.58	3.07	3.60	4.18	4.79
70.0	293.5	339.1	386.4	0.36	0.56	0.80	1.09	1.43	1.81	2.23	2.70	3.21	3.77	4.37	5.02
75.0	281.2	325.0	370.5	0.37	0.58	0.84	1.14	1.49	1.88	2.33	2.82	3.35	3.93	4.56	5.24

Pre-tension and Erection Sag/Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 90 m

Table A15 — PVC-covered Conductor

Pre-tension kgf (All Temps)			Sag (m) for span lengths (m) at all temperatures											
150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
800	950	1050	0.17	0.27	0.39	0.55	0.69	0.87	1.07	1.30	1.54	1.81	2.10	2.41

Erection Temp (°C)	Erection Tension (kgf)		Sag (m) for span lengths (m)												
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	882.0	1016.2	1152.6	0.16	0.25	0.35	0.48	0.63	0.80	0.98	1.19	1.41	1.66	1.92	2.21
5.0	807.9	930.2	1053.5	0.17	0.27	0.39	0.53	0.69	0.87	1.07	1.30	1.54	1.81	2.10	2.41
10.0	741.4	853.0	964.6	0.19	0.29	0.42	0.57	0.75	0.95	1.17	1.42	1.68	1.98	2.29	2.63
15.0	682.4	784.6	885.9	0.20	0.32	0.46	0.62	0.81	1.03	1.27	1.54	1.83	2.15	2.49	2.86
20.0	630.6	724.6	817.1	0.22	0.34	0.50	0.67	0.88	1.12	1.38	1.67	1.98	2.33	2.70	3.10
25.0	585.4	672.4	757.2	0.24	0.37	0.53	0.73	0.95	1.20	1.48	1.80	2.14	2.51	2.91	3.34
30.0	546.1	627.0	705.3	0.25	0.40	0.57	0.78	1.02	1.29	1.59	1.93	2.29	2.69	3.12	3.58
35.0	511.9	587.5	660.3	0.27	0.42	0.61	0.83	1.09	1.38	1.70	2.05	2.44	2.87	3.33	3.82
40.0	482.1	553.2	621.2	0.29	0.45	0.65	0.88	1.15	1.46	1.80	2.18	2.60	3.05	3.53	4.06
45.0	456.0	523.1	587.0	0.31	0.48	0.69	0.93	1.22	1.54	1.91	2.31	2.75	3.22	3.74	4.29
50.0	433.0	496.7	557.0	0.32	0.50	0.72	0.98	1.29	1.63	2.01	2.43	2.89	3.39	3.94	4.52
55.0	412.7	473.3	530.6	0.34	0.53	0.76	1.03	1.35	1.71	2.11	2.55	3.03	3.56	4.13	4.74
60.0	394.6	452.5	507.1	0.35	0.55	0.79	1.08	1.41	1.79	2.20	2.67	3.17	3.73	4.32	4.96
65.0	378.4	434.0	486.1	0.37	0.57	0.83	1.13	1.47	1.86	2.30	2.78	3.31	3.88	4.51	5.17
70.0	363.9	417.2	467.2	0.38	0.60	0.86	1.17	1.53	1.94	2.39	2.89	3.44	4.04	4.69	5.38
75.0	350.7	402.1	450.2	0.40	0.62	0.89	1.22	1.59	2.01	2.48	3.00	3.57	4.19	4.86	5.58

ESI

ELECTRICITY SUPPLY INDUSTRY

STANDARD 43—20

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11 kV AND 33 kV SINGLE CIRCUIT OVERHEAD LINES OF HEAVY CONSTRUCTION ON WOOD POLES

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11 kV AND 33 kV SINGLE CIRCUIT OVERHEAD LINES OF HEAVY CONSTRUCTION ON WOOD POLES

FOREWORD

In the past, 11 kV and 33 kV heavy duty overhead lines have been constructed to Specifications prepared by individual Area Boards. The advent of metrication and the coming into force of the Electricity (Overhead Lines) Regulations 1970 created the need for the revision of these Specifications and this Standard has been prepared to fulfil the needs of the Electricity Supply Industry.

1. SCOPE

This Standard covers 3-phase 11 kV and 33 kV overhead lines of heavy construction on wood poles, employing the following ACSR:

100 mm² (6/4.72 mm A1 + 7/1.57 mm St)
150 mm² (18/3.35 mm A1 + 1/3.35 mm St)
175 mm² (18/3.61 mm A1 + 1/3.61 mm St)
175 mm² (30/2.79 mm A1 + 7/2.79 mm St)
200 mm² (18/3.86 mm A1 + 1/3.86 mm St)

Other conductors may be used in conjunction with this Standard subject to users supplying their own erection data and verifying that the strength of components is adequate.

Lines to this Standard are designed to the conditions specified in The Electricity (Overhead Lines) Regulations 1970. Where these conditions may be exceeded due to geographical or weather conditions a modified form of construction as specified in Appendix A may be adopted.

2. DEFINITIONS

For the purpose of this Standard the following definitions apply:

Intermediate Support — A support in a straight run of a line on which the conductors are supported on pin or line post insulators.

Section Support — A support in a straight run of a line on which the conductors are made off on either side of the crossarm on tension insulator sets.

Pin Angle Support — A support at which a line deviates and the conductors are supported on pin or line post insulators.

Section Angle Support — A support at which a line deviates and the conductors are made off on either side of the crossarm on tension insulator sets.

Span — The horizontal distance between adjacent supports.

Wind Loading Span — The wind loading span associated with any support is half the sum of the spans adjacent to the support.

Average Span — The arithmetic average of a number of spans in a line or section of line.

Basic Span — The span length adopted for sag/tension calculations.

Recommended Span — The average span length in any section to which the line should be planned.

Maximum Span — The maximum span length permitted using normal conductor spacing.

Occasional Long Span — A span of line contained in a run of normal line construction but having a length above the maximum specified for normal span construction.

Sag — The vertical distance, under any system of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured mid-span.

Conductor Downpull — The vertical loading imposed by the conductors corresponding to a gradient measured between adjacent pole tops.

Pre-tensioning — The tension treatment applied to a conductor before final erection tension is established.

Overtensioning — Excess tension applied above normal tension at time of erection.

Auxiliary Equipment — Equipment other than that forming part of this line design which may be placed on supports, such as transformers, fusegear, etc.

3. REFERENCES

This Standard makes reference to the following documents:

BS 137, 'Insulators of Ceramic Material or Glass for Overhead Lines with a Nominal Voltage greater than 1000 V'.

BS 215, 'Aluminium Conductors and Aluminium Conductors, Steel Reinforced, for Overhead Power Transmission'.

Part 2, 'Aluminium Conductors, Steel-reinforced'.

BS 913, 'Wood Preservation by means of Pressure Creosoting'.

BS 1990, 'Wood Poles for Overhead Lines (Power and Telecommunication Lines)'.

BS 3288, 'Insulator and Conductor Fittings for Overhead Power Lines'.

BS 3452, 'Copper/Chrome Water-borne Wood Preservatives and their Application'.

BS 4072, 'Wood Preservation by means of Water-borne Copper/Chrome/Arsenic Compositions'.

BS 6485, 'PVC-covered Conductors for Overhead Power Lines'.

The Electricity (Overhead Lines) Regulations 1970, SI 1355.

*ESI Standard 43—89, 'Auxiliary Equipment Erected on Overhead Line Supports'.

ESI Standard 43—90, 'Anti-climbing Devices for H.V. Lines up to and including 132 kV'.

ESI Standard 43—91, 'Stay Strands and Stay Fittings for Overhead Lines'.

ESI Standard 43—92, 'Conductor Terminations, Joints and Insulation Binds for Overhead Lines up to and including 132 kV'.

ESI Standard 43—93, 'Line Insulators'.

ESI Standard 43—95, 'Steelwork for Overhead Lines'.

ESI Standard 43—96, 'Fasteners and Washers for Wood Pole Overhead Lines'.

*ESI Standard 43—97, 'Notices and Marker Posts for H.V. Lines up to and including 132 kV'.

*In course of preparation.

4. DESIGN DATA

This Standard is designed in accordance with the requirements of The Electricity (Overhead Lines) Regulations 1970.

4.1 Conductors

Conductor design tensions take into account the following conditions:

The minimum factors of safety on conductors shall be 2.0 on their calculated breaking loads.

The maximum working tension of conductors shall not exceed 2379 kgf (i.e. a factor of safety of 3 on 70 kN tension insulator fittings). The design tensions of aluminium based conductors at 5°C in still air shall not exceed 20 per cent of their calculated breaking load to obviate the risk of vibration.

The actual maximum working tensions and factors of safety are tabled below, together with the limiting design feature:

ACSR			M.W.T. (kgf)	F.O.S.	Limiting Design Feature
Size (mm ²)	Stranding (mm)	Type			
100	6/4.72 + 7/1.57	Bare PVC	1459 1501	2.29 2.22	Vibration Limit
150	18 + 1/3.35	Bare PVC	1628 1632	2.24 2.23	Vibration Limit
175	18 + 1/3.61	Bare PVC	1794 1793	2.34 2.34	Vibration Limit
175	30 + 7/2.79	Bare PVC	2379 2379	3.42 3.42	Tension Insulator Fittings
200	18 + 1/3.86	Bare PVC	1959 1958	2.42 2.42	Vibration Limit

4.2 Crossarm Assemblies

The minimum factor of safety on all crossarm assemblies is 2.5 on the ultimate tensile strength of the steel.

Crossarm assemblies in general are designed to support conductor downpull on each side of the support corresponding to a gradient of 1 in 10 on the maximum span length.

Exceptions are:

- 4.2.1 As shown in Tables 1, 2 and 3 for intermediate and pin angle crossarm.
- 4.2.2 Limit of 1 in 13 for the section angle crossarm for angles of deviation above 60° with 175 mm² (30 + 7/2.79 mm) ACSR.

4.3 Supports

The stresses created in intermediate supports are bending stresses due to windage on conductors, the loading points being 340 mm and 420 mm above pole tops for outer and centre conductors respectively. The stresses created in stayed supports are crippling stresses due to stay tension and vertical conductor loads acting at the pole top. The distribution of stresses in the limbs of 'H' supports on which all stays are adjusted to equal tension is considered to be as follows:

Stays on outer limb of angle support	All vertical components of stay tension plus half vertical conductor load taken on outer limb.
One stay on outer limb and two stays on inner limb of angle support.	60 per cent of stress in inner limb. 40 per cent of stress in outer limb.
Stayed equally on each limb	50 per cent of stress in each limb.

The strength of supports is derived from information contained in BS 1990 with respect to red fir poles.

The factor of safety for supports in normal conditions is 2.5.

Permissible strut loadings are given in Table 10.

5. MATERIALS

All components used in the construction of lines to this Standard shall comply with British Standards and/or ESI Standards indicated.

5.1 Conductors

ACSR	—	BS 215, Part 2
PVC-covered conductors (Type 16)	—	BS 6485

During stranding all inner wires shall be coated with an approved grease in such manner that the interstices between wires are effectively filled with grease within the circle circumscribing the penultimate layer of the complete conductor.

For extra protection of bare conductors in corrosive areas and when specified by the purchaser all wires shall be greased and the conductor subsequently wiped to remove excess grease from the outer surface whilst maintaining grease filling in the inter-wire crevices.

5.2 Wood Poles

Poles shall be European redwood complying with the requirements of BS 1990. They shall be fabricated to Fig. 1 and 2, all holes being drilled to template, the centre line of the template being placed along the effective centre line of the pole. Pole markings shall be made on the same side of the pole as that on which the drilling templates have been used. Pole tops shall be evenly quartered, the angle at the apex being 120°. When fully seasoned and fabricated, poles shall be creosoted in accordance with BS 913 by Rueping Process. An alternative method of wood preservative may be applied in accordance with BS 3452 or BS 4072 if specified by the purchaser. Where 'H' poles are required, the limbs shall be suitably matched and marked.

5.3 Crossarm Steelwork

Crossarm steelwork shall comply with ESI Standard 43—95.

5.4 Insulators

String insulator units, pin and line post insulators shall comply with BS 137 and with ESI Standard 43—93.

5.5 Insulator Fittings

Insulator fittings shall comply with BS 3288.

5.6 Conductor Terminations, Joints and Binders

Conductor terminations, joints and binders shall comply with ESI Standard 43—92.

5.7 Stays

Stay strand and stay fittings shall comply with ESI Standard 43—91.

5.8 Fasteners and Washers

Fasteners and washers shall comply with ESI Standard 43—96.

6. TECHNICAL REQUIREMENTS FOR THE ERECTION OF LINES

6.1 Span Lengths

Recommended and maximum span lengths for normal construction shall be 130 m and 150 m respectively for all conductors.

The occasional long span shall be 150 m to 180 m in length and provide a minimum ratio of 0.4 between mid-span spacing and still air design sag at 50°C.

6.2 Supports

Supports shall be single or 'H' type poles as specified in Table 1 and erected in accordance with Fig. 3—10. Where an occasional long span is the first or last span of a line, an 'H' terminal may be used as one support subject to the ratio limitation detailed at 6.1.

6.3 Maximum Angles of Line Deviation

The maximum angles of line deviation on pin and line post insulators are given in Table 2 (ACSR bare) and Table 3 (ACSR PVC-covered). These tables also indicate the changeover point for the crossarm from single to double member.

The maximum angles of line deviation on tension insulator strings are given in Table 1.

6.4 Wind Loading Spans

Wind loading spans are given in the following tables:

Table 4	—	Single Poles	—	ACSR Bare
Table 5	—	Single Poles	—	ACSR PVC-covered
Table 6	—	'H' Poles	—	ACSR Bare
Table 7	—	'H' Poles	—	ACSR PVC-covered

The tables could also apply to pin angle and section angle poles having an angle of deviation below 5°. Poles are assumed to have minimum top dimensions in accordance with BS 1990.

6.5 Staying Arrangements and Strut Loading for Angle and Terminal Supports

Staying arrangements are given in Table 8 (ACSR Bare) and Table 9 (ACSR PVC-covered).

Stay assemblies shall be to ESI Standard 43—91.

The strut loading imposed on angle and terminal supports shall be obtained from Nomograms 1—10. These loadings shall then be related to the safe working loads of the poles by reference to Table 10. When out-of-balance tension exists on a section pole, balancing stays should be fitted as required.

6.6 Conductor Design and Erection Sags and Tensions

Conductor design sags and tensions are given in Tables 13—18.

Ordinates for plotting uplift curves and ground clearance curves are given in Table 11 (ACSR Bare) and Table 12 (ACSR PVC-covered).

Ordinates for plotting uplift curves are based on conductor design tensions.

Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long-term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR	—	20 per cent reduction
6 + 7 ACSR	—	10 per cent reduction
30 + 7 ACSR	—	5 per cent reduction

The conductors shall be pre-tensioned for a minimum time of 60 minutes. At 15-minute intervals during this period the tension or sag shall be adjusted in order to maintain the correct value.

The conductors shall be then finally erected with an overtension equivalent to 10 per cent at 15°C.

Conductor pre-tension and erection sags and tensions are given in Tables 19—24. Erection figures include the appropriate value of overtension.

6.7 Use of ACSR PVC-covered

Technical data is provided in this Standard covering the erection of full section(s) of line conductored with ACSR PVC-covered. When a single span of ACSR PVC-covered is erected in a section of line the span shall not exceed 60 per cent of the average span.

6.8 Provision of Danger Plates

Each support on a line to this Standard shall be provided with a danger plate to ESI Standard 43—97. The plate shall be mounted approximately 3 m above ground level.

6.9 Precautions Against Access

The types of support requiring precautions against access and the methods to be employed are detailed in ESI Standard 43—90.

6.10 Minimum Height of Line Conductors and Clearance

The minimum height of line conductors shall comply with the Electricity (Overhead Lines) Regulation, 1970. Data are given in this Standard for conductors operating at maximum temperatures of 50°C and 75°C.

6.11 Erection of Auxiliary Equipment

Where a support carries auxiliary equipment the construction of the support shall be in accordance with ESI Standard 43—89, which also contains details of the earthing of these supports.

No allowance has been made in the basic support consideration for the additional mechanical loadings which would be imposed by auxiliary equipment and additional conductors.

6.12 Relationships with other Bodies

Where lines to this Standard are erected over or alongside the plant of the Post Office or British Rail the provisions agreed shall be complied with. Additionally, the requirements of the British Waterways Board should be noted.

Table 1 — Support Types and Duties

Duty	Assembly Fig. No.	Conductor Size mm ²	Deviation Angle		Reference for Downpull, Angles and Stays Table No.
			Maximum	Limit on H Poles stayed on one limb	
Straight line. Small deviation angles. Limited downpull.	3 Single Pole	100	0 to 6°	—	2, 3, 8 & 9
		150	0 to 5°	—	
		175(18/1)	0 to 4°	—	
		175(30/7)	0 to 3°	—	
		200	0 to 4°	—	
Straight line. Small deviation angles. 1:10 downpull	4 Single Pole	100	5° to 6°	—	2, 3, 8 & 9
		150	4° to 5°	—	
		175(18/1)	3° to 4°	—	
		175(30/7)	2° to 3°	—	
		200	3° to 4°	—	
Straight line section. Medium deviation angles 1:10 downpull	5 Single Pole	100	60°	—	8 & 9
		150	30°	—	
		175(18/1)	30°	—	
		175(30/7)	30°	—	
		200	30°	—	
Heavy deviation angles 1:10 downpull	6 'H' Pole	100	60°	60°	8 & 9
		150	60°	60°	
		175(18/1)	60°	53°	
		175(30/7)	60°	38°	
		200	60°	48°	
Extra heavy deviation angles 1:10 downpull 1:13 downpull 175(30/7) above 60°	7 'H' Pole	100	75°	68°	8 & 9
		150	75°	60°	
		175(18/1)	75°	53°	
		175(30/7)	75°	38°	
		200	75°	48°	
Occasional long span. Straight line on pin or line post insulators. Straight line section. Heavy deviation angles. 1:10 downpull	8 'H' pole	100	60°	60°	8 & 9
		150	60°	58°	
		175(18/1)	60°	51°	
		175(30/7)	60°	36°	
		200	60°	46°	
Termination for all conductors 1:10 downpull Terminal for occasional long span if adjacent support is Fig. 8 construction and mid- span spacing complies with ratio specified in 6.1.	9 'H' Pole	100	—	—	8 & 9
		150			
		175(18/1)			
		175(30/7)			
		200			
Termination for 100 mm ² ACSR 1:10 downpull	10 Single Pole	100	—	—	8 & 9

Downpull Limits and Maximum Angles of Line Deviation on Pin and Line Post Insulators

Table 2 — Bare Conductor

ACSR				Maximum Angle of Line Deviation (°)	
Size (mm ²)	Stranding (mm)	Wind Span (m)	Gradient	Single Member Crossarm	Double Member Crossarm
100	6/4.72 + 7/1.57	130	0 — 1:30	6	6
			1:30 — 1:20	6	6
			1:20 — 1:10	3	6
		150	0 — 1:30	5	5
			1:30 — 1:20	5	5
			1:20 — 1:10	*	5
150	18 + 1/3.35	130	0 — 1:30	5	5
			1:30 — 1:20	5	5
			1:20 — 1:10	*	5
		150	0 — 1:30	4	4
			1:30 — 1:20	4	4
			1:20 — 1:10	*	4
175	18 + 1/3.61	130	0 — 1:30	4	4
			1:30 — 1:20	4	4
			1:20 — 1:10	*	4
		150	0 — 1:30	3	3
			1:30 — 1:20	3	3
			1:20 — 1:10	*	3
175	30 + 7/2.79	130	0 — 1:30	3	3
			1:30 — 1:20	*	3
			1:20 — 1:10	*	3
		150	0 — 1:30	2	2
			1:30 — 1:20	*	2
			1:20 — 1:10	*	2
200	18 + 1/3.86	130	0 — 1:30	4	4
			1:30 — 1:20	4	4
			1:20 — 1:10	*	4
		150	0 — 1:30	3	3
			1:30 — 1:20	2	3
			1:20 — 1:10	*	3

* Use double member crossarm (Fig. 4) for intermediate or pin angle conditions.

Downpull Limits and Maximum Angles of Line Deviation on Pin and Line Post Insulators.

Table 3 – PVC-covered Conductor

ACSR				Maximum Angle of Line Deviation (°)	
Size (mm ²)	Stranding (mm)	Wind Span (m)	Gradient	Single Member Crossarm	Double Member Crossarm
100	6/4.72 + 7/1.57	130	0 – 1:30	5	5
			1:30 – 1:20	5	5
			1:20 – 1:10	*	5
		150	0 – 1:30	4	4
			1:30 – 1:20	4	4
			1:20 – 1:10	*	4
150	18 + 1/3.35	130	0 – 1:30	4	4
			1:30 – 1:20	4	4
			1:20 – 1:10	*	4
		150	0 – 1:30	3	3
			1:30 – 1:20	3	3
			1:20 – 1:10	*	3
175	18 + 1/3.61	130	0 – 1:30	4	4
			1:30 – 1:20	4	4
			1:20 – 1:10	*	4
		150	0 – 1:30	3	3
			1:30 – 1:20	0	3
			1:20 – 1:10	*	3
175	30 + 7/2.79	130	0 – 1:30	2	3
			1:30 – 1:20	*	3
			1:20 – 1:10	*	3
		150	0 – 1:30	*	2
			1:30 – 1:20	*	2
			1:20 – 1:10	*	2
200	18 + 1/3.86	130	0 – 1:30	3	3
			1:30 – 1:20	1	3
			1:20 – 1:10	*	3
		150	0 – 1:30	2	2
			1:30 – 1:20	*	2
			1:20 – 1:10	*	2

* Use double member crossarm (Fig. 4) for intermediate or pin angle conditions.

Wind Loading Spans for Single Poles

Table 4 — Bare Conductor

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100 mm ² ACSR 6/4.72 + 7/1.57 mm	3 x 150 mm ² ACSR 18/3.35 + 1/3.35 mm	3 x 175 mm ² ACSR 18/3.61 + 1/3.61 mm	3 x 175 mm ² ACSR 30/2.79 + 7/2.79 mm	3 x 200 mm ² ACSR 18/3.86 + 1/3.86 mm
8.5	1.8	215 (M)	69	64	61	59	59
		265 (S)	133	123	119	114	115
		275	149	139	134	129	129
		285	150	150	150	144	145
		295	—	—	—	150	150
9.0	1.8	220 (M)	69	64	61	59	59
		275 (S)	139	129	124	120	120
		285	150	144	139	134	135
		295	—	150	150	149	150
		305	—	—	—	150	—
9.5	1.8	225 (M)	69	64	61	59	59
		280 (S)	137	127	123	118	119
		290	150	142	137	132	133
		300	—	150	150	146	147
		310	—	—	—	150	150
10.0	1.8	230 (M)	69	64	61	59	59
		285 (S)	136	126	122	117	118
		295	150	141	136	130	131
		305	—	150	149	144	145
		315	—	—	150	150	150
10.5	1.8	235 (M)	69	64	62	59	60
		290 (S)	135	125	121	116	117
		300	150	139	134	129	130
		310	—	150	147	141	142
		320	—	—	150	150	150
11.0	1.8	240 (M)	69	64	62	60	60
		295 (S)	134	125	120	116	116
		305	148	138	133	128	128
		310	150	150	145	140	140
		325	—	—	150	150	150
11.5	1.8	245 (M)	69	64	62	60	60
		300 (S)	134	124	120	115	116
		310	147	136	132	126	127
		320	150	148	143	138	139
		330	—	150	150	149	150
		340	—	—	—	150	—

Table 4 (Cont'd.)

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100mm ² ACSR 6/4.72 + 7/1.57mm	3 x 150mm ² ACSR 18/3.35 + 1/3.35mm	3 x 175mm ² ACSR 18/3.61 + 1/3.61mm	3 x 175mm ² ACSR 30/2.79 + 7/2.79mm	3 x 200mm ² ACSR 18/3.86 + 1/3.86mm
12.0	1.8	250 (M)	70	65	62	60	60
		305 (S)	133	123	119	114	115
		315	146	135	130	125	126
		325	150	147	142	136	137
		335	—	150	150	147	148
		345	—	—	—	150	150
13.0	1.8	260 (M)	71	66	64	61	62
		320 (S)	139	129	124	120	120
		330	150	140	135	130	131
		340	—	150	146	141	142
		350	—	—	150	150	150
14.0	1.8	275 (M)	76	71	68	66	66
		335 (S)	144	133	129	124	124
		345	150	144	139	133	134
		355	—	150	149	143	144
		365	—	—	150	150	150
15.0	2.1	290 (M)	82	76	73	70	71
		350 (S)	148	137	132	127	128
		360	150	147	141	136	137
		370	—	150	150	145	146
		380	—	—	—	150	150
16.0	2.1	305 (M)	88	81	78	75	76
		365 (S)	150	143	138	133	133
		375	—	150	147	141	142
		385	—	—	150	150	150
17.0	2.1	320 (M)	95	88	85	82	82
		375 (S)	150	142	137	131	132
		385	—	150	145	140	141
		395	—	—	150	148	149
		405	—	—	—	150	150
18.0	2.1	330 (M)	96	89	86	82	83
		390 (S)	150	145	140	134	135
		400	—	150	148	142	143
		410	—	—	150	150	150
20.0	2.1	360 (M)	104	96	93	89	90
		415 (S)	150	146	141	136	136
		425	—	150	148	143	143
		435	—	—	150	149	150
		445	—	—	—	150	—

NOTES:

1. (M) or (S) against a pole diameter indicates that the pole is standard medium or stout pole in accordance with BS 1990.
2. Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Wind Loading Spans for Single Poles

Table 5 – PVC-covered Conductor

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100mm ² ACSR 6/4.72 + 7/1.57mm	3 x 150mm ² ACSR 18/3.35 + 1/3.35mm	3 x 175mm ² ACSR 18/3.61 + 1/3.61mm	3 x 175mm ² ACSR 30/2.79 + 7/2.79mm	3 x 200mm ² ACSR 18/3.86 + 1/3.86mm
8.5	1.8	215 (M)	61	57	55	53	54
		265 (S)	118	111	107	103	104
		275	133	125	121	116	117
		285	149	140	135	130	131
		295	150	150	150	144	146
		305			-	150	150
9.0	1.8	220 (M)	61	57	55	53	54
		275 (S)	124	116	112	108	109
		285	139	130	126	121	122
		295	150	144	140	135	136
		305		150	150	148	149
		315				150	150
9.5	1.8	225 (M)	61	57	55	53	54
		280 (S)	122	114	111	107	108
		290	137	128	124	120	120
		300	150	141	137	132	133
		310		150	150	144	146
		320				150	150
10.0	1.8	230 (M)	61	57	55	53	54
		285 (S)	121	113	110	106	107
		295	135	126	122	118	119
		305	149	139	135	130	131
		315	150	150	147	142	143
		325	-		150	150	150
10.5	1.8	235 (M)	61	57	56	54	54
		290 (S)	120	113	109	105	106
		300	134	125	121	117	117
		310	147	137	133	128	129
		320	150	149	144	139	140
		330		150	150	150	150
11.0	1.8	240 (M)	62	58	56	54	54
		295 (S)	120	112	108	104	105
		305	132	124	120	115	116
		315	144	135	131	126	127
		325	150	146	142	137	138
		335		150	150	147	148
11.5	1.8	345	-	-	-	150	150
		245 (M)	62	58	56	54	54
		300 (S)	119	111	108	104	105
		310	131	122	119	114	115
		320	143	133	129	125	125
		330	150	144	140	135	136
12.0	1.8	340	-	150	150	144	146
		350	-	-	-	150	150
		250 (M)	62	58	56	54	55
		305 (S)	118	111	107	103	104
		315	130	121	118	113	114
		325	141	132	128	123	124
12.0	1.8	335	150	142	138	133	134
		345	-	150	147	142	143
		355	-	-	150	150	150

(Cont'd)

Table 5 (Cont'd.)

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100mm ² ACSR 6/4.72 + 7/1.57mm	3 x 150mm ² ACSR 18/3.35 + 1/3.35mm	3 x 175mm ² ACSR 18/3.61 + 1/3.61mm	3 x 175mm ² ACSR 30/2.79 + 7/2.79mm	3 x 200mm ² ACSR 18/3.86 + 1/3.86mm
13.0	1.8	260 (M)	63	59	57	55	56
		320 (S)	124	116	112	108	109
		330	135	126	122	118	119
		340	146	136	132	127	128
		350	150	146	141	137	137
		360	—	150	150	146	147
		370	—	—	—	150	150
14.0	1.8	275 (M)	68	64	62	59	60
		335 (S)	128	120	116	112	113
		345	138	129	125	121	121
		355	148	138	134	129	130
		365	150	147	143	138	139
		375	—	150	150	146	147
		385	—	—	—	150	150
15.0	2.1	290 (M)	73	68	66	64	64
		350 (S)	132	123	119	115	116
		360	141	132	128	123	124
		370	150	140	136	131	132
		380	—	149	144	139	140
		390	—	150	150	147	148
		400	—	—	—	150	150
16.0	2.1	305 (M)	78	73	71	68	69
		365 (S)	137	128	124	120	121
		375	146	137	133	128	129
		385	150	145	141	136	137
		395	—	150	149	143	144
		405	—	—	150	150	150
17.0	2.1	320 (M)	85	79	77	74	74
		375 (S)	136	127	123	119	120
		385	145	135	131	126	127
		395	150	143	139	134	135
		405	—	150	146	141	142
		415	—	—	150	148	149
		425	—	—	—	150	150
18.0	2.1	330 (M)	85	80	77	74	75
		390 (S)	139	130	126	121	122
		400	147	137	133	128	129
		410	150	145	140	135	136
		420	—	150	147	142	143
		430	—	—	150	149	150
		440	—	—	—	150	—
20.0	2.1	360 (M)	93	87	84	81	81
		415 (S)	141	131	127	123	124
		425	148	138	134	129	130
		435	150	145	140	135	136
		445	—	150	146	141	142
		455	—	—	150	147	148
		465	—	—	—	150	150

NOTES:

1. (M) or (S) against a pole diameter indicates that the pole is the standard medium or stout pole in accordance with BS 1990.
2. Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Wind Loading Spans for 'H' Poles

Table 6 — Bare Conductor

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100mm ² ACSR 6/4.72 + 7/1.57mm	3 x 150mm ² ACSR 18/3.35 + 1/3.35mm	3 x 175mm ² ACSR 18/3.61 + 1/3.61mm	3 x 175mm ² ACSR 30/2.79 + 7/2.79mm	3 x 200mm ² ACSR 18/3.86 + 1/3.86mm
8.5	1.8	215 (M) 265 (S)	150 180	139 180	135 180	129 180	130 180
9.0	1.8	220 (M) 275 (S)	150 180	139 180	134 180	129 180	130 180
9.5	1.8	225 (M) 280 (S)	149 180	139 180	134 180	129 180	129 180
10.0	1.8	230 (M) 285 (S)	150 180	139 180	134 180	129 180	129 180
10.5	1.8	235 (M) 290 (S)	150 180	139 180	134 180	129 180	130 180
11.0	1.8	240 (M) 295 (S)	151 180	140 180	135 180	130 180	130 180
11.5	1.8	245 (M) 300 (S)	151 180	140 180	135 180	130 180	131 180
12.0	1.8	250 (M) 305 (S)	151 180	140 180	136 180	130 180	131 180
13.0	1.8	260 (M) 320 (S)	155 180	143 180	138 180	133 180	134 180
14.0	1.8	275 (M) 335 (S)	165 180	153 180	148 180	142 180	143 180
15.0	2.1	290 (M) 350 (S)	177 180	164 180	159 180	152 180	153 180
16.0	2.1	305 (M) 365 (S)	180 —	176 180	169 180	163 180	164 180
17.0	2.1	320 (M) 375 (S)	180 —	180 —	180 —	176 180	177 180
18.0	2.1	330 (M) 390 (S)	180 —	180 —	180 —	178 180	179 180
20.0	2.1	360 (M)	180	180	180	180	180

NOTES:

1. (M) or (S) against a pole diameter indicates that the pole is the standard medium or stout pole in accordance with BS 1990.
2. Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Wind Loading Spans for 'H' Poles

Table 7 — PVC-covered Conductor

Pole Details			Maximum Wind Loading Spans (m)				
Length (m)	Planting Depth (m)	Dia. at 1.5m from butt (mm)	3 x 100mm ² ACSR 6/4.72 + 7/1.57mm	3 x 150mm ² ACSR 18/3.35 + 1/3.35mm	3 x 175mm ² ACSR 18/3.61 + 1/3.61mm	3 x 175mm ² ACSR 30/2.79 + 7/2.79mm	3 x 200mm ² ACSR 18/3.86 + 1/3.86mm
8.5	1.8	215 (M) 265 (S)	134 180	125 180	121 180	117 180	118 180
9.0	1.8	220 (M) 275 (S)	133 180	125 180	121 180	116 180	117 180
9.5	1.8	225 (M) 280 (S)	133 180	124 180	121 180	116 180	117 180
10.0	1.8	230 (M) 285 (S)	133 180	125 180	121 180	116 180	117 180
10.5	1.8	235 (M) 290 (S)	134 180	125 180	121 180	117 180	118 180
11.0	1.8	240 (M) 295 (S)	134 180	125 180	121 180	117 180	118 180
11.5	1.8	245 (M) 300 (S)	135 180	126 180	122 180	118 180	118 180
12.0	1.8	250 (M) 305 (S)	135 180	126 180	122 180	118 180	119 180
13.0	1.8	260 (M) 320 (S)	138 180	129 180	125 180	120 180	121 180
14.0	1.8	275 (M) 335 (S)	147 180	138 180	133 180	129 180	130 180
15.0	2.1	290 (M) 350 (S)	158 180	148 180	143 180	138 180	139 180
16.0	2.1	305 (M) 365 (S)	169 180	158 180	153 180	147 180	148 180
17.0	2.1	320 (M) 375 (S)	180 —	171 180	165 180	159 180	160 180
18.0	2.1	330 (M) 390 (S)	180 —	172 180	166 180	161 180	162 180
20.0	2.1	360 (M) 415 (S)	180 —	180 —	180 —	174 180	175 180

NOTES:

- (M) or (S) against a pole diameter indicates that the pole is the standard medium or stout pole in accordance with BS 1990.
- Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Stay Details (Excluding strut loading considerations)

Table 8 — Bare Conductor

Conductors	Angle of Stay Slope	Maximum Angles of Line Deviation (°)															
		Grade 700 Stay Strand								Grade 1150 Stay Strand							
		7/4.00 mm Stays				19/3.55 mm Stays				7/3.25 mm Stays				7/4.00 mm Stays			
		1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
3 x 100mm ² ACSR 6/4.72 + 7/1.57 mm	30°	9	24	41	58	22	50	75	—	10	26	44	62	19	45	75	—
	35°	11	29	49	70	26	61	75	—	12	31	53	75	24	54	75	—
	40°	14	34	57	75	31	68	75	—	15	36	61	75	27	63	75	—
	45°	16	39	64	75	34	68	75	—	17	41	69	75	31	68	75	—
3 x 150 mm ² ACSR 18/3.35 + 1/3.35 mm	30°	7	21	36	51	19	44	75	—	8	22	39	54	17	40	67	75
	35°	10	25	43	61	23	53	75	—	11	27	46	65	21	48	75	—
	40°	12	30	50	71	27	60	75	—	13	32	53	75	24	55	75	—
	45°	14	34	57	75	30	60	75	—	15	36	60	75	27	60	75	—
3 x 175 mm ² ACSR 18/3.61 + 1/3.61 mm	30°	6	19	32	45	17	40	66	75	7	20	35	49	15	36	60	75
	35°	8	23	39	55	21	48	75	—	9	24	41	58	18	43	72	75
	40°	10	27	45	63	24	53	75	—	11	29	48	68	21	50	75	—
	45°	12	30	51	72	27	53	75	—	13	33	54	75	24	53	75	—
3 x 175 mm ² ACSR 30/2.79 + 7/2.79 mm	30°	5	14	24	34	13	29	48	68	5	15	26	36	11	26	44	61
	35°	6	17	29	40	15	35	58	75	7	18	31	43	14	32	52	73
	40°	8	20	33	46	18	38	67	75	8	21	35	50	16	37	60	75
	45°	9	22	37	52	20	38	75	—	10	24	40	56	18	38	68	75
3 x 200 mm ² ACSR 18/3.86 + 1/3.86 mm	30°	6	17	29	41	15	36	60	75	6	18	31	44	14	32	54	75
	35°	8	21	35	49	19	43	72	75	8	22	37	53	17	39	64	75
	40°	9	24	41	57	22	48	75	—	10	26	43	61	19	45	75	—
	45°	11	27	46	65	25	48	75	—	12	29	49	70	22	48	75	—

Table 9 — PVC-covered Conductor

Conductors	Angle of Stay Slope	Maximum Angles of Line Deviation (°)															
		Grade 700 Stay Strand								Grade 1150 Stay Strand							
		7/4.00 mm Stays				19/3.55 mm Stays				7/3.25 mm Stays				7/4.00 mm Stays			
		1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
3 x 100 mm ² ACSR 6/4.72 + 7/1.57 mm	30°	7	22	39	54	20	48	75	—	8	24	42	59	18	43	73	75
	35°	10	27	47	67	24	58	75	—	11	29	50	71	21	52	75	—
	40°	12	32	54	75	28	67	75	—	13	34	58	75	25	60	75	—
	45°	14	37	61	75	32	68	75	—	15	39	66	75	29	68	75	—
3 x 150 mm ² ACSR 18/3.35 + 1/3.35 mm	30°	6	20	35	50	18	43	73	75	7	21	38	53	16	39	66	75
	35°	8	24	42	60	22	52	75	—	9	26	45	64	19	47	75	—
	40°	10	29	49	70	25	60	75	—	11	31	52	75	23	54	75	—
	45°	12	33	55	75	29	60	75	—	13	35	59	75	26	60	75	—
3 x 175 mm ² ACSR 18/3.61 + 1/3.61 mm	30°	5	18	31	45	16	39	66	75	6	19	34	48	14	35	59	75
	35°	7	22	38	54	19	47	75	—	8	24	41	58	17	42	71	75
	40°	9	26	44	63	23	53	75	—	10	28	47	67	20	49	75	—
	45°	11	30	50	71	26	53	75	—	12	32	53	75	23	53	75	—
3 x 175 mm ² ACSR 30/2.79 + 7/2.79 mm	30°	4	13	23	33	12	29	48	67	4	14	25	35	10	26	43	60
	35°	5	16	28	39	14	34	57	75	6	17	30	42	13	31	51	71
	40°	6	19	32	46	17	38	66	75	7	21	35	49	15	36	59	75
	45°	8	22	37	52	19	38	75	—	9	23	39	55	17	38	67	75
3 x 200 mm ² ACSR 18/3.86 + 1/3.86 mm	30°	5	16	28	40	14	35	59	75	5	17	31	43	12	32	53	75
	35°	6	20	34	49	17	42	71	75	7	21	37	52	15	38	64	75
	40°	8	23	40	57	21	48	75	—	9	25	43	61	18	44	74	75
	45°	10	27	45	64	23	48	75	—	11	29	48	69	21	48	75	—

Notes on Tables 8 & 9

1. Terminal stay requirements shall be matched to 60° angle of line deviation.
2. Maximum angles of line deviation shall be reduced by 2° for occasional long span requirements.
3. Reference must be made to Table 1 for limiting angles of deviation.

Table 10 — Permissible Strut Loading of Poles

Dia. at 1.5 m (mm)	Pole Class (BS 1990)	Pole Planting Depth (m)	Pole Length (m)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)
215	M	1.8	8.5	150	3388	165	4199	180	5148
220			9.0	150	3049	165	3774	180	4620
225			9.5	150	2769	165	3423	180	4185
230			10.0	150	2536	165	3129	185	4076
235			10.5	150	2338	170	3082	190	3990
240			11.0	150	2169	170	2853	190	3688
245			11.5	150	2024	170	2657	190	3428
250			12.0	150	1898	170	2487	190	3203
260			13.0	160	1936	180	2505	200	3191
275			14.0	160	1809	180	2327	205	3123
290		2.1	15.0	165	1872	185	2394	210	3193
305			16.0	170	1887	190	2392	215	3159
320			17.0	180	2026	200	2539	220	3143
330			18.0	180	1886	200	2356	220	2909
360			20.0	180	1765	205	2301	230	2952
265	S	1.8	8.5	190	8308	215	11029	240	14370
275			9.0	190	7664	215	10138	240	13169
280			9.5	190	6903	215	9121	245	12442
285			10.0	190	6270	215	8273	245	11269
290			10.5	190	5736	220	7969	250	10797
295			11.0	190	5282	220	7325	250	9910
300			11.5	190	4891	220	6771	250	9147
305			12.0	190	4553	220	6292	250	8487
320			13.0	195	4351	225	5953	255	7961
335			14.0	195	3988	225	5426	255	7221
350		2.1	15.0	195	3801	225	5167	260	7193
365			16.0	200	3736	230	5032	265	6939
375			17.0	200	3424	230	4595	265	6314
390			18.0	200	3247	230	4334	265	5925
415			20.0	200	2901	230	3840	265	5206
				Poles with top dia. equal to min. specified in BS 1990		Poles with mean value of top dia.		Poles with top dia. equal to max. specified in BS 1990	

For notes on Table 10 see Page 25

NOTES ON TABLE 10 — PERMISSIBLE STRUT LOADING OF POLES

1. The values of strut loading are safe working loads.
2. The table shall be used in conjunction with Nomograms 1—10.
3. The weight of any pole mounted equipment shall be added to the value of strut loading taken from the appropriate nomogram before reference is made to the table.
4. Pole diameters to match wind loading spans shall be adopted for stayed poles not having additional pole mounted equipment where the angle of deviation is below 5° .
5. The maximum allowable downpull for 175 mm² (30 + 7/2.79 mm) ACSR shall be 1 in 13 for angles of deviation above 60° .
6. The limbs of 'H' poles shall be selected as follows:
 - (a) The total strut loading shall be derived from the appropriate nomogram.
 - (b) The total strut loading shall be apportioned as follows:

(i) Stays on outer limb only	—	100 per cent
(ii) Stayed equally on each limb	—	50 per cent
(iii) Two stays on inner limb and one stay on outer limb	—	60 per cent
 - (c) Both limbs of 'H' pole shall be of the same size, matched and designed to withstand apportioned strut loading.

Ordinates for Plotting Uplift and Ground Clearance Curves — ACSR Basic Span 130 m

Table 11 — Bare Conductor

Span (m) (2X)	Sag (m)								
	100mm ² 6/4.72 + 7/1.57mm			175mm ² 30 + 7/2.79mm			150mm ² , 175mm ² , 200mm ² 18 + 1/3.35mm 18 + 1/3.61mm 18 + 1/3.86mm		
	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C
40	0.10	0.25	0.28	0.11	0.22	0.26	0.12	0.28	0.33
80	0.40	0.93	1.12	0.43	0.87	1.05	0.48	1.13	1.31
120	0.91	2.09	2.52	0.98	1.97	2.36	1.09	2.54	2.95
160	1.62	3.71	4.48	1.74	3.49	4.20	1.94	4.51	5.24
200	2.53	5.80	7.00	2.72	5.46	6.56	3.02	7.05	8.18
240	3.64	8.35	10.08	3.91	7.86	9.45	4.35	10.15	11.78
280	4.95	11.37	13.72	5.33	10.70	12.86	5.93	13.82	16.03
320	6.47	14.85	17.92	6.96	13.97	16.79	7.74	18.05	20.94
360	8.10	18.80	22.68	8.80	17.69	21.25	9.80	22.84	26.50
400	10.11	23.21	28.00	10.87	21.84	26.23	12.10	28.20	32.72
440	12.23	28.08	33.87	13.15	26.42	31.74	14.64	34.12	39.59
480	14.56	33.42	40.31	15.65	31.44	37.78	17.42	40.61	47.12
520	17.08	39.22	47.31	18.37	36.90	44.34	20.44	47.66	55.30
560	19.81	45.48	54.87	21.30	42.80	51.42	23.71	55.27	64.13
600	22.74	52.21	62.99	24.45	49.13	59.03	27.21	63.45	73.62
640	25.88	59.40	71.66	27.82	55.89	67.16	30.96	72.19	83.76
680	29.21	67.06	80.90	31.41	63.10	75.82	34.95	81.50	94.56

NOTE: Ordinates for plotting uplift curves are based on conductor design tensions. Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR — 20% reduction
6 + 7 ACSR — 10% reduction
30 + 7 ACSR — 5% reduction

Ordinates for Plotting Uplift and Ground Clearance Curves — ACSR Basic Span 130m

Table 12 — PVC-covered Conductor

Span (m) (2X)	Sag (m)								
	100mm ² 6/4.72 + 7/1.57mm			175mm ² 30 + 7/2.79mm			150mm ² , 175mm ² , 200mm ² 18 + 1/3.35mm 18 + 1/3.61mm 18 + 1/3.86mm		
	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C
40	0.15	0.27	0.32	0.14	0.25	0.29	0.17	0.33	0.37
80	0.59	1.09	1.26	0.57	1.00	1.16	0.67	1.31	1.47
120	1.32	2.46	2.84	1.29	2.25	2.61	1.51	2.95	3.31
160	2.35	4.38	5.05	2.29	4.01	4.65	2.68	5.24	5.88
200	3.67	6.84	7.89	3.58	6.26	7.26	4.18	8.19	9.19
240	5.28	9.84	11.36	5.16	9.01	10.45	6.03	11.79	13.23
280	7.19	13.40	15.46	7.02	12.26	14.22	8.20	16.05	18.01
320	9.39	17.50	20.20	9.17	16.02	18.58	10.71	20.96	23.52
360	11.88	22.15	25.56	11.61	20.27	23.51	13.56	26.53	29.77
400	14.67	27.34	31.56	14.33	25.03	29.03	16.74	32.75	36.75
440	17.75	33.08	38.18	17.34	30.28	35.12	20.25	39.63	44.47
480	21.12	39.37	45.44	20.64	36.04	41.80	24.10	47.16	52.92
520	24.79	46.21	53.33	24.22	42.30	49.05	28.29	55.35	62.11
560	28.75	53.59	61.85	28.09	49.05	56.89	32.81	64.19	72.04
600	33.00	61.52	71.00	32.25	56.31	65.31	37.66	73.69	82.69
640	37.55	69.99	80.78	36.69	64.07	74.30	42.85	83.84	94.09
680	42.39	79.02	91.19	41.42	72.33	83.88	48.37	94.65	106.22

NOTE: Ordinates for plotting uplift curves are based on conductor design tensions. Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR — 20% reduction
6 + 7 ACSR — 10% reduction
30 + 7 ACSR — 5% reduction

Design Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR -- Basic Span 130 m

Table 13 -- Bare Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	722.4	0.11	0.17	0.25	0.34	0.45	0.56	0.70	0.84	1.00	1.18	1.36	1.57	1.78	2.01	2.25
5.0	666.9	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.91	1.09	1.27	1.48	1.70	1.93	2.18	2.44
10.0	616.6	0.13	0.20	0.29	0.40	0.52	0.66	0.82	0.99	1.17	1.38	1.60	1.83	2.09	2.36	2.64
15.0	571.4	0.14	0.22	0.32	0.43	0.56	0.71	0.88	1.06	1.27	1.49	1.72	1.98	2.25	2.54	2.85
20.0	531.2	0.15	0.24	0.34	0.46	0.61	0.77	0.95	1.14	1.36	1.60	1.85	2.13	2.42	2.73	3.06
25.0	495.7	0.16	0.25	0.37	0.50	0.65	0.82	1.01	1.23	1.46	1.71	1.99	2.28	2.59	2.93	3.28
30.0	464.5	0.17	0.27	0.39	0.53	0.69	0.88	1.08	1.31	1.56	1.83	2.12	2.43	2.77	3.13	3.51
35.0	436.9	0.18	0.29	0.42	0.56	0.74	0.93	1.15	1.39	1.66	1.94	2.25	2.59	2.94	3.32	3.73
40.0	412.6	0.20	0.31	0.44	0.60	0.78	0.99	1.22	1.47	1.75	2.06	2.39	2.74	3.12	3.52	3.95
45.0	391.2	0.21	0.32	0.46	0.63	0.82	1.04	1.28	1.55	1.85	2.17	2.52	2.89	3.29	3.71	4.16
50.0	372.2	0.22	0.34	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.65	3.04	3.46	3.90	4.37
55.0	355.2	0.23	0.36	0.51	0.69	0.91	1.15	1.41	1.71	2.04	2.39	2.77	3.18	3.62	4.09	4.58
60.0	340.0	0.24	0.37	0.53	0.72	0.95	1.20	1.48	1.79	2.13	2.50	2.90	3.33	3.78	4.27	4.79
65.0	326.4	0.25	0.39	0.56	0.75	0.99	1.25	1.54	1.86	2.22	2.60	3.02	3.46	3.94	4.45	4.99
70.0	314.1	0.26	0.40	0.58	0.78	1.02	1.30	1.60	1.94	2.30	2.70	3.14	3.60	4.10	4.62	5.18
75.0	302.9	0.27	0.42	0.60	0.81	1.06	1.34	1.66	2.01	2.39	2.80	3.25	3.73	4.25	4.79	5.37

Design Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR — Basic Span 130 m

Table 14 — PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	707.2	0.16	0.25	0.35	0.48	0.63	0.80	0.99	1.19	1.42	1.67	1.93	2.22	2.53	2.85	3.20
5.0	666.9	0.17	0.26	0.38	0.51	0.67	0.85	1.05	1.27	1.51	1.77	2.05	2.35	2.68	3.02	3.39
10.0	630.4	0.18	0.28	0.40	0.54	0.71	0.90	1.11	1.34	1.59	1.87	2.17	2.49	2.83	3.20	3.58
15.0	597.4	0.19	0.29	0.42	0.57	0.75	0.95	1.17	1.41	1.68	1.97	2.29	2.63	2.99	3.37	3.78
20.0	567.6	0.20	0.31	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	2.41	2.76	3.15	3.55	3.98
25.0	540.7	0.21	0.32	0.47	0.63	0.83	1.04	1.29	1.56	1.86	2.18	2.53	2.90	3.30	3.73	4.18
30.0	516.5	0.22	0.34	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.65	3.04	3.46	3.90	4.38
35.0	494.5	0.23	0.35	0.51	0.69	0.90	1.14	1.41	1.71	2.03	2.38	2.76	3.17	3.61	4.08	4.57
40.0	474.5	0.24	0.37	0.53	0.72	0.94	1.19	1.47	1.78	2.12	2.48	2.88	3.31	3.76	4.25	4.76
45.0	456.4	0.25	0.38	0.55	0.75	0.98	1.24	1.53	1.85	2.20	2.58	3.00	3.44	3.91	4.42	4.95
50.0	439.8	0.25	0.40	0.57	0.78	1.01	1.28	1.59	1.92	2.28	2.68	3.11	3.57	4.06	4.58	5.14
55.0	424.7	0.26	0.41	0.59	0.80	1.05	1.33	1.64	1.99	2.37	2.78	3.22	3.70	4.20	4.75	5.32
60.0	410.8	0.27	0.43	0.61	0.83	1.09	1.38	1.70	2.05	2.45	2.87	3.33	3.82	4.35	4.91	5.50
65.0	398.0	0.28	0.44	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.96	3.44	3.94	4.49	5.07	5.68
70.0	386.2	0.29	0.45	0.65	0.89	1.16	1.46	1.81	2.19	2.60	3.05	3.54	4.06	4.62	5.22	5.85
75.0	375.2	0.30	0.47	0.67	0.91	1.19	1.51	1.86	2.25	2.68	3.14	3.64	4.18	4.76	5.37	6.02

Design Sag/Tension Tables for 175 mm² (30 + 7/2.79) ACSR — Basic Span 130 m

Table 15 — Bare Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)															
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0	
0.0	1480.5	0.12	0.18	0.27	0.36	0.47	0.60	0.74	0.89	1.06	1.25	1.45	1.66	1.89	2.13	2.39	
5.0	1382.6	0.13	0.20	0.28	0.39	0.51	0.64	0.79	0.96	1.14	1.34	1.55	1.78	2.02	2.28	2.56	
10.0	1292.9	0.14	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.22	1.43	1.66	1.90	2.16	2.44	2.74	
15.0	1211.4	0.14	0.23	0.32	0.44	0.58	0.73	0.90	1.09	1.30	1.52	1.77	2.03	2.31	2.61	2.92	
20.0	1137.6	0.15	0.24	0.35	0.47	0.61	0.78	0.96	1.16	1.38	1.62	1.88	2.16	2.46	2.78	3.11	
25.0	1071.3	0.16	0.25	0.37	0.50	0.65	0.83	1.02	1.23	1.47	1.72	2.00	2.29	2.61	2.95	3.30	
30.0	1011.8	0.17	0.27	0.39	0.53	0.69	0.87	1.08	1.31	1.55	1.82	2.12	2.43	2.76	3.12	3.50	
35.0	958.5	0.18	0.28	0.41	0.56	0.73	0.92	1.14	1.38	1.64	1.93	2.23	2.56	2.92	3.29	3.69	
40.0	910.7	0.19	0.30	0.43	0.59	0.77	0.97	1.20	1.45	1.73	2.03	2.35	2.70	3.07	3.47	3.89	
45.0	867.7	0.20	0.31	0.45	0.62	0.81	1.02	1.26	1.52	1.81	2.13	2.47	2.83	3.22	3.64	4.08	
50.0	829.1	0.21	0.33	0.47	0.65	0.84	1.07	1.32	1.59	1.90	2.23	2.58	2.96	3.37	3.81	4.27	
55.0	794.3	0.22	0.34	0.50	0.67	0.88	1.11	1.38	1.66	1.98	2.32	2.70	3.09	3.52	3.97	4.46	
60.0	762.8	0.23	0.36	0.52	0.70	0.92	1.16	1.43	1.73	2.06	2.42	2.81	3.22	3.67	4.14	4.64	
65.0	734.2	0.24	0.37	0.54	0.73	0.95	1.21	1.49	1.80	2.14	2.51	2.92	3.35	3.81	4.30	4.82	
70.0	708.1	0.25	0.39	0.56	0.76	0.99	1.25	1.54	1.87	2.22	2.61	3.02	3.47	3.95	4.46	5.00	
75.0	684.2	0.26	0.40	0.57	0.78	1.02	1.29	1.60	1.93	2.30	2.70	3.13	3.59	4.09	4.61	5.17	

Design Sag/Tension Tables for 175 mm² (30 + 7/2.79) ACSR — Basic Span 130 m

Table 16 — PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	1410.2	0.15	0.24	0.35	0.47	0.61	0.78	0.96	1.16	1.38	1.62	1.88	2.16	2.46	2.77	3.11
5.0	1334.8	0.16	0.25	0.36	0.50	0.65	0.82	1.01	1.23	1.46	1.71	1.99	2.28	2.59	2.93	3.28
10.0	1266.0	0.17	0.27	0.38	0.52	0.68	0.87	1.07	1.29	1.54	1.81	2.09	2.40	2.73	3.09	3.46
15.0	1203.3	0.18	0.28	0.40	0.55	0.72	0.91	1.12	1.36	1.62	1.90	2.20	2.53	2.88	3.25	3.64
20.0	1146.2	0.19	0.29	0.42	0.58	0.76	0.96	1.18	1.43	1.70	1.99	2.31	2.65	3.02	3.41	3.82
25.0	1094.4	0.20	0.31	0.44	0.61	0.79	1.00	1.24	1.50	1.78	2.09	2.42	2.78	3.16	3.57	4.00
30.0	1047.2	0.21	0.32	0.46	0.63	0.83	1.05	1.29	1.56	1.86	2.18	2.53	2.91	3.31	3.73	4.18
35.0	1004.2	0.22	0.34	0.48	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.64	3.03	3.45	3.89	4.36
40.0	965.0	0.22	0.35	0.50	0.69	0.90	1.14	1.40	1.70	2.02	2.37	2.75	3.15	3.59	4.05	4.54
45.0	929.2	0.23	0.36	0.52	0.71	0.93	1.18	1.46	1.76	2.10	2.46	2.85	3.28	3.73	4.21	4.72
50.0	896.4	0.24	0.38	0.54	0.74	0.97	1.22	1.51	1.83	2.17	2.55	2.96	3.39	3.86	4.36	4.89
55.0	866.2	0.25	0.39	0.56	0.77	1.00	1.26	1.56	1.89	2.25	2.64	3.06	3.51	4.00	4.51	5.06
60.0	838.5	0.26	0.40	0.58	0.79	1.03	1.31	1.61	1.95	2.32	2.73	3.16	3.63	4.13	4.66	5.23
65.0	812.8	0.27	0.42	0.60	0.82	1.06	1.35	1.66	2.01	2.40	2.81	3.26	3.74	4.26	4.81	5.39
70.0	789.1	0.27	0.43	0.62	0.84	1.10	1.39	1.71	2.07	2.47	2.90	3.36	3.86	4.39	4.95	5.55
75.0	767.1	0.28	0.44	0.63	0.86	1.13	1.43	1.76	2.13	2.54	2.98	3.46	3.97	4.51	5.10	5.71

Design Sag/Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 130 m

Table 17 — Bare Conductor

Temp (°C)	Tension (kgf)			Sag (m) for span lengths (m)														
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	787.9	906.9	1026.7	0.13	0.21	0.30	0.41	0.53	0.68	0.83	1.01	1.20	1.41	1.63	1.88	2.13	2.41	2.70
5.0	728.1	838.2	949.3	0.14	0.23	0.33	0.44	0.58	0.73	0.90	1.09	1.30	1.52	1.77	2.03	2.31	2.61	2.92
10.0	675.1	777.4	881.0	0.16	0.24	0.35	0.48	0.62	0.79	0.97	1.18	1.40	1.64	1.91	2.19	2.49	2.81	3.15
15.0	628.5	724.0	820.9	0.17	0.26	0.38	0.51	0.67	0.85	1.04	1.26	1.50	1.77	2.05	2.35	2.67	3.02	3.38
20.0	587.7	677.3	768.3	0.18	0.28	0.40	0.55	0.71	0.90	1.12	1.35	1.61	1.89	2.19	2.51	2.86	3.23	3.62
25.0	552.0	636.3	722.3	0.19	0.30	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	2.33	2.67	3.04	3.43	3.85
30.0	520.6	600.3	681.8	0.20	0.32	0.45	0.62	0.81	1.02	1.26	1.52	1.81	2.13	2.47	2.83	3.22	3.64	4.08
35.0	493.0	568.7	646.2	0.21	0.33	0.48	0.65	0.85	1.08	1.33	1.61	1.91	2.25	2.61	2.99	3.40	3.84	4.31
40.0	468.6	540.7	614.7	0.22	0.35	0.50	0.69	0.90	1.13	1.40	1.69	2.01	2.36	2.74	3.15	3.58	4.04	4.53
45.0	446.9	515.8	586.7	0.23	0.37	0.53	0.72	0.94	1.19	1.47	1.77	2.11	2.48	2.87	3.30	3.75	4.24	4.75
50.0	427.6	493.7	561.6	0.25	0.38	0.55	0.75	0.98	1.24	1.53	1.85	2.21	2.59	3.00	3.45	3.92	4.43	4.96
55.0	410.2	473.7	539.2	0.26	0.40	0.58	0.78	1.02	1.29	1.60	1.93	2.30	2.70	3.13	3.59	4.09	4.61	5.17
60.0	394.6	455.8	518.9	0.27	0.42	0.60	0.81	1.06	1.34	1.66	2.01	2.39	2.80	3.25	3.73	4.25	4.80	5.38
65.0	380.5	439.5	500.5	0.28	0.43	0.62	0.84	1.10	1.39	1.72	2.08	2.48	2.91	3.37	3.87	4.40	4.97	5.57
70.0	367.6	424.7	483.7	0.29	0.45	0.64	0.87	1.14	1.44	1.78	2.15	2.56	3.01	3.49	4.01	4.56	5.15	5.77
75.0	355.8	411.2	468.4	0.29	0.46	0.66	0.90	1.18	1.49	1.84	2.23	2.65	3.11	3.61	4.14	4.71	5.32	5.96

Design/Sag Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 130 m

Table 18 — PVC-covered Conductor

Temp (°C)	Tension (kgf)			Sag (m) for span lengths (m)														
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	770.3	887.2	1006.1	0.18	0.28	0.41	0.55	0.72	0.91	1.12	1.36	1.62	1.90	2.20	2.53	2.88	3.25	3.64
5.0	728.1	838.2	949.3	0.19	0.30	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	2.33	2.68	3.05	3.44	3.86
10.0	690.2	794.3	898.6	0.20	0.31	0.45	0.62	0.80	1.02	1.26	1.52	1.81	2.12	2.46	2.83	3.21	3.63	4.07
15.0	656.3	755.0	853.2	0.21	0.33	0.48	0.65	0.85	1.07	1.32	1.60	1.90	2.23	2.59	2.97	3.38	3.82	4.28
20.0	625.9	719.7	812.5	0.22	0.35	0.50	0.68	0.89	1.12	1.39	1.68	2.00	2.34	2.72	3.12	3.55	4.01	4.49
25.0	598.4	687.9	776.0	0.23	0.36	0.52	0.71	0.93	1.17	1.45	1.75	2.09	2.45	2.84	3.26	3.71	4.19	4.70
30.0	573.7	659.3	743.1	0.24	0.38	0.55	0.74	0.97	1.23	1.51	1.83	2.18	2.56	2.97	3.40	3.87	4.37	4.90
35.0	551.2	633.4	713.4	0.25	0.39	0.57	0.77	1.01	1.28	1.57	1.91	2.27	2.66	3.09	3.54	4.03	4.55	5.10
40.0	530.8	609.8	686.4	0.26	0.41	0.59	0.80	1.05	1.32	1.64	1.98	2.36	2.76	3.21	3.68	4.19	4.73	5.30
45.0	512.2	588.4	661.9	0.27	0.42	0.61	0.83	1.09	1.37	1.70	2.05	2.44	2.87	3.32	3.81	4.34	4.90	5.49
50.0	495.2	568.7	639.5	0.28	0.44	0.63	0.86	1.12	1.42	1.75	2.12	2.53	2.96	3.44	3.95	4.49	5.07	5.68
55.0	479.6	550.7	618.9	0.29	0.45	0.65	0.89	1.16	1.47	1.81	2.19	2.61	3.06	3.55	4.08	4.64	5.23	5.87
60.0	465.2	534.1	600.0	0.30	0.47	0.67	0.92	1.20	1.51	1.87	2.26	2.69	3.16	3.66	4.20	4.78	5.40	6.05
65.0	451.9	518.8	582.6	0.31	0.48	0.69	0.94	1.23	1.56	1.92	2.33	2.77	3.25	3.77	4.33	4.92	5.56	6.23
70.0	439.5	504.6	566.4	0.32	0.49	0.71	0.97	1.27	1.60	1.98	2.39	2.85	3.34	3.87	4.45	5.06	5.71	6.41
75.0	428.1	491.4	551.4	0.33	0.51	0.73	0.99	1.30	1.64	2.03	2.46	2.92	3.43	3.98	4.57	5.20	5.87	6.58

Pre-tension and Erection Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR — Basic Span 130 m

Table 19 — Bare Conductor

Pre-tension kgf (All Temps)	Sag (m) for span lengths (m) at all temperatures														
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
750	0.11	0.17	0.25	0.33	0.44	0.55	0.68	0.83	0.98	1.15	1.34	1.53	1.75	1.97	2.21

Erection Temp. (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	798.4	0.10	0.16	0.23	0.31	0.40	0.51	0.63	0.76	0.91	1.06	1.23	1.42	1.61	1.82	2.04
5.0	736.9	0.11	0.17	0.25	0.33	0.44	0.55	0.68	0.83	0.98	1.15	1.34	1.53	1.75	1.97	2.21
10.0	680.2	0.12	0.18	0.27	0.36	0.47	0.60	0.74	0.89	1.06	1.25	1.45	1.66	1.89	2.14	2.39
15.0	628.5	0.13	0.20	0.29	0.39	0.51	0.65	0.80	0.97	1.15	1.35	1.57	1.80	2.05	2.31	2.59
20.0	582.1	0.14	0.22	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.46	1.69	1.94	2.21	2.49	2.80
25.0	540.8	0.15	0.23	0.34	0.46	0.59	0.75	0.93	1.12	1.34	1.57	1.82	2.09	2.38	2.69	3.01
30.0	504.1	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.68	1.95	2.24	2.55	2.88	3.23
35.0	471.8	0.17	0.27	0.39	0.52	0.68	0.86	1.06	1.29	1.53	1.80	2.09	2.40	2.73	3.08	3.45
40.0	443.4	0.18	0.29	0.41	0.56	0.73	0.92	1.13	1.37	1.63	1.92	2.22	2.55	2.90	3.28	3.67
45.0	418.4	0.19	0.30	0.44	0.59	0.77	0.97	1.20	1.45	1.73	2.03	2.35	2.70	3.07	3.47	3.89
50.0	396.3	0.20	0.32	0.46	0.62	0.81	1.03	1.27	1.53	1.83	2.14	2.49	2.85	3.25	3.66	4.11
55.0	376.7	0.21	0.34	0.48	0.65	0.85	1.08	1.33	1.61	1.92	2.25	2.61	3.00	3.42	3.86	4.32
60.0	359.2	0.23	0.35	0.51	0.69	0.90	1.13	1.40	1.69	2.01	2.36	2.74	3.15	3.58	4.04	4.53
65.0	343.6	0.24	0.37	0.53	0.72	0.94	1.18	1.46	1.77	2.11	2.47	2.87	3.29	3.74	4.23	4.74
70.0	329.6	0.25	0.38	0.55	0.75	0.98	1.23	1.52	1.84	2.20	2.58	2.99	3.43	3.90	4.41	4.94
75.0	317.0	0.26	0.40	0.57	0.78	1.01	1.28	1.59	1.92	2.28	2.68	3.11	3.57	4.06	4.58	5.14

Pre-tension and Erection Tables for 175 mm² (30 + 7/2.79) ACSR - Basic Span 130 m

Table 21 - Bare Conductor

Pre-tension kgf (All Temps)	Sag (m) for span lengths (m) at all temperatures														
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
1500	0.11	0.18	0.26	0.35	0.46	0.58	0.72	0.87	1.03	1.21	1.40	1.61	1.83	2.07	2.32

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	1636.7	0.11	0.17	0.24	0.33	0.43	0.54	0.67	0.81	0.96	1.13	1.31	1.50	1.71	1.93	2.16
5.0	1527.4	0.11	0.18	0.26	0.35	0.46	0.58	0.72	0.87	1.03	1.21	1.40	1.61	1.83	2.07	2.32
10.0	1425.9	0.12	0.19	0.28	0.38	0.49	0.62	0.77	0.93	1.10	1.29	1.50	1.72	1.96	2.21	2.48
15.0	1332.5	0.13	0.20	0.30	0.40	0.52	0.66	0.82	0.99	1.18	1.39	1.61	1.84	2.10	2.37	2.66
20.0	1247.3	0.14	0.22	0.32	0.43	0.56	0.71	0.88	1.06	1.26	1.48	1.72	1.97	2.24	2.53	2.84
25.0	1170.0	0.15	0.23	0.34	0.46	0.60	0.76	0.93	1.13	1.34	1.58	1.83	2.10	2.39	2.70	3.03
30.0	1100.4	0.16	0.25	0.36	0.49	0.64	0.80	0.99	1.20	1.43	1.68	1.95	2.23	2.54	2.87	3.22
35.0	1037.9	0.17	0.26	0.38	0.52	0.67	0.85	1.05	1.27	1.52	1.78	2.06	2.37	2.69	3.04	3.41
40.0	981.9	0.18	0.28	0.40	0.55	0.71	0.90	1.11	1.35	1.60	1.88	2.18	2.50	2.85	3.22	3.61
45.0	931.6	0.19	0.29	0.42	0.57	0.75	0.95	1.17	1.42	1.69	1.98	2.30	2.64	3.00	3.39	3.80
50.0	886.6	0.20	0.31	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	2.42	2.77	3.15	3.56	3.99
55.0	846.1	0.21	0.32	0.46	0.63	0.83	1.05	1.29	1.56	1.86	2.18	2.53	2.91	3.31	3.73	4.18
60.0	809.6	0.22	0.34	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.64	3.04	3.45	3.90	4.37
65.0	776.7	0.23	0.35	0.51	0.69	0.90	1.14	1.41	1.70	2.03	2.38	2.76	3.16	3.60	4.07	4.56
70.0	746.8	0.23	0.37	0.53	0.72	0.94	1.18	1.46	1.77	2.11	2.47	2.87	3.29	3.75	4.23	4.74
75.0	719.6	0.24	0.38	0.55	0.74	0.97	1.23	1.52	1.84	2.19	2.57	2.98	3.42	3.89	4.39	4.92

Pre-tension and Erection Tables for 175 mm² (30 + 7/2.79) ACSR — Basic Span 130 m

Table 22 — PVC-covered Conductor

Pre-tension kgf (All Temps)	Sag (m) for span lengths (m) at all temperatures														
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
1500	0.15	0.23	0.33	0.45	0.59	0.74	0.91	1.11	1.32	1.55	1.79	2.06	2.34	2.64	2.96

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)														
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	1567.3	0.14	0.22	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.46	1.69	1.94	2.21	2.49	2.80
5.0	1479.1	0.15	0.23	0.33	0.45	0.59	0.74	0.91	1.11	1.32	1.55	1.79	2.06	2.34	2.64	2.96
10.0	1397.9	0.15	0.24	0.35	0.47	0.62	0.78	0.97	1.17	1.39	1.64	1.90	2.18	2.48	2.80	3.13
15.0	1323.6	0.16	0.26	0.37	0.50	0.65	0.83	1.02	1.24	1.47	1.73	2.00	2.30	2.62	2.95	3.31
20.0	1255.8	0.17	0.27	0.39	0.53	0.69	0.87	1.08	1.30	1.55	1.82	2.11	2.42	2.76	3.11	3.49
25.0	1194.0	0.18	0.28	0.41	0.56	0.72	0.92	1.13	1.37	1.63	1.91	2.22	2.55	2.90	3.27	3.67
30.0	1137.8	0.19	0.30	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	2.33	2.67	3.04	3.44	3.85
35.0	1086.7	0.20	0.31	0.45	0.61	0.80	1.01	1.24	1.51	1.79	2.10	2.44	2.80	3.19	3.60	4.03
40.0	1040.2	0.21	0.33	0.47	0.64	0.83	1.05	1.30	1.57	1.87	2.20	2.55	2.93	3.33	3.76	4.21
45.0	997.8	0.22	0.34	0.49	0.66	0.87	1.10	1.36	1.64	1.95	2.29	2.66	3.05	3.47	3.92	4.39
50.0	959.2	0.23	0.35	0.51	0.69	0.90	1.14	1.41	1.71	2.03	2.38	2.76	3.17	3.61	4.08	4.57
55.0	923.8	0.23	0.37	0.53	0.72	0.94	1.19	1.46	1.77	2.11	2.47	2.87	3.29	3.75	4.23	4.74
60.0	891.5	0.24	0.38	0.55	0.74	0.97	1.23	1.52	1.84	2.18	2.56	2.97	3.41	3.88	4.38	4.92
65.0	861.7	0.25	0.39	0.56	0.77	1.00	1.27	1.57	1.90	2.26	2.65	3.08	3.53	4.02	4.54	5.09
70.0	834.3	0.26	0.41	0.58	0.79	1.04	1.31	1.62	1.96	2.33	2.74	3.18	3.65	4.15	4.69	5.25
75.0	809.0	0.27	0.42	0.60	0.82	1.07	1.35	1.67	2.02	2.41	2.83	3.28	3.76	4.28	4.83	5.42

Pre-tension and Erection Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 130 m

Table 23 — Bare Conductor

Pre-tension kgf (All Temps)		Sag (m) for span lengths (m) at all temperatures															
150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
800	950	1050	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	1.59	1.83	2.08	2.35	2.63

Erection Temp. (°C)	Erection Tension (kgf)		Sag (m) for span lengths (m)															
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	878.0	1010.9	1145.0	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.91	1.08	1.26	1.47	1.68	1.92	2.16	2.42
5.0	808.7	931.2	1054.9	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	1.59	1.83	2.08	2.35	2.63
10.0	746.5	859.8	974.4	0.14	0.22	0.32	0.43	0.56	0.71	0.88	1.06	1.27	1.49	1.72	1.98	2.25	2.54	2.85
15.0	691.4	796.4	903.0	0.15	0.24	0.34	0.47	0.61	0.77	0.95	1.15	1.37	1.60	1.86	2.14	2.43	2.74	3.08
20.0	642.8	740.7	840.3	0.16	0.26	0.37	0.50	0.65	0.83	1.02	1.24	1.47	1.73	2.00	2.30	2.61	2.95	3.31
25.0	600.2	691.9	785.3	0.18	0.27	0.39	0.54	0.70	0.89	1.09	1.32	1.57	1.85	2.14	2.46	2.80	3.16	3.54
30.0	562.9	649.1	737.1	0.19	0.29	0.42	0.57	0.75	0.94	1.17	1.41	1.68	1.97	2.28	2.62	2.98	3.37	3.77
35.0	530.2	611.6	694.9	0.20	0.31	0.45	0.61	0.79	1.00	1.24	1.50	1.78	2.09	2.42	2.78	3.17	3.57	4.01
40.0	501.5	578.6	657.7	0.21	0.33	0.47	0.64	0.84	1.06	1.31	1.58	1.88	2.21	2.56	2.94	3.35	3.78	4.23
45.0	476.1	549.5	624.9	0.22	0.34	0.50	0.67	0.88	1.11	1.38	1.67	1.98	2.33	2.70	3.10	3.52	3.98	4.46
50.0	453.6	523.7	595.7	0.23	0.36	0.52	0.71	0.92	1.17	1.44	1.75	2.08	2.44	2.83	3.25	3.70	4.17	4.68
55.0	433.6	500.6	569.8	0.24	0.38	0.54	0.74	0.97	1.22	1.51	1.83	2.18	2.55	2.96	3.40	3.87	4.37	4.89
60.0	415.6	480.0	546.5	0.25	0.39	0.57	0.77	1.01	1.28	1.58	1.91	2.27	2.66	3.09	3.54	4.03	4.55	5.10
65.0	399.5	461.5	525.5	0.26	0.41	0.59	0.80	1.05	1.33	1.64	1.98	2.36	2.77	3.21	3.69	4.20	4.74	5.31
70.0	384.9	444.7	506.5	0.27	0.43	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.87	3.33	3.83	4.35	4.92	5.51
75.0	371.6	429.4	489.2	0.28	0.44	0.63	0.86	1.13	1.43	1.76	2.13	2.54	2.98	3.45	3.96	4.51	5.09	5.71

Pre-tension and Erection Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR -- Basic Span 130 m

Table 24 -- PVC-covered Conductor

Pre-tension kgf (All Temps)			Sag (m) for span lengths (m) at all temperatures														
150 mm ² 175 mm ² 200 mm ²			40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
800	950	1050	0.17	0.27	0.39	0.52	0.68	0.87	1.07	1.29	1.54	1.81	2.10	2.41	2.74	3.09	3.47

Erection Temp (°C)	Erection Tension (kgf)		Sag (m) for span lengths (m)														
	150 mm ²	175 mm ² 200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0
0.0	861.2	992.1 1125.2	0.16	0.25	0.36	0.49	0.64	0.81	1.01	1.22	1.45	1.70	1.97	2.26	2.57	2.91	3.26
5.0	809.7	932.3 1056.0	0.17	0.27	0.39	0.52	0.68	0.87	1.07	1.29	1.54	1.81	2.10	2.41	2.74	3.09	3.47
10.0	763.4	878.6 994.0	0.18	0.28	0.41	0.56	0.73	0.92	1.14	1.37	1.63	1.92	2.23	2.55	2.91	3.28	3.68
15.0	722.0	830.5 938.5	0.19	0.30	0.43	0.59	0.77	0.97	1.20	1.45	1.73	2.03	2.35	2.70	3.07	3.47	3.89
20.0	684.8	787.4 888.9	0.20	0.32	0.46	0.62	0.81	1.03	1.27	1.53	1.82	2.14	2.48	2.85	3.24	3.66	4.10
25.0	651.4	748.8 844.5	0.21	0.33	0.48	0.65	0.85	1.08	1.33	1.61	1.92	2.25	2.61	3.00	3.41	3.85	4.32
30.0	621.4	714.1 804.7	0.22	0.35	0.50	0.68	0.89	1.13	1.40	1.69	2.01	2.36	2.74	3.14	3.58	4.04	4.53
35.0	594.4	682.9 769.0	0.23	0.37	0.53	0.72	0.93	1.18	1.46	1.77	2.10	2.47	2.86	3.29	3.74	4.22	4.73
40.0	570.1	654.8 736.8	0.24	0.38	0.55	0.75	0.97	1.23	1.52	1.84	2.19	2.57	2.99	3.43	3.90	4.40	4.94
45.0	548.0	629.3 707.6	0.25	0.40	0.57	0.78	1.01	1.28	1.59	1.92	2.28	2.68	3.11	3.57	4.06	4.58	5.14
50.0	527.9	606.1 681.2	0.26	0.41	0.59	0.81	1.05	1.33	1.65	1.99	2.37	2.78	3.23	3.70	4.21	4.76	5.33
55.0	509.5	584.9 657.1	0.27	0.43	0.61	0.84	1.09	1.38	1.71	2.06	2.46	2.88	3.34	3.84	4.37	4.93	5.53
60.0	492.7	565.6 635.1	0.28	0.44	0.64	0.86	1.13	1.43	1.76	2.13	2.54	2.98	3.46	3.97	4.51	5.10	5.71
65.0	477.3	547.8 614.9	0.29	0.46	0.66	0.89	1.17	1.47	1.82	2.20	2.62	3.08	3.57	4.10	4.66	5.26	5.90
70.0	463.1	531.4 596.3	0.30	0.47	0.68	0.92	1.20	1.52	1.88	2.27	2.70	3.17	3.68	4.22	4.81	5.42	6.08
75.0	449.9	516.3 579.2	0.31	0.48	0.70	0.95	1.24	1.56	1.93	2.34	2.78	3.27	3.79	4.35	4.95	5.58	6.26

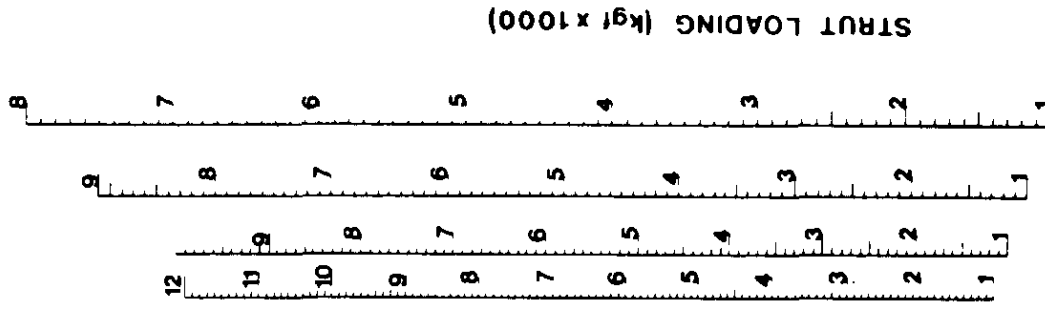
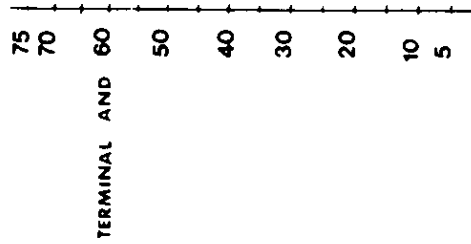
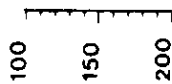
NOMOGRAM I

100mm'(6/4.72+7/1.57)ACSR.

BARE

LEVEL GROUND

FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf
1 : 30	290
1 : 20	440
1 : 10	870



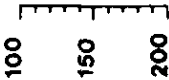
WIND
SPAN
(m)

ANGLE OF
DEVIATION
(°)

30° 35° 40° 45°
STAY SLOPE

NOMOGRAM 2
150mm' (18/3.35 +1/3.35) ACSR
BARE
LEVEL GROUND

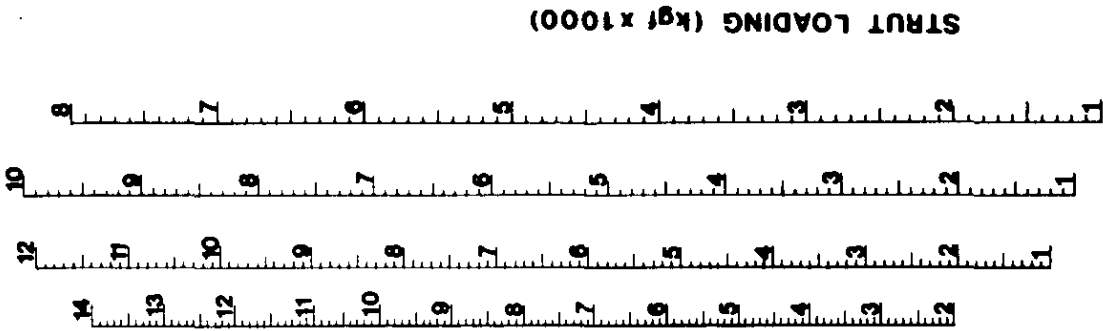
FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf
1 : 30	330
1 : 20	490
1 : 10	970



**WIND
SPAN
(m)**



**ANGLE OF
DEVIATION
(θ)**

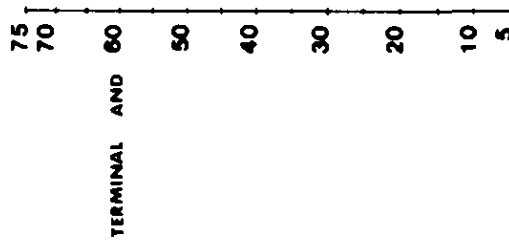
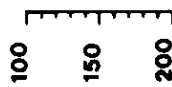


NOMOGRAM 3

175 mm² (18/3·61 + 1/3·61) ACSR.

BARE
 LEVEL GROUND

FOR DOWNPULL CONDITIONS		
DECLINATION	ADDITIONAL kgf	
1 : 30	360	
1 : 20	540	
1 : 10	1070	

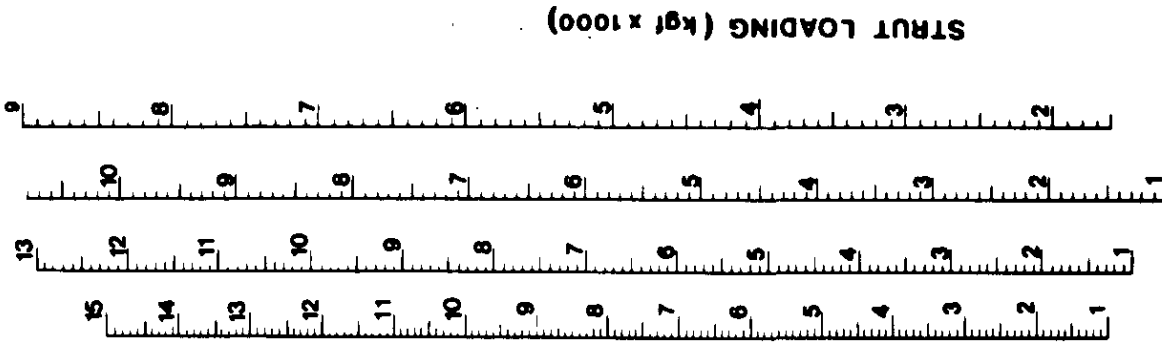


WIND
 SPAN
 (m)

ANGLE OF
 DEVIATION
 (°)

30° 35° 40° 45°
 STAY SLOPE

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NOMOGRAM 4

175 mm² (30/2.79 + 7/2.79) ACSR

BARE

LEVEL GROUND

FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf
1 : 30	480
1 : 20	710
1 : 13	1100
1 : 10	1420

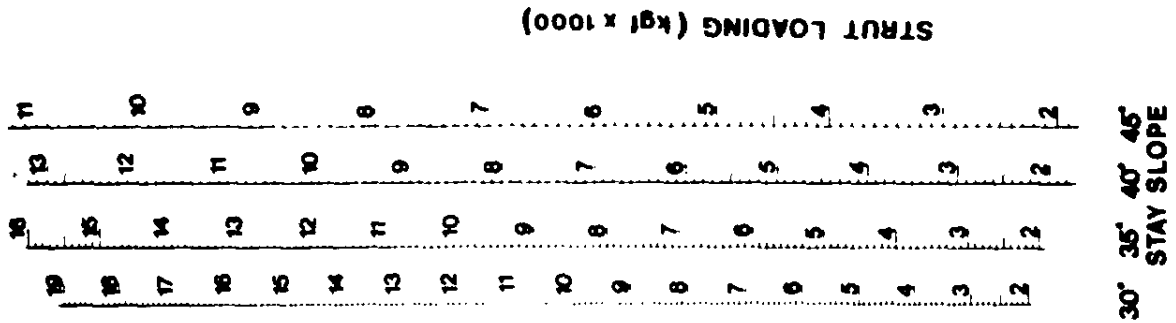
100
150
200

WIND
SPAN
(m)

75
70
60
50
40
30
20
10
5

TERMINAL AND

ANGLE OF
DEVIATION
(°)



NOMOGRAM 5 **200mm' (18/3.86 + 1/3.86) ACSR** **BARE** **LEVEL GROUND**

FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf
1 : 30	390
1 : 20	590
1 : 10	1170

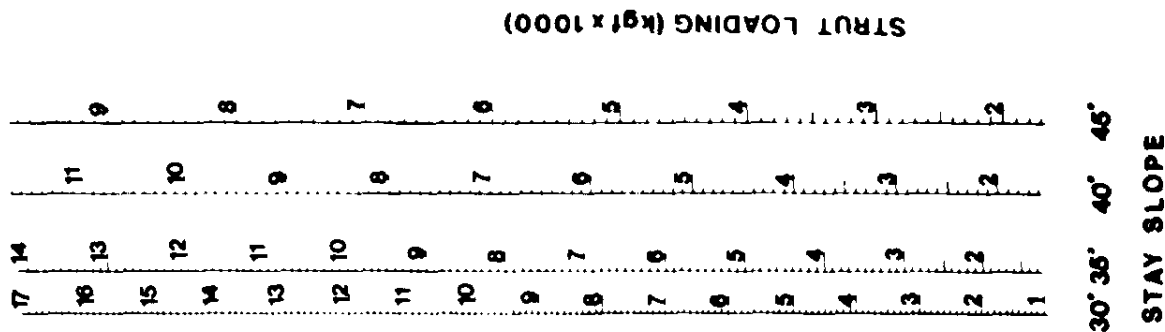
100
150
200

75
70
60
50
40
30
20
10
5

TERMINAL AND

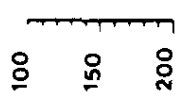
WIND
SPAN
(m)

ANGLE OF
DEVIATION
(°)



NOMOGRAM 6
 100mm" (6/4.72 + 7/1.57) ACSR
 PVC COVERED
 LEVEL GROUND

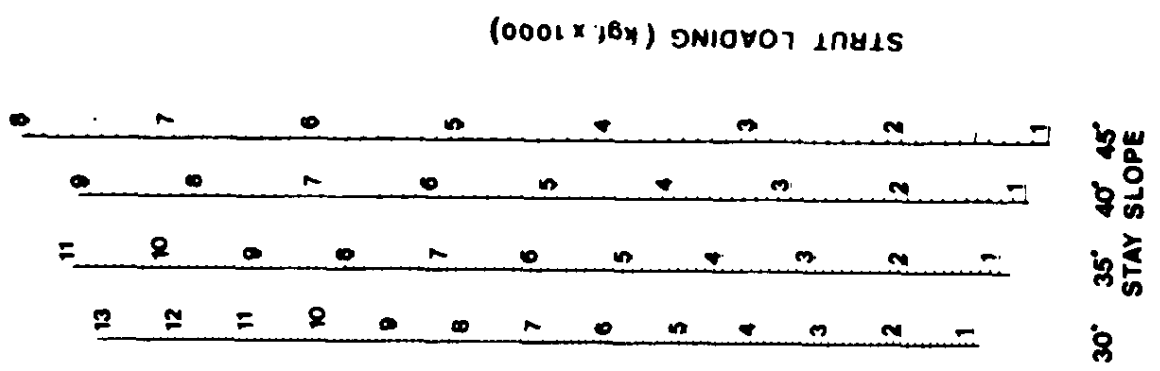
FOR DOWNPULL CONDITIONS		
DECLINATION	ADDITIONAL kgf.	
1 : 30	300	
1 : 20	450	
1 : 10	900	



WIND
SPAN
(m)

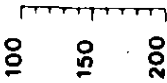


ANGLE OF
DEVIATION
(°)

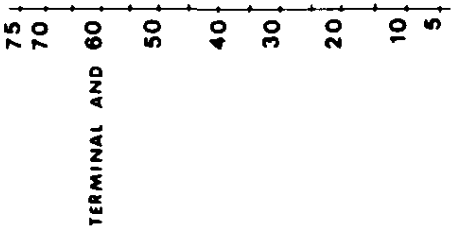


NOMOGRAM 7
150mm' (18/ 3.35+ 1/3.35) ACSR
PVC COVERED
LEVEL GROUND

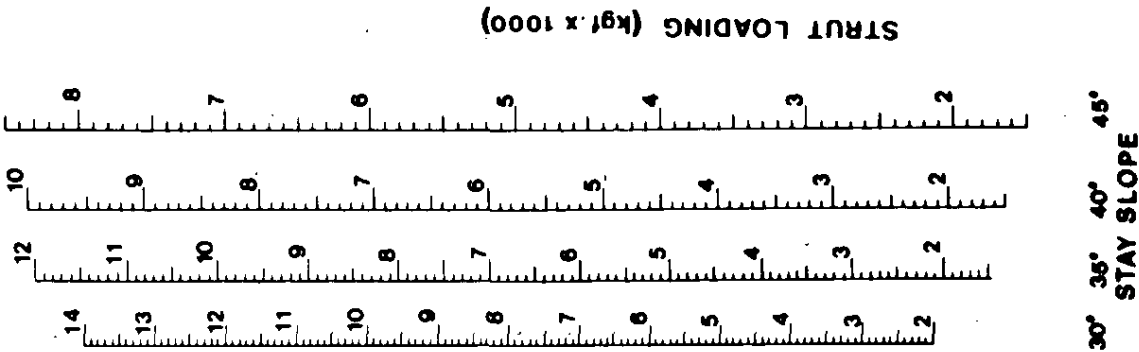
FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf.
1 : 30	330
1 : 20	490
1 : 10	980



WIND
SPAN
(m)

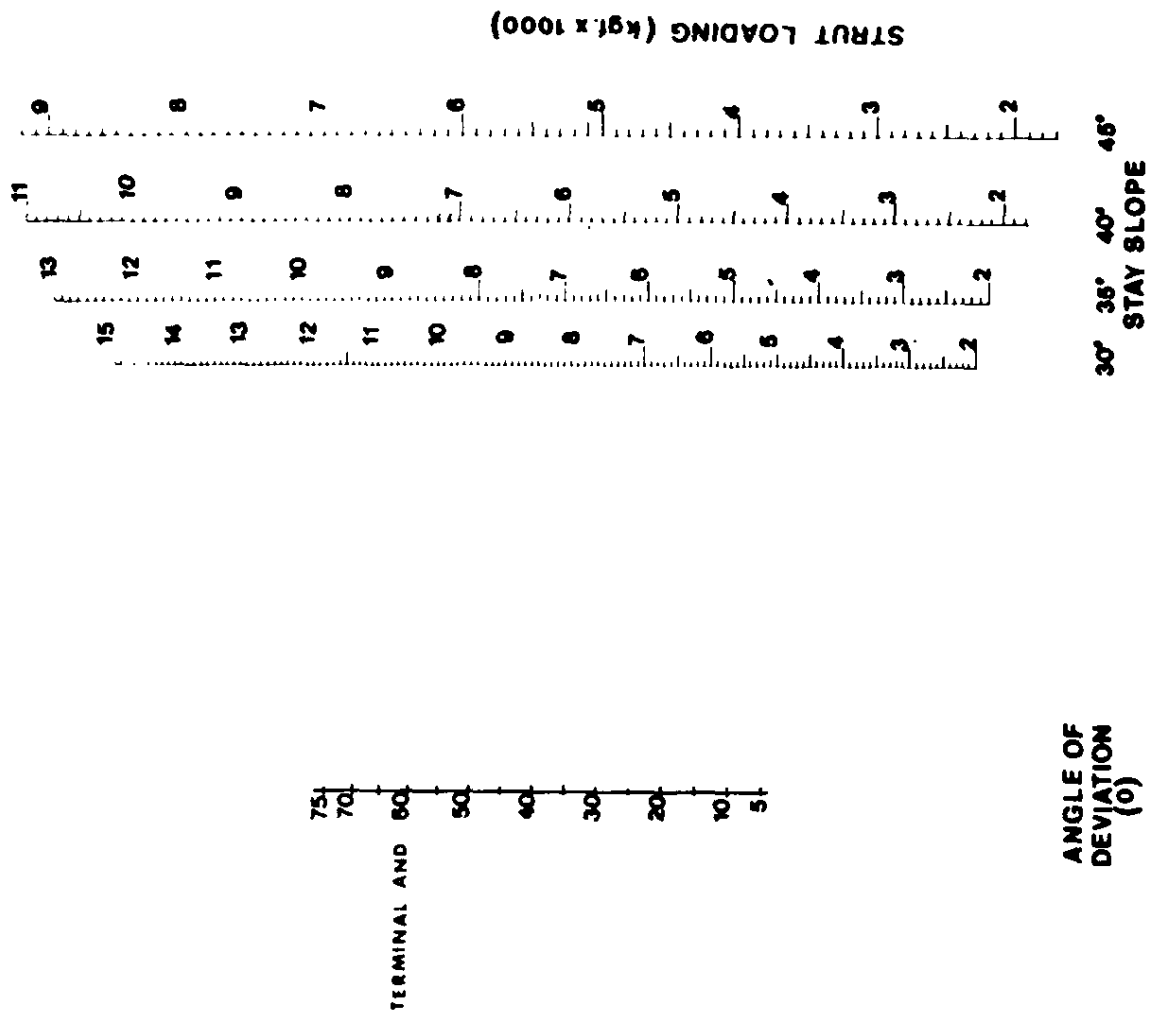


ANGLE OF
DEVIATION
(°)



NOMOGRAM 8
175mm² (18/3.61/ +1/3.61) ACSR
PVC COVERED
LEVEL GROUND

FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf.
1 : 30	360
1 : 20	540
1 : 10	1070



NOMOGRAM 9

175mm² (30/2.79 + 7/2.79) ACSR
 PVC COVERED

LEVEL GROUND

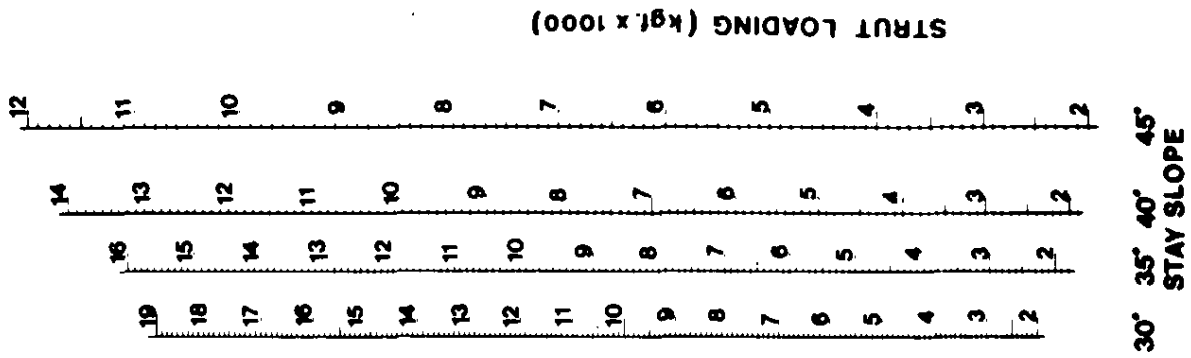
FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf.
1 : 30	480
1 : 20	710
1 : 13	1100
1 : 10	1420

100
 150
 200

75
 70
 60
 50
 40
 30
 20
 10
 5
 TERMINAL AND

WIND
 SPAN
 (m)

ANGLE OF
 DEVIATION
 (°)



NOMOGRAM 10

200mm' (18/3.86 + 1/3.86) ACSR

PVC COVERED

LEVEL GROUND

FOR DOWNPULL CONDITIONS	
DECLINATION	ADDITIONAL kgf.
1 : 30	390
1 : 20	590
1 : 10	1170

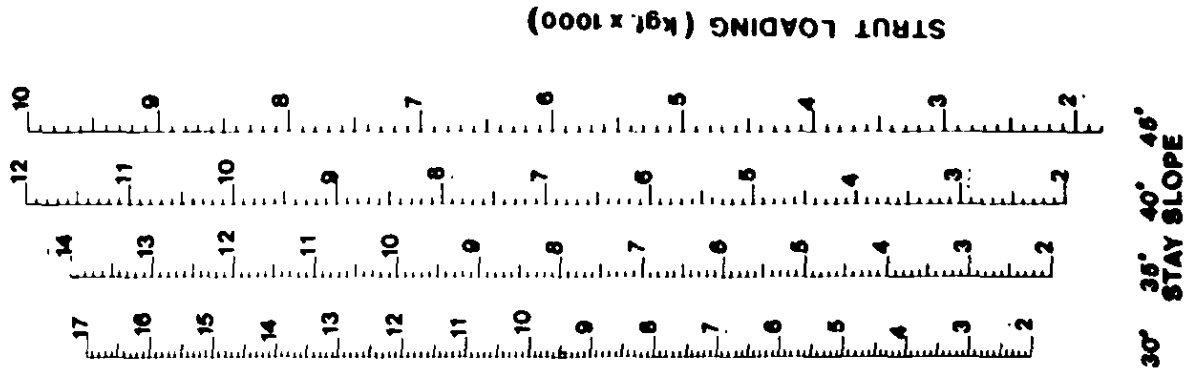
100
150
200

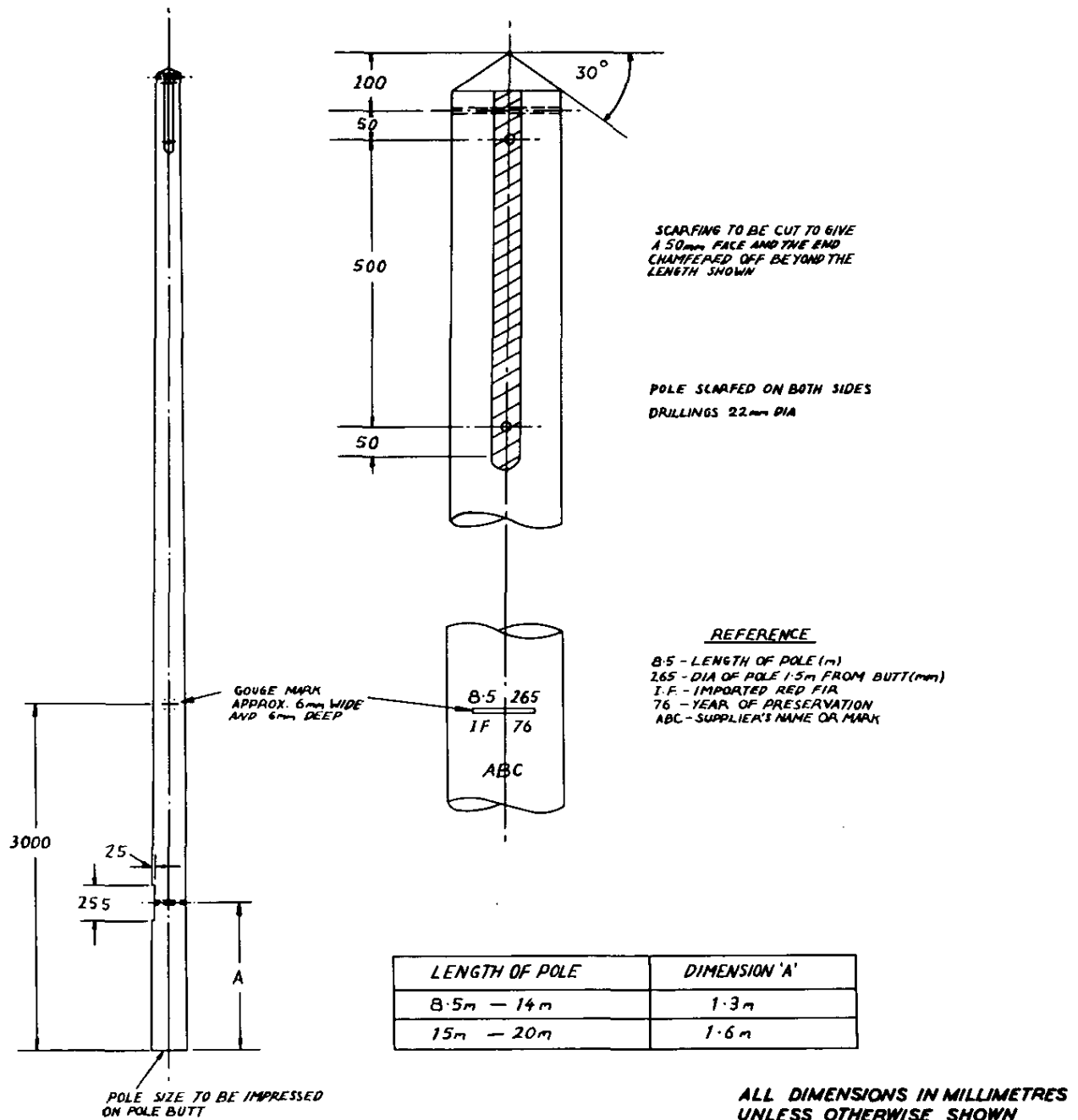
75
70
60
50
40
30
20
10
5

TERMINAL AND

WIND
SPAN
(m)

ANGLE OF
DEVIATION
(°)





89/314

Fig. 1 Single pole drilling, scarfing and marking
(ESI Drawing 432001)

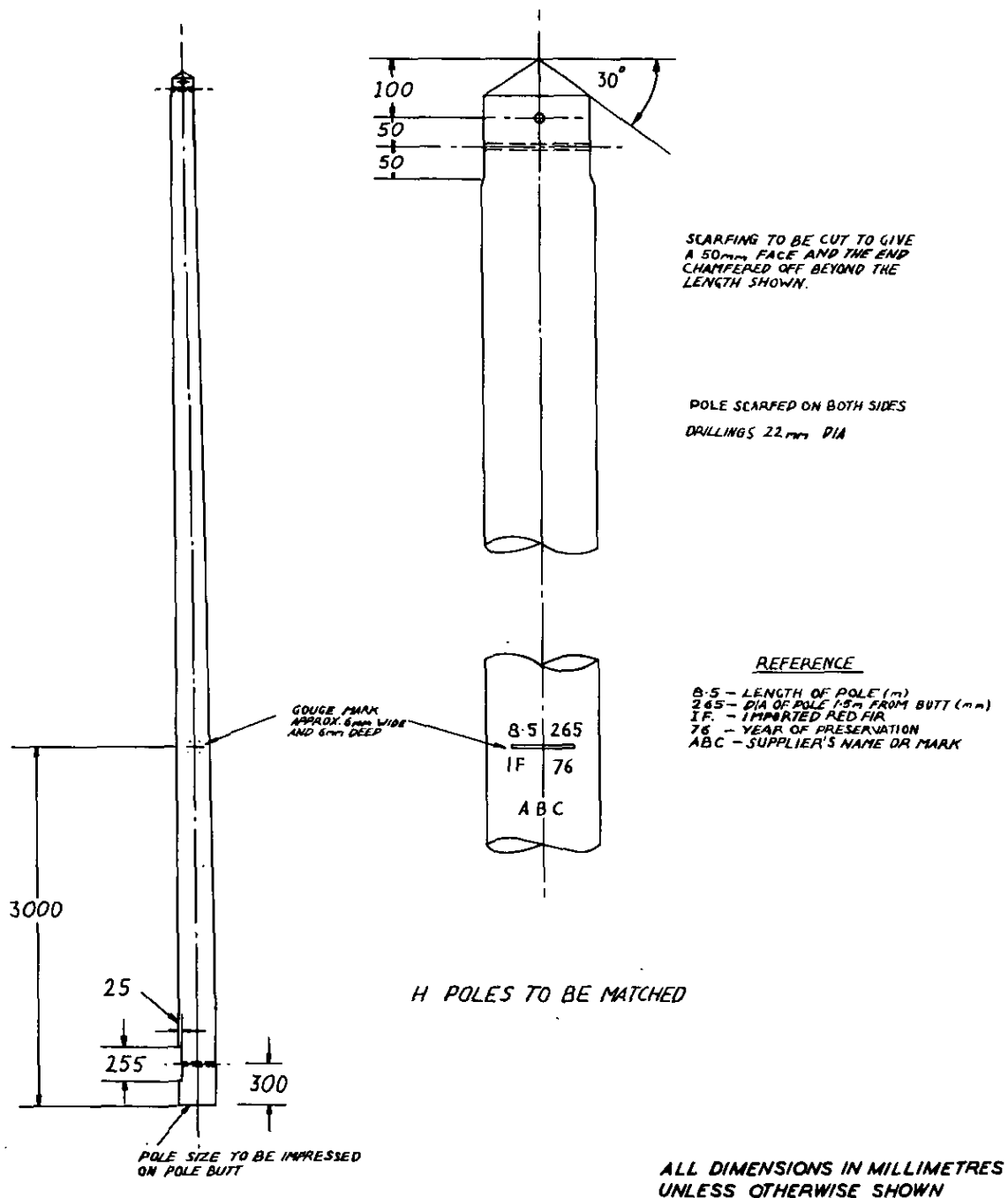
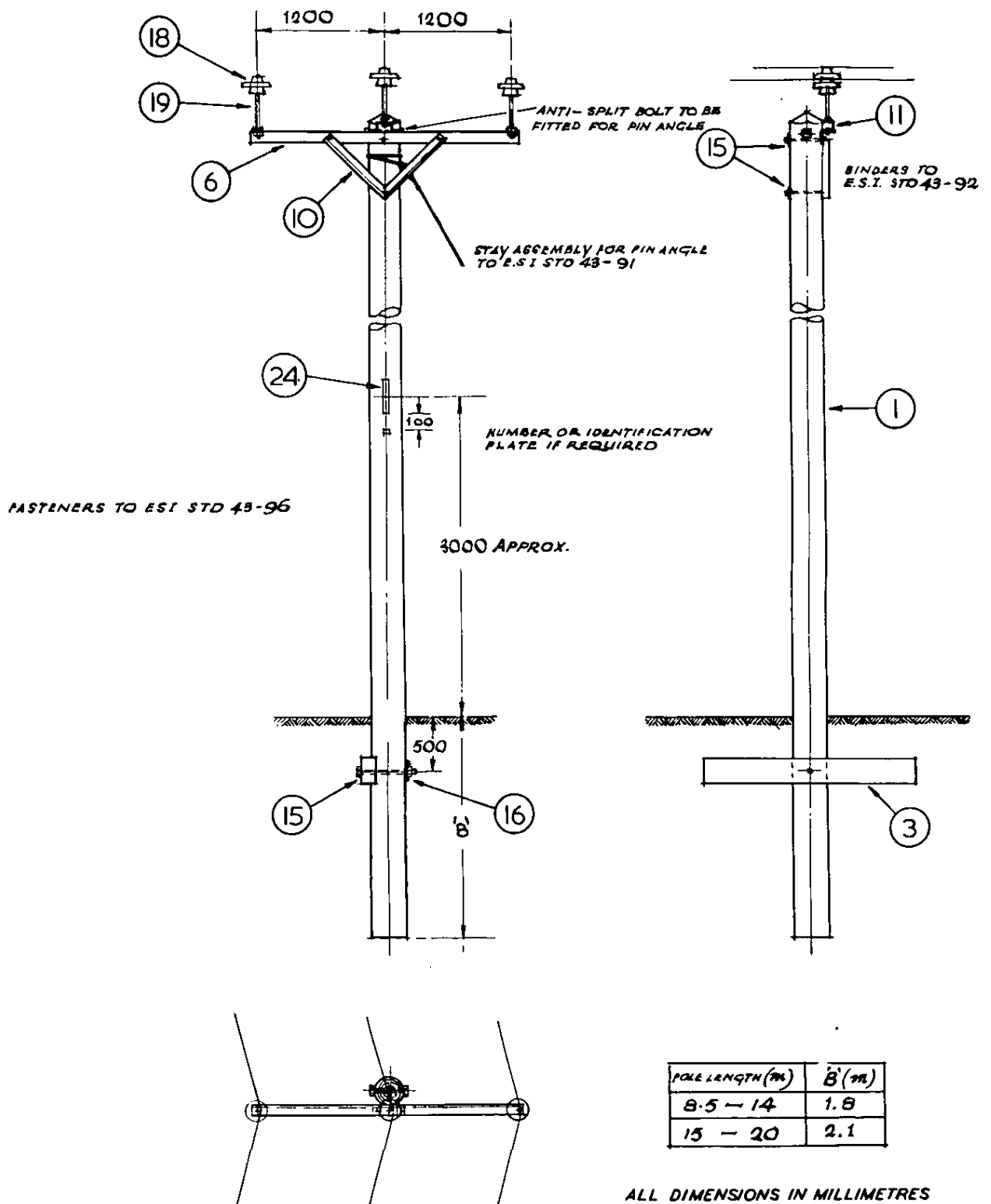


Fig. 2 'H' pole member drilling, scarfing and marking
(ESI Drawing 432002)



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Fig. 3 General assembly — intermediate or pin angle pole
(Single member crossarm)
(ESI Drawing 432003)

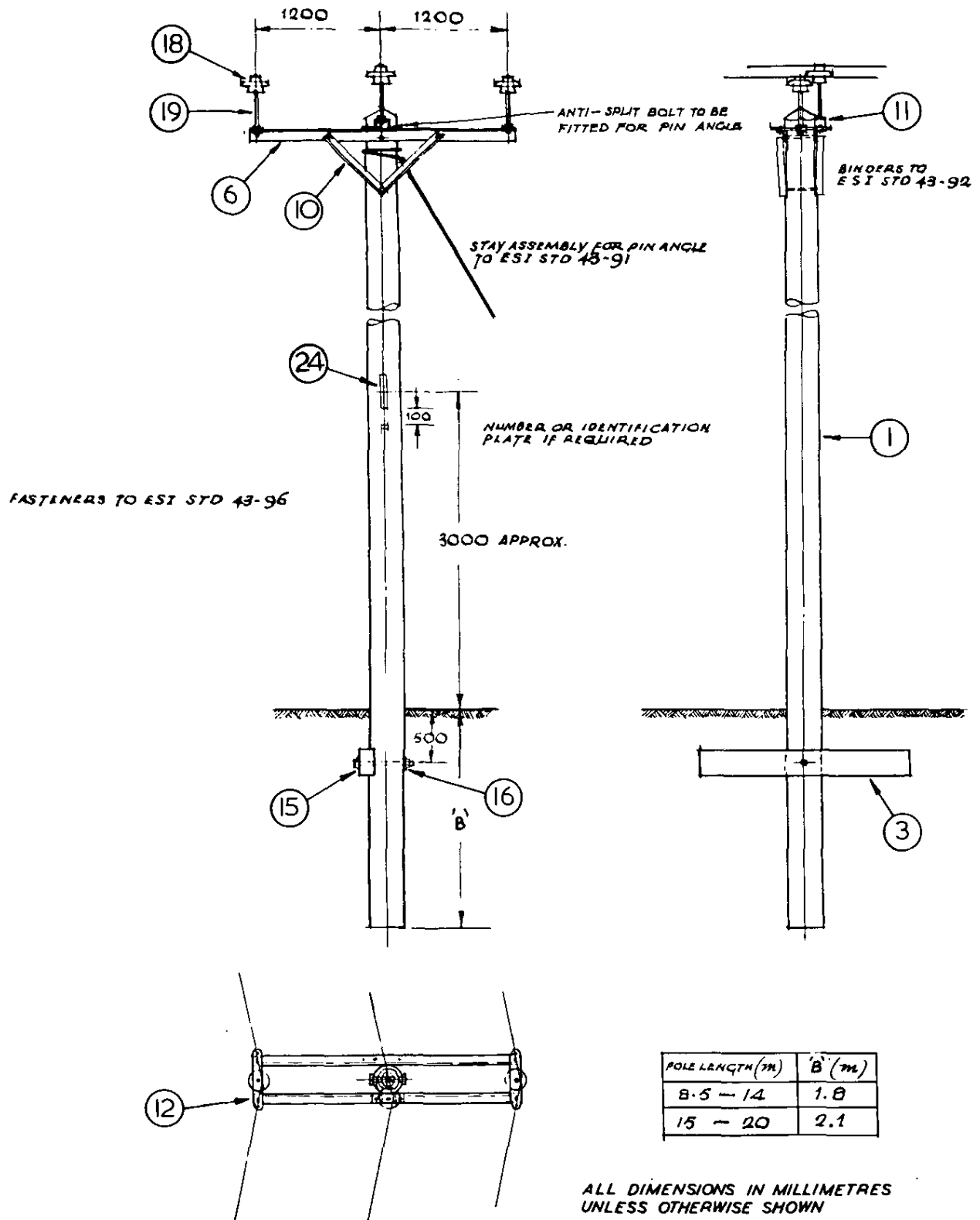
Material List for Fig. 3

Intermediate or Pin Angle Pole
(Single Member Crossarm)

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref.	M 20 Bolts
1	1	Wood Pole	432001	2 or 3 *
3	1	Kicking Block	439103 Type 2	1
6	1	Crossarm Member	439519	—
10	2	Crossarm Strut	439526	2
11	1	Insulator Bracket	439518	2
15	3	Square Flat Washer	439604	—
16	1	Square Curved Washer	439605	—
18	3	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —
19	3	11 kV Pin (50 mm shank) 33 kV Pin (50 mm shank)	BS 3288 Ref. 29 BS 3288 Ref. 31	— —
24	1	Danger Plate	439701	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Pin Angle — (i) *Anti-split bolt with two square curved washers to be fitted.
(ii) Additional round washers and locknuts required for bonding stay strand to pole top bolts.



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Fig. 4 General assembly – intermediate or pin angle pole
(Double member crossarm)
(ESI Drawing 432004)

Material List for Fig. 4

Intermediate or Pin Angle Pole
(Double Member Crossarm)

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
1	1	Wood Pole	432001	2 or 3*	—
3	1	Kicking Block	439103 Type 2	1	—
6	2	Crossarm Member	439519	—	—
10	4	Crossarm Strut	439526	4	—
11	1	Insulator Bracket	439518	2	—
12	2	Section Strap	439525	4	8
15	1	Square Flat Washer	439604	—	—
16	1	Square Curved Washer	439605	—	—
18	3	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —	— — —
19	3	11 kV Pin (50 mm shank) 33 kV Pin (50 mm shank)	BS 3288 Ref. 29 BS 3288 Ref. 31	— —	— —
24	1	Danger Plate	439701	—	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Pin Angle — (i) *Anti-split bolt with two square curved washers to be fitted.
(ii) Additional round washers and locknuts required for bonding stay strand to pole top bolts.

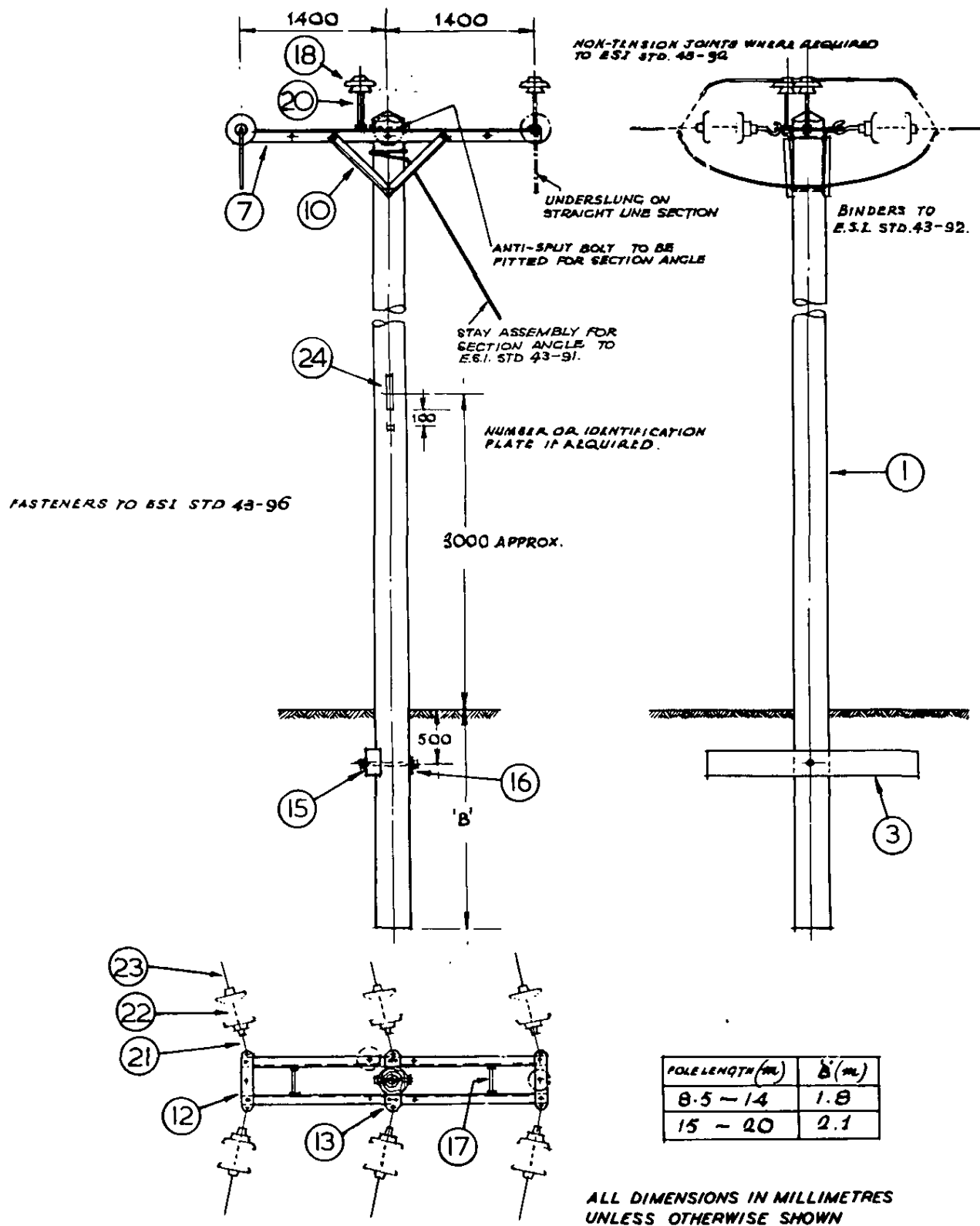


Fig. 5 General assembly - single pole straight line section or section angle (see Table 1)
(ESI Drawing 432005)

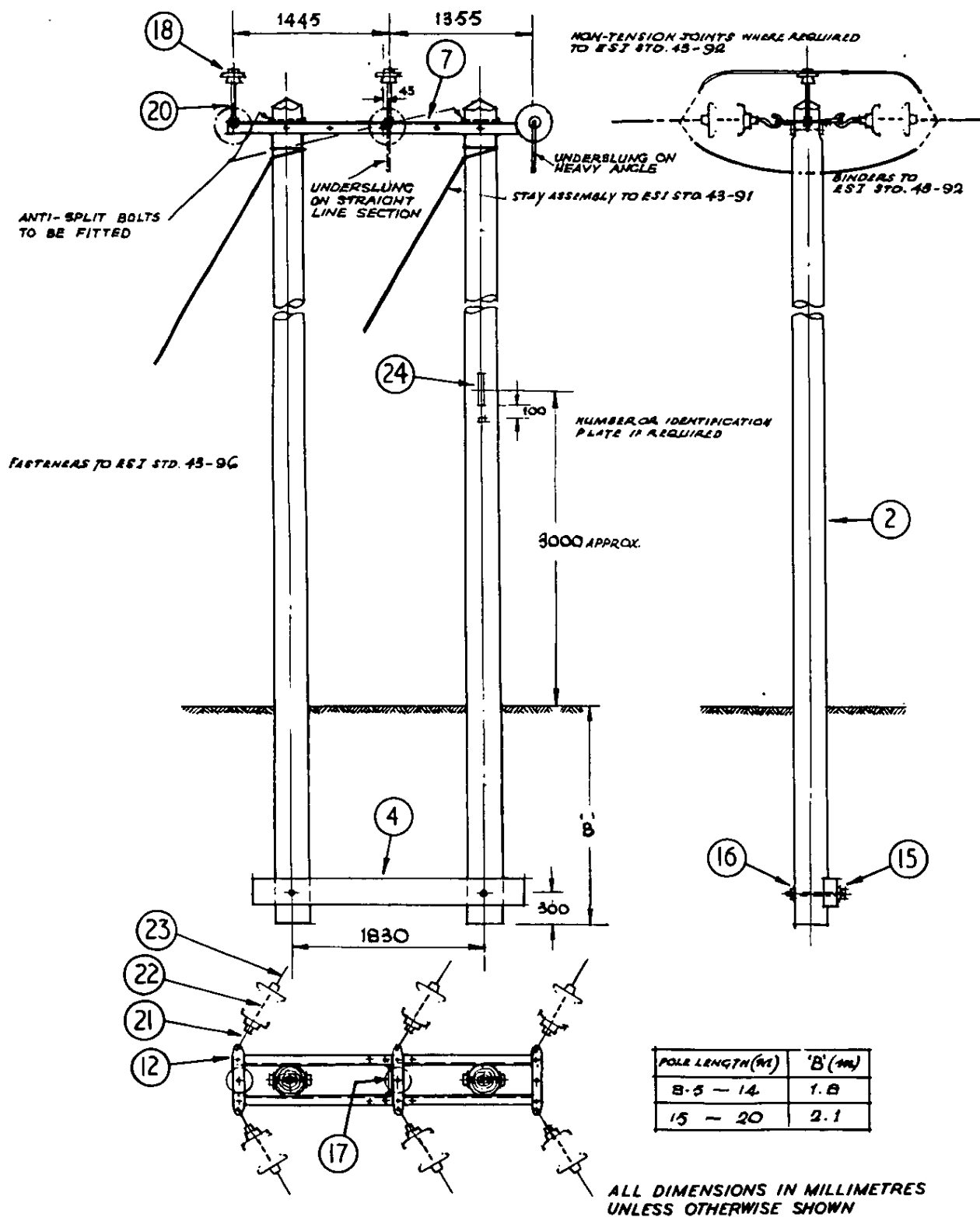
Material List for Fig. 5

Single Pole Straight Line Section or Section Angle

Item No.	Quantity	Description	ESI Std., Dwg. No or BS Ref	M 20 Bolts	Round Washers
1	1	Wood Pole	432001	2 or 3*	—
3	1	Kicking Block	439103 Type 2	1	—
7	2	Crossarm Member	439520	—	—
10	4	Crossarm Strut	439526	4	—
12	2	Section Strap	439525	4	4
13	2	Terminal Strap	439528	4	—
15	1	Square Flat Washer	439604	—	—
16	1	Square Curved Washer	439605	—	—
17	2	Tie Rod	439608 Type 3	—	—
18	1 or 2	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —	— — —
20	1 or 2	Pilot Pin (50 mm shank)	BS 3288 Ref. 50	—	—
21	6	Ball-ended Hook	BS 3288 Ref. 15/32	—	—
22	6	Insulator String	43-93	—	—
23	6	Conductor Termination	43-92	—	—
24	1	Danger Plate	439701	—	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Section Angle — (i) *Anti-split bolt with two square curved washers to be fitted.
(ii) Additional round washers and locknuts required for bonding stay strand to pole top bolts.



89/319

Fig. 6 General assembly - 'H' pole section angle up to 60°
(ESI Drawing 432006)

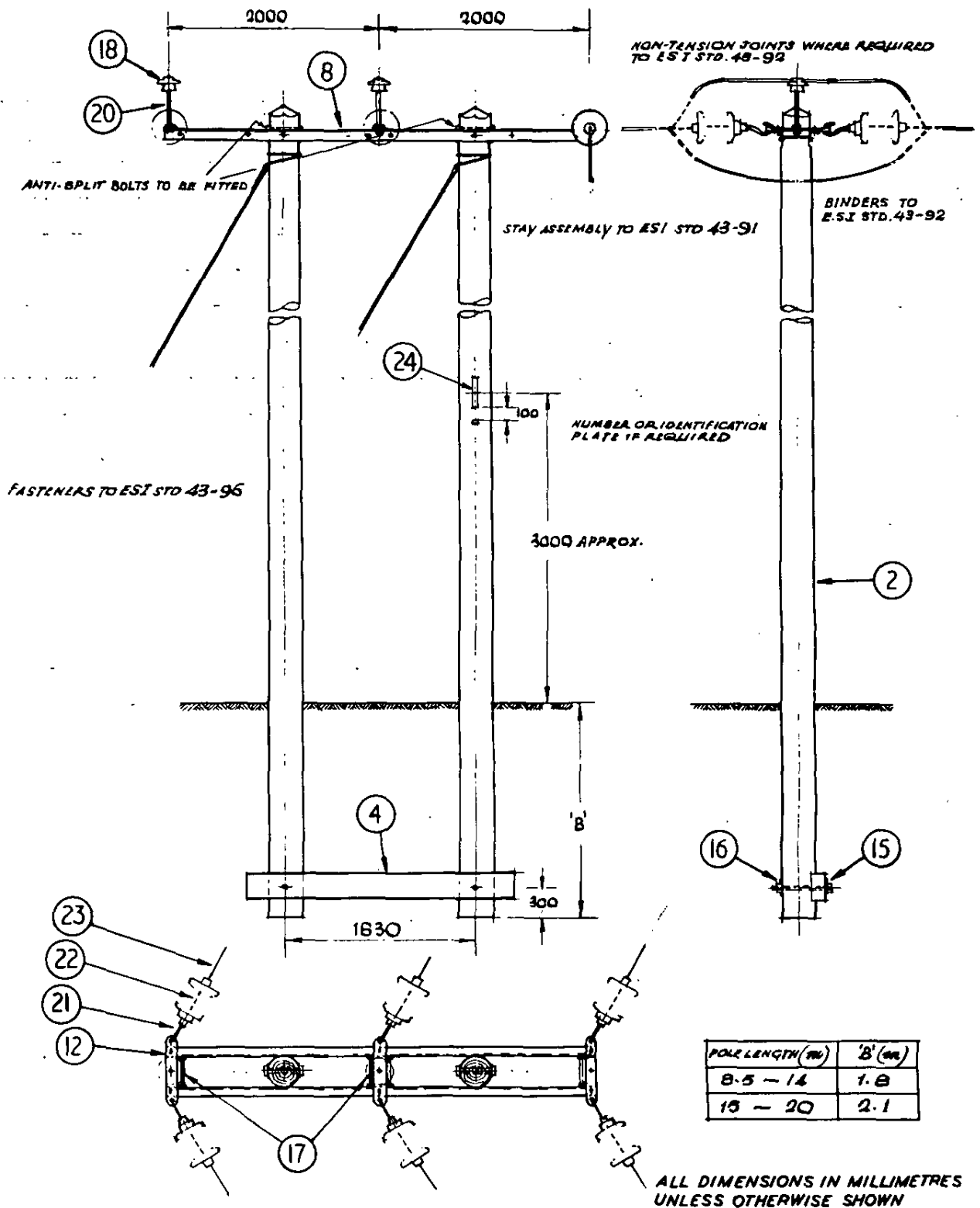
Material List for Fig. 6

'H' Pole Section Angle up to 60°

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
2	2	Wood Pole	432002	4 *	—
4	1	Brace Block	439112 Type 1	2	—
7	2	Crossarm Member	439520	—	—
12	3	Section Strap	439525	6	6
15	2	Square Flat Washer	439604	—	—
16	6	Square Curved Washer	439605	—	—
17	1	Tie Rod	439608 Type 3	—	—
18	2 or 3	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —	— — —
20	2 or 3	Pilot Pin (50 mm shank)	BS 3288 Ref. 50	—	—
21	6	Ball-ended Hook	BS 3288 Ref. 15/32	—	—
22	6	Insulator String	43-93	—	—
23	6	Conductor Termination	43-92	—	—
24	1	Danger Plate	439701	—	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Additional round washers and locknuts required for bonding stay strand to pole top bolts.
3. *Pole top bolts include two anti-split bolts each fitted with two square curved washers.



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Fig. 7 General assembly - 'H' pole section angle between 60° and 75°
(ESI Drawing 432007)

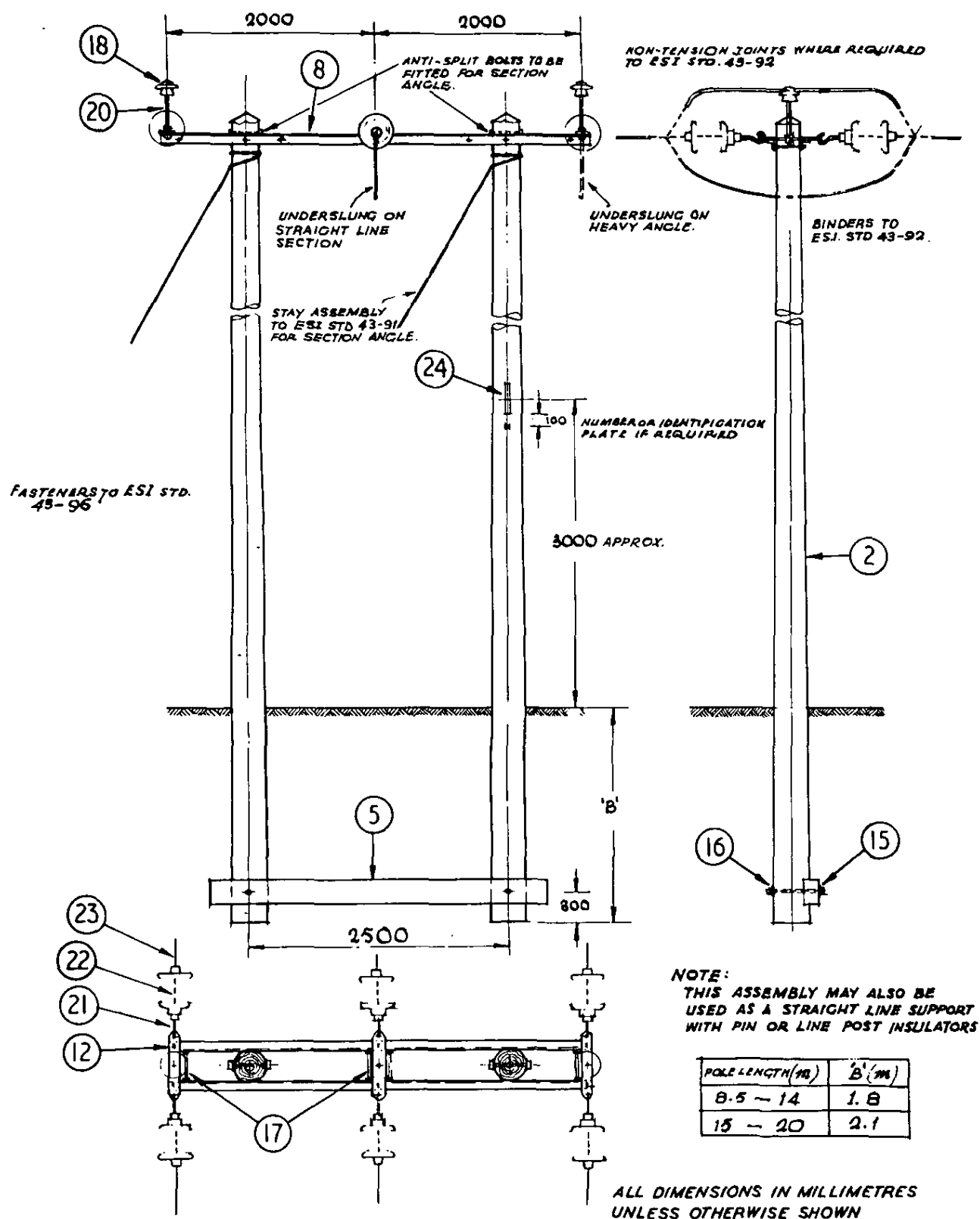
Material List for Fig. 7

'H' Pole Section Angle between 60° and 75°

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref	M 20 Bolts	Round Washers
2	2	Wood Pole	432002	4 *	—
4	1	Brace Block	439112 Type 1	2	—
8	2	Crossarm Member	439521	—	—
12	3	Section Strap	439525	6	6
15	2	Square Flat Washer	439604	—	—
16	6	Square Curved Washer	439605	—	—
17	4	Tie Rod	439608 Type 3	—	—
18	2	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —	— — —
20	2	Pilot Pin (55 mm shank)	BS 3288 Ref. 50	—	—
21	6	Ball-ended Hook	BS 3288 Ref. 15/32	—	—
22	6	Insulator String	43-93	—	—
23	6	Conductor Termination	43-92	—	—
24	1	Danger Plate	439701	—	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Additional round washers and locknuts required for bonding stay strand to pole top bolts.
3. *Pole top bolts include two anti-split bolts each fitted with two square curved washers.



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Fig. 8 General assembly - 'H' pole straight line section or section angle up to 60° for occasional long span (ESI Drawing 432008)

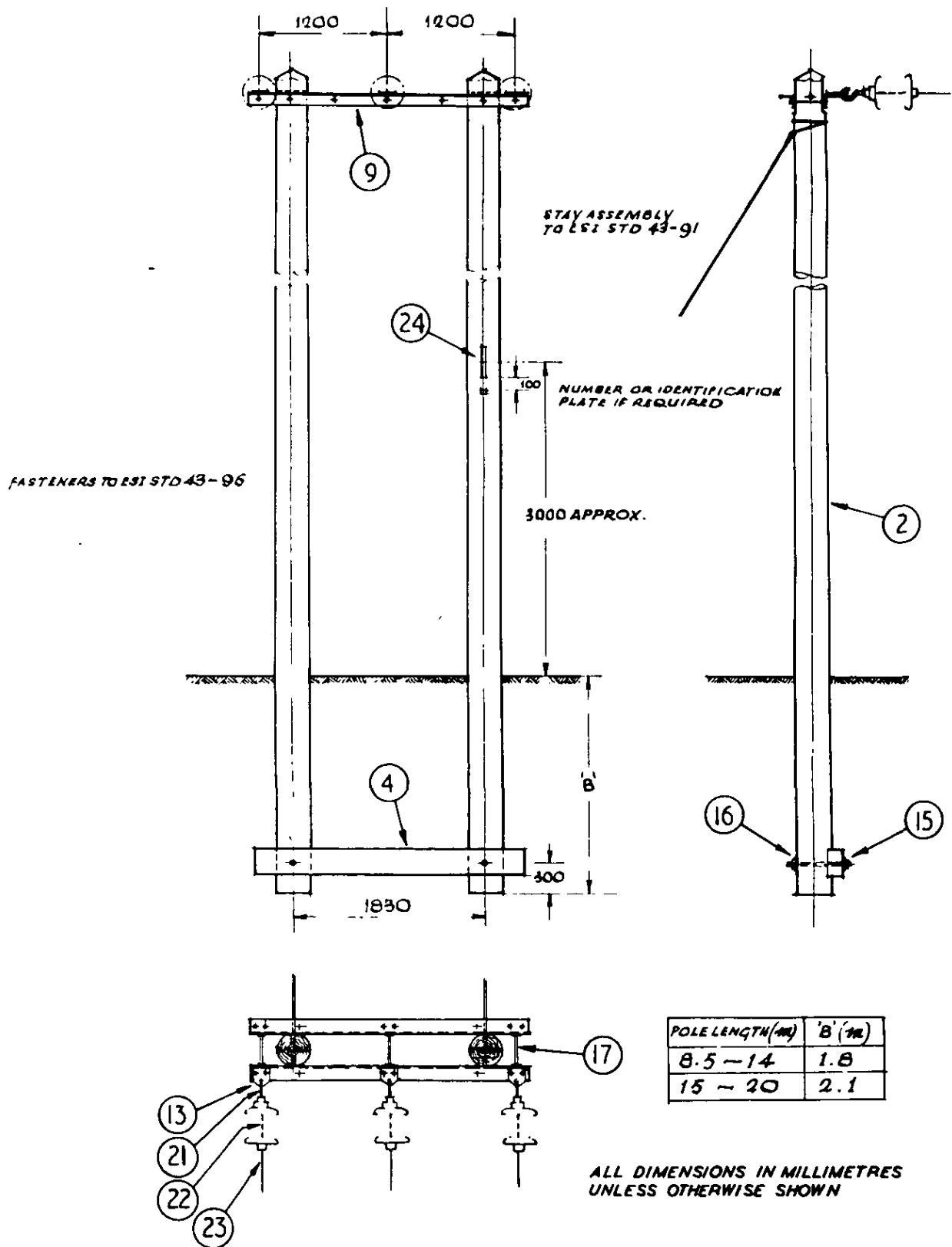
Material List for Fig. 8

'H' Pole Straight Line Section or
Section Angle up to 60° for Occasional Long Span

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref	M 20 Bolts	Round Washers
2	2	Wood Pole	432002	2 or 4*	--
5	1	Brace Block	439112 Type 2	2	~
8	2	Crossarm Member	439521	—	~
12	3	Section Strap	439525	6	6
15	2	Square Flat Washer	439604	—	—
16	2	Square Curved Washer	439605	—	—
17	4	Tie Rod	439608 Type 3	—	—
18	2 or 3	11 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	— — —	-- -- --
20	2 or 3	Pilot Pin (50 mm shank)	BS 3288 Ref. 50	--	—
21	6	Ball-ended Hook	BS 3288 Ref. 15/32	—	—
22	6	Insulator String	43-93	—	—
23	6	Conductor Termination	43-92	—	—
24	1	Danger Plate	439701	--	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Section Angle — (i) *Two anti-split bolts, each fitted with two square curved washers.
(ii) Additional round washers and locknuts required for bonding stay strand to pole top bolts.
When used as a straight line support with pin or line post insulators, items 20-23 are replaced by item 19(3 off).



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Fig. 9 General assembly — 'H' pole terminal
(ESI Drawing 432009)

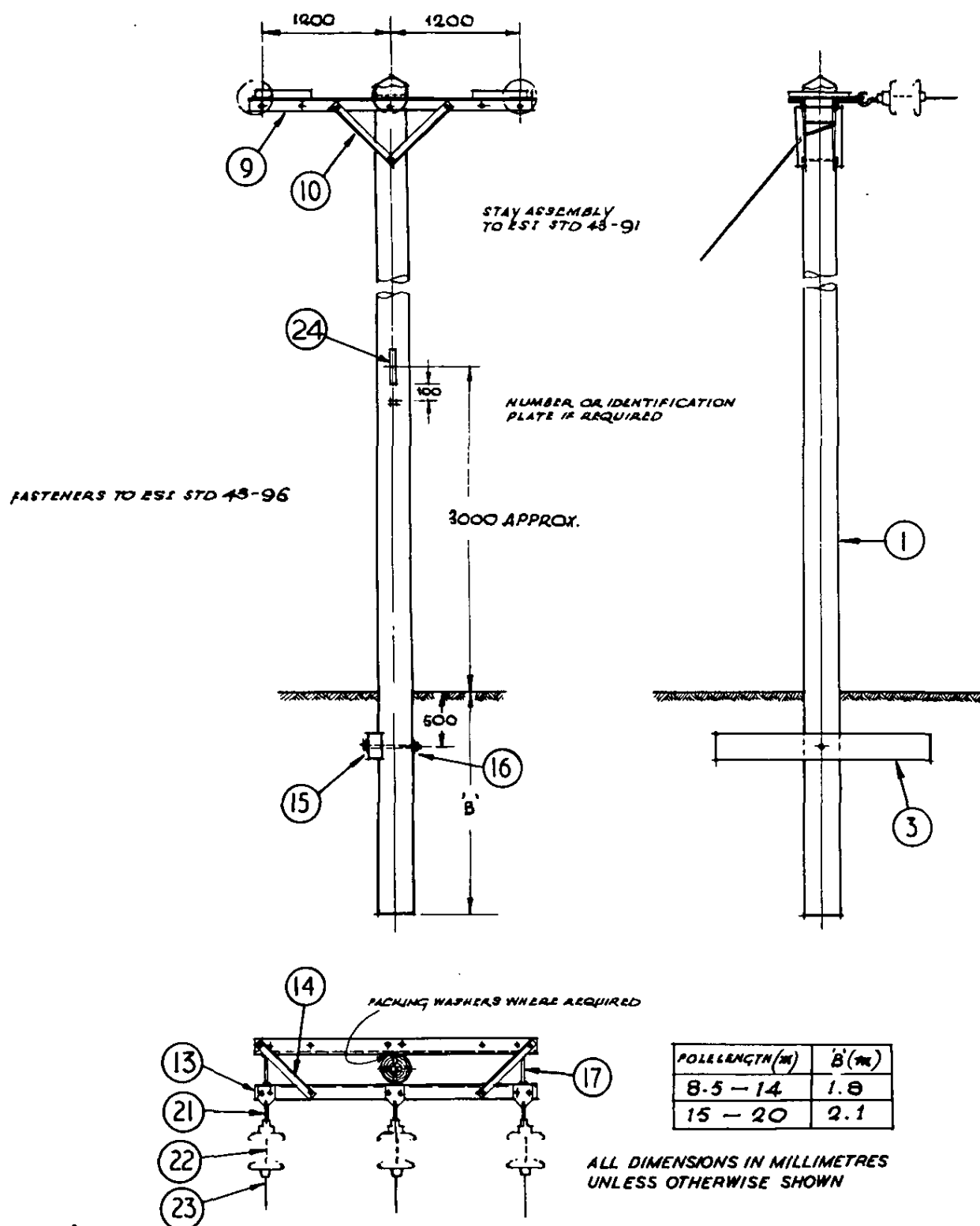
Material List for Fig. 9

'H' Pole Terminal

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref.	M 20 Bolts
2	2	Wood Pole	432002	2
4	1	Brace Block	439112 Type 1	2
9	2	Crossarm Member	439522	--
13	3	Terminal Strap	439528	6
15	2	Square Flat Washer	439604	—
16	2	Square Curved Washer	439605	
17	3	Tie Rod	439608 Type 3	--
21	3	Ball-ended Hook	BS 3288 Ref. 15/32	--
22	3	Insulator String	43-93	—
23	3	Conductor Termination	43-92	—
24	1	Danger Plate	439701	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Additional round washers and locknuts required for bonding stay strand to pole top bolts.



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Fig. 10 General assembly — single pole terminal
(ESI Drawing 432010)

Material List for Fig. 10

Single Pole Terminal

Item No.	Quantity	Description	ESI Std., Dwg. No. or BS Ref.	M 20 Bolts
1	1	Wood Pole	432001	2
3	1	Kicking Block	439103 Type 2	1
9	2	Crossarm Member	439522	—
10	4	Crossarm Strut	439526	4
13	3	Terminal Strap	439528	6
14	2	Crossarm Brace	439515	4
15	1	Square Flat Washer	439604	—
16	1	Square Curved Washer	439605	—
17	2	Tie Rod	439608 Type 3	—
21	3	Ball-ended Hook	BS 3288 Ref. 15/32	—
22	3	Insulator String	43-93	—
23	3	Conductor Termination	43-92	—
24	1	Danger plate	439701	—

NOTES:

1. All bolts 60 mm long with the exception of pole bolts.
2. Additional round washers and locknuts required for bonding stay strand to pole top bolts.

APPENDIX A

LINES IN SEVERE ENVIRONMENTS

The purpose of this Appendix is to specify modifications to this Standard to cater for lines in situations where the design conditions specified in The Electricity (Overhead Lines) Regulations, 1970 may be exceeded due to geographical or weather conditions. A definition of such situations is not given because it is best determined by individual Electricity Boards based upon local knowledge and operating experience.

A1. SPAN LENGTHS

Recommended and maximum span lengths for normal construction shall be 90 m and 105 m respectively for all conductors.

The occasional long span shall be 130 m to 150 m in length and provide a minimum ratio of 0.5 between mid-span spacing and still air design sag at 50°C.

A2. SUPPORTS

A2.1 Section Angle and Terminal Supports

Requirements shall be determined by the method laid down in 6.5 on page 12, but, to provide for increased factor of safety of 3.5 on poles, the loadings so determined shall be related to Table A1 of this Appendix with necessary reference to notes on Table 10 of this Standard.

A2.2 Intermediate and Section Supports

When determining the diameters of intermediate and section supports, the wind loading span shall be that shown in Tables 4—7, Pages 17—22, multiplied by 0.70.

A3. STAYS

It is recommended that the largest angle of stay slope possible up to a maximum of 45° be adopted.

A4. CONDUCTOR DESIGN AND ERECTION SAGS AND TENSIONS

Ordinates for plotting uplift curves and ground clearance curves are given in Table A2 (bare conductor) and Table A3 (PVC-covered conductor). These are based on a basic span of 90 m.

Conductor design sags and tensions are given in Tables A4—A9.

Conductor pre-tension and erection sags and tensions are given in Tables A10—A15.

Table A1 — Permissible Strut Loading of Poles — Factor of Safety 3.5

Dia. at 1.5 m (mm)	Pole Class (BS 1990)	Pole Planting Depth (m)	Pole Length (m)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)	Pole Top Dia. (mm)	Single Pole Max. Strut Loading (kgf)
215	M	1.8	8.5	150	2420	160	2999	180	3677
220			9.0	150	2178	165	2695	180	3300
225			9.5	150	1978	165	2445	180	2989
230			10.0	150	1811	165	2235	185	2911
235			10.5	150	1670	170	2201	190	2850
240			11.0	150	1550	170	2039	190	2635
245			11.5	150	1446	170	1898	190	2449
250			12.0	150	1356	170	1777	190	2288
260			13.0	160	1383	180	1790	200	280
275			14.0	160	1292	180	1663	205	2231
290		2.1	15.0	165	1337	185	1710	210	2281
305			16.0	170	1348	190	1709	215	2256
320			17.0	180	1447	200	1813	220	2245
330			18.0	180	1347	200	1683	220	2078
360			20.0	180	1261	205	1644	230	2108
265	S	1.8	8.5	190	5934	215	7878	240	10264
275			9.0	190	5474	215	7242	240	9407
280			9.5	190	4931	215	6515	245	8887
285			10.0	190	4479	215	5909	245	8049
290			10.5	190	4097	220	5692	250	7712
295			11.0	190	3773	220	5232	250	7079
300			11.5	190	3494	220	4837	250	6534
305			12.0	190	3252	220	4494	250	6062
320			13.0	195	3108	225	4252	255	5686
335			14.0	195	2849	225	3876	255	5158
350		2.1	15.0	195	2716	225	3691	260	5138
365			16.0	200	2669	230	3594	265	4956
375			17.0	200	2446	230	3282	265	4510
390			18.0	200	2319	230	3096	265	4232
415			20.0	200	2072	230	2743	265	3719
				Poles with top dia. equal to min. specified in BS 1990		Poles with mean value of top dia.		Poles with top dia. equal to max. specified in BS 1990	

For notes on Table A1 see page 25

Ordinates for Plotting Uplift and Ground Clearance Curves — ACSR Basic Span 90 m

Table A2 — Bare Conductor

Span (m) (2X)	Sag (m)								
	100mm ² 6/4.72 + 7/1.57mm			175mm ² 30 + 7/2.79mm			150mm ² , 175mm ² , 200mm ² 18 + 1.35mm 18 + 1/3.61mm 18 + 1/3.86mm		
	-6° C (Still Air)	50° C	75° C	-6° C (Still Air)	50° C	75° C	-6° C (Still Air)	50° C	75° C
40	0.10	0.29	0.37	0.09	0.24	0.31	0.11	0.35	0.42
80	0.39	1.15	1.47	0.36	0.95	1.25	0.46	1.39	1.69
120	0.88	2.60	3.30	0.81	2.12	2.80	1.03	3.12	3.79
160	1.56	4.61	5.86	1.44	3.78	4.98	1.83	5.55	6.74
200	2.43	7.21	9.16	2.26	5.90	7.78	2.86	8.68	10.53
240	3.51	10.38	13.19	3.25	8.50	11.21	4.12	12.48	15.16
280	4.77	14.12	17.95	4.42	11.56	15.25	5.61	16.98	20.63
320	6.23	18.45	23.45	5.78	15.10	19.92	7.32	22.18	26.95
360	7.89	23.35	29.68	7.31	19.11	25.22	9.27	28.07	34.11
400	9.74	28.82	36.64	9.03	23.60	31.13	11.44	34.65	42.11
440	11.78	34.88	44.33	10.93	28.55	37.67	13.84	41.93	50.95
480	14.02	41.51	52.76	13.00	33.98	44.83	16.47	49.90	60.64
520	16.46	48.71	61.92	15.26	39.88	52.61	19.33	58.57	71.16
560	19.09	56.49	71.81	17.70	46.25	61.02	22.42	67.92	82.53
600	21.91	64.85	82.43	20.32	53.09	70.04	25.74	77.97	94.74
640	24.93	73.79	93.79	23.11	60.41	79.69	29.28	88.71	107.79
680	28.14	83.30	105.88	26.09	68.20	89.96	33.06	100.15	121.69

NOTE: Ordinates for plotting uplift curves are based on conductor design tensions. Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR — 20% reduction
6 + 7 ACSR — 10% reduction
30 + 7 ACSR — 5% reduction

Ordinates for Plotting Uplift and Ground Clearance Curves —ACSR Basic Span 90 m

Table A3 — PVC-covered Conductor

Span (m) (2X)	Span (m)								
	100mm ² 6/4.72 + 7/1.57mm			175mm ² 30 + 7/2.79mm			150mm ² , 175mm ² , 200mm ² 18 + 1/3.35mm 18 + 1/3.61mm 18 + 1/3.86mm		
	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C
40	0.14	0.33	0.40	0.12	0.27	0.34	0.16	0.39	0.46
80	0.56	1.32	1.60	0.47	1.07	1.35	0.63	1.56	1.83
120	1.26	2.96	3.60	1.05	2.41	3.04	1.41	3.51	4.12
160	2.23	5.27	6.39	1.87	4.29	5.40	2.51	6.24	7.32
200	3.49	8.23	9.99	2.92	6.70	8.43	3.93	9.75	11.44
240	5.03	11.85	14.38	4.20	9.64	12.14	5.66	14.04	16.47
280	6.84	16.13	19.57	5.72	13.12	16.53	7.70	19.11	22.42
320	8.94	21.07	25.56	7.47	17.14	21.59	10.06	24.96	29.29
360	11.31	26.67	32.35	9.46	21.70	27.32	12.73	31.58	37.06
400	13.96	32.92	39.94	11.68	26.78	33.73	15.71	38.99	45.76
440	16.90	39.84	48.33	14.13	32.41	40.82	19.01	47.18	55.37
480	20.11	47.41	57.51	16.81	38.57	48.57	22.63	56.15	65.89
520	23.60	55.64	67.50	19.73	45.26	57.01	26.56	65.90	77.33
560	27.37	64.53	78.28	22.89	52.49	66.11	30.80	76.42	89.68
600	31.42	74.08	89.86	26.27	60.26	75.90	35.36	87.73	102.95
640	35.75	84.28	102.24	29.89	68.56	86.35	40.23	99.82	117.14
680	40.36	95.15	115.42	33.75	77.40	97.48	45.41	112.69	132.24

NOTE: Ordinates for plotting uplift curves are based on conductor design tensions. Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR — 20% reduction
 6 + 7 ACSR — 10% reduction
 30 + 7 ACSR — 5% reduction

Design Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR — Basic Span 90 m

Table A4 — Bare Conductor

Temp.	Tension	Sag (m) for span lengths (m)											
(°C)	(kgf)	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	736.6	0.11	0.17	0.25	0.33	0.44	0.55	0.68	0.83	0.98	1.15	1.34	1.53
5.0	666.9	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.91	1.09	1.27	1.48	1.70
10.0	602.3	0.13	0.21	0.30	0.41	0.53	0.68	0.83	1.01	1.20	1.41	1.64	1.88
15.0	543.6	0.15	0.23	0.33	0.45	0.59	0.75	0.92	1.12	1.33	1.56	1.81	2.08
20.0	491.3	0.16	0.26	0.37	0.50	0.65	0.83	1.02	1.24	1.47	1.73	2.00	2.30
25.0	445.7	0.18	0.28	0.41	0.55	0.72	0.91	1.13	1.36	1.62	1.91	2.21	2.54
30.0	406.4	0.20	0.31	0.45	0.61	0.79	1.00	1.24	1.50	1.78	2.09	2.42	2.78
35.0	372.9	0.22	0.34	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.64	3.03
40.0	344.4	0.23	0.36	0.53	0.71	0.93	1.18	1.46	1.77	2.10	2.47	2.86	3.28
45.0	320.2	0.25	0.39	0.57	0.77	1.00	1.27	1.57	1.90	2.26	2.65	3.08	3.53
50.0	299.5	0.27	0.42	0.60	0.82	1.07	1.36	1.68	2.03	2.42	2.84	3.29	3.78
55.0	281.6	0.29	0.45	0.64	0.87	1.14	1.45	1.78	2.16	2.57	3.02	3.50	4.01
60.0	266.2	0.30	0.47	0.68	0.92	1.21	1.53	1.89	2.28	2.72	3.19	3.70	4.25
65.0	252.8	0.32	0.50	0.72	0.97	1.27	1.61	1.99	2.41	2.86	3.36	3.90	4.47
70.0	240.9	0.33	0.52	0.75	1.02	1.33	1.69	2.09	2.52	3.00	3.52	4.09	4.69
75.0	230.4	0.35	0.55	0.79	1.07	1.40	1.77	2.18	2.64	3.14	3.69	4.27	4.91

Table A5 —PVC-covered Conductor

Temp.	Tension	Sag (m) for span lengths (m)											
(°C)	(kgf)	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	724.1	0.15	0.24	0.35	0.47	0.62	0.78	0.96	1.17	1.39	1.63	1.89	2.17
5.0	666.9	0.17	0.26	0.38	0.51	0.67	0.85	1.05	1.27	1.51	1.77	2.05	2.35
10.0	614.9	0.18	0.28	0.41	0.56	0.73	0.92	1.13	1.37	1.63	1.92	2.22	2.55
15.0	568.2	0.20	0.31	0.44	0.60	0.79	0.99	1.23	1.49	1.77	2.07	2.41	2.76
20.0	526.8	0.21	0.33	0.48	0.65	0.85	1.07	1.32	1.60	1.91	2.24	2.60	2.98
25.0	490.2	0.23	0.36	0.51	0.70	0.91	1.15	1.42	1.72	2.05	2.40	2.79	3.20
30.0	458.1	0.24	0.38	0.55	0.75	0.97	1.23	1.52	1.84	2.19	2.57	2.98	3.43
35.0	429.8	0.26	0.41	0.58	0.80	1.04	1.31	1.62	1.96	2.34	2.74	3.18	3.65
40.0	405.1	0.28	0.43	0.62	0.84	1.10	1.39	1.72	2.08	2.48	2.91	3.37	3.87
45.0	383.3	0.29	0.45	0.66	0.89	1.16	1.47	1.82	2.20	2.62	3.08	3.57	4.09
50.0	364.0	0.31	0.48	0.69	0.94	1.23	1.55	1.92	2.32	2.76	3.24	3.76	4.31
55.0	346.9	0.32	0.50	0.72	0.99	1.29	1.63	2.01	2.43	2.90	3.40	3.94	4.52
60.0	331.7	0.34	0.53	0.76	1.03	1.35	1.70	2.10	2.54	3.03	3.55	4.12	4.73
65.0	318.0	0.35	0.55	0.79	1.07	1.40	1.78	2.19	2.65	3.16	3.71	4.30	4.94
70.0	305.7	0.37	0.57	0.82	1.12	1.46	1.85	2.28	2.76	3.29	3.86	4.47	5.13
75.0	294.5	0.38	0.59	0.85	1.16	1.52	1.92	2.37	2.87	3.41	4.00	4.64	5.33

Design Sag/Tension Tables for 175 mm² (30 + 7/2.79) ACSR – Basic Span 90 m

Table A6 – Bare Conductor

Temp. (°C)	Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	1764.8	0.10	0.15	0.22	0.30	0.40	0.50	0.62	0.75	0.89	1.05	1.21	1.39
5.0	1627.4	0.11	0.17	0.24	0.33	0.43	0.54	0.67	0.81	0.97	1.13	1.32	1.51
10.0	1496.1	0.12	0.18	0.26	0.36	0.47	0.59	0.73	0.88	1.05	1.23	1.43	1.64
15.0	1372.1	0.13	0.20	0.29	0.39	0.51	0.64	0.80	0.96	1.15	1.35	1.56	1.79
20.0	1256.5	0.14	0.22	0.31	0.43	0.56	0.70	0.87	1.05	1.25	1.47	1.70	1.96
25.0	1150.4	0.15	0.24	0.34	0.47	0.61	0.77	0.95	1.15	1.37	1.60	1.86	2.14
30.0	1054.5	0.17	0.26	0.37	0.51	0.66	0.84	1.04	1.25	1.49	1.75	2.03	2.33
35.0	969.1	0.18	0.28	0.41	0.55	0.72	0.91	1.13	1.36	1.62	1.91	2.21	2.54
40.0	893.8	0.20	0.31	0.44	0.60	0.78	0.99	1.22	1.48	1.76	2.07	2.40	2.75
45.0	828.0	0.21	0.33	0.48	0.65	0.84	1.07	1.32	1.60	1.90	2.23	2.59	2.97
50.0	770.7	0.23	0.35	0.51	0.69	0.91	1.15	1.42	1.72	2.04	2.40	2.78	3.19
55.0	720.9	0.24	0.38	0.55	0.74	0.97	1.23	1.52	1.83	2.18	2.56	2.97	3.41
60.0	677.6	0.26	0.40	0.58	0.79	1.03	1.31	1.61	1.95	2.32	2.72	3.16	3.63
65.0	639.7	0.27	0.43	0.61	0.84	1.09	1.38	1.71	2.07	2.46	2.89	3.35	3.84
70.0	606.4	0.29	0.45	0.65	0.88	1.15	1.46	1.80	2.18	2.59	3.04	3.53	4.05
75.0	577.1	0.30	0.47	0.68	0.93	1.21	1.53	1.89	2.29	2.73	3.20	3.71	4.26

Table A7 – PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	1697.0	0.13	0.20	0.29	0.39	0.51	0.65	0.80	0.96	1.15	1.35	1.56	1.79
5.0	1573.9	0.14	0.21	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.45	1.68	1.93
10.0	1458.2	0.15	0.23	0.33	0.45	0.59	0.75	0.93	1.12	1.34	1.57	1.82	2.09
15.0	1350.8	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.69	1.96	2.25
20.0	1252.0	0.17	0.27	0.39	0.53	0.69	0.87	1.08	1.31	1.56	1.83	2.12	2.43
25.0	1162.4	0.19	0.29	0.42	0.57	0.74	0.94	1.16	1.41	1.68	1.97	2.28	2.62
30.0	1081.7	0.20	0.31	0.45	0.61	0.80	1.01	1.25	1.51	1.80	2.11	2.45	2.81
35.0	1009.6	0.21	0.33	0.48	0.66	0.86	1.09	1.34	1.62	1.93	2.26	2.63	3.01
40.0	945.5	0.23	0.36	0.51	0.70	0.92	1.16	1.43	1.73	2.06	2.42	2.80	3.22
45.0	888.8	0.24	0.38	0.55	0.75	0.97	1.23	1.52	1.84	2.19	2.57	2.98	3.42
50.0	838.5	0.26	0.40	0.58	0.79	1.03	1.31	1.61	1.95	2.32	2.73	3.16	3.63
55.0	794.0	0.27	0.43	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.88	3.34	3.83
60.0	754.4	0.29	0.45	0.65	0.88	1.15	1.45	1.79	2.17	2.58	3.03	3.51	4.03
65.0	719.0	0.30	0.47	0.68	0.92	1.20	1.52	1.88	2.28	2.71	3.18	3.69	4.23
70.0	687.4	0.31	0.49	0.71	0.96	1.26	1.59	1.97	2.38	2.83	3.32	3.86	4.43
75.0	659.0	0.33	0.51	0.74	1.01	1.31	1.66	2.05	2.48	2.96	3.47	4.02	4.62

Design Sag/Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 90 m

Table A8 — Bare Conductor

Temp (°C)	Tension (kgf)			Sag (m) for span lengths (m)											
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	810.0	932.7	1056.3	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	1.59	1.82
5.0	728.1	838.2	949.3	0.14	0.23	0.32	0.44	0.58	0.73	0.90	1.09	1.30	1.52	1.77	2.03
10.0	654.8	753.8	854.0	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.70	1.97	2.26
15.0	590.7	680.1	770.8	0.18	0.28	0.40	0.54	0.71	0.90	1.11	1.35	1.60	1.88	2.18	2.50
20.0	535.5	616.8	699.5	0.20	0.31	0.44	0.60	0.78	0.99	1.23	1.48	1.77	2.07	2.40	2.76
25.0	488.7	563.1	639.0	0.21	0.34	0.48	0.66	0.86	1.09	1.34	1.63	1.93	2.27	2.63	3.02
30.0	449.1	517.7	587.8	0.23	0.37	0.53	0.72	0.93	1.18	1.46	1.77	2.10	2.47	2.86	3.29
35.0	415.6	479.3	544.6	0.25	0.39	0.57	0.77	1.01	1.28	1.58	1.91	2.27	2.67	3.09	3.55
40.0	387.3	446.8	508.0	0.27	0.42	0.61	0.83	1.08	1.37	1.69	2.05	2.44	2.86	3.32	3.81
45.0	363.1	419.0	476.6	0.29	0.45	0.65	0.88	1.16	1.46	1.80	2.18	2.60	3.05	3.54	4.06
50.0	342.2	395.1	449.7	0.31	0.48	0.69	0.94	1.22	1.55	1.91	2.32	2.76	3.23	3.75	4.31
55.0	324.2	374.4	426.2	0.32	0.51	0.73	0.99	1.29	1.64	2.02	2.44	2.91	3.41	3.96	4.55
60.0	308.4	356.2	405.7	0.34	0.53	0.76	1.04	1.36	1.72	2.12	2.57	3.06	3.59	4.16	4.78
65.0	294.4	340.2	387.6	0.36	0.56	0.80	1.09	1.42	1.80	2.22	2.69	3.20	3.76	4.36	5.00
70.0	282.1	326.0	371.5	0.37	0.58	0.84	1.14	1.48	1.88	2.32	2.81	3.34	3.92	4.55	5.22
75.0	271.0	313.3	357.1	0.39	0.60	0.87	1.18	1.55	1.96	2.41	2.92	3.48	4.08	4.73	5.43

Design Sag/Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 90 m

Table A9 — PVC-covered Conductor

Temp (°C)	Tension (kgf)			Sag (m) for span lengths (m)											
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	793.0	913.6	1036.4	0.17	0.27	0.39	0.54	0.70	0.88	1.09	1.32	1.57	1.85	2.14	2.46
5.0	728.1	838.2	949.3	0.19	0.30	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	2.33	2.68
10.0	670.7	771.6	872.5	0.21	0.32	0.47	0.63	0.83	1.05	1.29	1.56	1.86	2.18	2.53	2.91
15.0	620.3	713.3	805.4	0.22	0.35	0.50	0.69	0.90	1.13	1.40	1.69	2.01	2.36	2.74	3.15
20.0	576.5	662.5	747.1	0.24	0.38	0.54	0.74	0.96	1.22	1.51	1.82	2.17	2.54	2.95	3.39
25.0	538.3	618.4	696.5	0.26	0.40	0.58	0.79	1.03	1.31	1.61	1.95	2.32	2.73	3.16	3.63
30.0	505.2	580.1	652.7	0.28	0.43	0.62	0.84	1.10	1.39	1.72	2.08	2.48	2.91	3.37	3.87
35.0	476.2	546.7	614.5	0.29	0.46	0.66	0.89	1.17	1.48	1.82	2.21	2.63	3.08	3.58	4.11
40.0	450.8	517.4	581.2	0.31	0.48	0.69	0.94	1.23	1.56	1.93	2.33	2.78	3.26	3.78	4.34
45.0	428.4	491.6	551.9	0.32	0.51	0.73	0.99	1.30	1.64	2.03	2.45	2.92	3.43	3.98	4.57
50.0	408.6	468.8	526.0	0.34	0.53	0.77	1.04	1.36	1.72	2.13	2.57	3.06	3.60	4.17	4.79
55.0	391.0	448.5	503.0	0.36	0.56	0.80	1.09	1.42	1.80	2.22	2.69	3.20	3.76	4.36	5.00
60.0	375.2	430.4	482.4	0.37	0.58	0.83	1.14	1.48	1.88	2.32	2.80	3.34	3.92	4.54	5.21
65.0	361.0	414.0	463.9	0.39	0.60	0.87	1.18	1.54	1.95	2.41	2.92	3.47	4.07	4.72	5.42
70.0	348.1	399.2	447.2	0.40	0.62	0.90	1.22	1.60	2.02	2.50	3.02	3.60	4.22	4.90	5.62
75.0	336.4	385.8	432.0	0.41	0.65	0.93	1.27	1.65	2.09	2.59	3.13	3.72	4.37	5.07	5.82

Pre-tension and Erection Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR — Basic Span 90 m

Table A10 — Bare Conductor

Pre-tension kgf (All Temps.)	Sag (m) for span lengths (m) at all temperatures											
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
750	0.11	0.17	0.25	0.34	0.44	0.56	0.69	0.83	0.99	1.16	1.35	1.55

Erection Temp. (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	805.1	0.10	0.16	0.22	0.31	0.40	0.51	0.62	0.76	0.90	1.05	1.22	1.40
5.0	731.5	0.11	0.17	0.25	0.34	0.44	0.56	0.69	0.83	0.99	1.16	1.35	1.55
10.0	662.1	0.12	0.19	0.27	0.37	0.49	0.61	0.76	0.92	1.09	1.28	1.49	1.71
15.0	597.9	0.13	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.21	1.42	1.55	1.89
20.0	539.7	0.15	0.23	0.34	0.46	0.60	0.75	0.93	1.13	1.34	1.57	1.82	2.09
25.0	487.9	0.16	0.26	0.37	0.50	0.66	0.83	1.03	1.25	1.48	1.74	2.02	2.32
30.0	442.7	0.18	0.28	0.41	0.56	0.73	0.92	1.14	1.37	1.63	1.92	2.22	2.55
35.0	403.9	0.20	0.31	0.45	0.61	0.80	1.01	1.24	1.51	1.79	2.10	2.44	2.80
40.0	370.7	0.22	0.34	0.49	0.66	0.87	1.10	1.36	1.64	1.95	2.29	2.66	3.05
45.0	342.6	0.23	0.37	0.53	0.72	0.94	1.19	1.47	1.77	2.11	2.48	2.88	3.30
50.0	318.6	0.25	0.39	0.57	0.77	1.01	1.28	1.58	1.91	2.27	2.67	3.09	3.55
55.0	298.1	0.27	0.42	0.61	0.83	1.08	1.37	1.69	2.04	2.43	2.85	3.30	3.79
60.0	280.5	0.29	0.45	0.64	0.88	1.15	1.45	1.79	2.17	2.58	3.03	3.51	4.03
65.0	265.2	0.30	0.47	0.68	0.93	1.21	1.53	1.89	2.29	2.73	3.20	3.71	4.26
70.0	251.9	0.32	0.50	0.72	0.98	1.28	1.62	2.00	2.41	2.87	3.37	3.91	4.49
75.0	240.1	0.33	0.52	0.75	1.03	1.34	1.69	2.09	2.53	3.01	3.54	4.10	4.71

Pre-tension and Erection Sag/Tension Tables for 100 mm² (6/4.72 + 7/1.57) ACSR — Basic Span 90 m

Table A11 — PVC-covered Conductor

Pre-tension kgf (All Temps.)	Sag (m) for span lengths (m) at all temperatures											
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
750	0.15	0.24	0.34	0.46	0.61	0.77	0.95	1.15	1.36	1.60	1.86	2.13

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	799.5	0.14	0.22	0.31	0.43	0.56	0.71	0.87	1.06	1.26	1.47	1.71	1.96
5.0	736.4	0.15	0.24	0.34	0.46	0.61	0.77	0.95	1.15	1.36	1.60	1.86	2.13
10.0	678.1	0.16	0.26	0.37	0.50	0.66	0.83	1.03	1.24	1.48	1.74	2.02	2.31
15.0	625.1	0.18	0.28	0.40	0.55	0.71	0.90	1.12	1.35	1.61	1.89	2.19	2.51
20.0	577.3	0.19	0.30	0.43	0.59	0.77	0.98	1.21	1.46	1.74	2.04	2.37	2.72
25.0	534.8	0.21	0.33	0.47	0.64	0.83	1.06	1.30	1.58	1.88	2.20	2.56	2.93
30.0	497.3	0.22	0.35	0.50	0.69	0.90	1.14	1.40	1.70	2.02	2.37	2.75	3.16
35.0	464.3	0.24	0.38	0.54	0.74	0.96	1.22	1.50	1.82	2.16	2.54	2.94	3.38
40.0	435.3	0.26	0.40	0.58	0.79	1.03	1.30	1.60	1.94	2.31	2.71	3.14	3.61
45.0	409.9	0.27	0.43	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.88	3.34	3.83
50.0	387.5	0.29	0.45	0.65	0.88	1.15	1.46	1.80	2.18	2.59	3.04	3.53	4.05
55.0	367.8	0.30	0.47	0.68	0.93	1.21	1.54	1.90	2.29	2.73	3.21	3.72	4.27
60.0	350.2	0.32	0.50	0.72	0.98	1.27	1.61	1.99	2.41	2.87	3.37	3.90	4.48
65.0	334.6	0.33	0.52	0.75	1.02	1.33	1.69	2.08	2.52	3.00	3.52	4.09	4.69
70.0	320.7	0.35	0.54	0.78	1.07	1.39	1.76	2.18	2.63	3.13	3.68	4.26	4.89
75.0	308.1	0.36	0.57	0.82	1.11	1.45	1.83	2.26	2.74	3.26	3.83	4.44	5.09

Pre-tension and Erection Sag/Tension Tables for 175 mm² (30 + 7/2.19) ACSR — Basic Span 90 m.

Table A12 — Bare Conductor.

Pre-tension kgf (All Temps.)	Sag (m) for span lengths (m) at all temperatures											
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
1800	0.10	0.15	0.22	0.30	0.39	0.50	0.61	0.74	0.88	1.04	1.20	1.38

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	1921.9	0.09	0.14	0.20	0.28	0.36	0.46	0.57	0.69	0.82	0.96	1.11	1.28
5.0	1779.2	0.10	0.15	0.22	0.30	0.39	0.50	0.61	0.74	0.88	1.04	1.20	1.38
10.0	1641.3	0.11	0.17	0.24	0.33	0.43	0.54	0.67	0.81	0.96	1.12	1.30	1.50
15.0	1509.3	0.12	0.18	0.26	0.35	0.46	0.59	0.72	0.88	1.04	1.22	1.42	1.63
20.0	1384.4	0.13	0.20	0.28	0.39	0.51	0.64	0.79	0.95	1.14	1.33	1.55	1.78
25.0	1267.9	0.14	0.22	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.46	1.69	1.94
30.0	1160.8	0.15	0.24	0.34	0.46	0.60	0.76	0.94	1.14	1.36	1.59	1.84	2.12
35.0	1063.9	0.16	0.26	0.37	0.50	0.66	0.83	1.03	1.24	1.48	1.74	2.01	2.31
40.0	977.3	0.18	0.28	0.40	0.55	0.72	0.91	1.12	1.35	1.61	1.89	2.19	2.52
45.0	901.0	0.19	0.30	0.44	0.59	0.78	0.98	1.21	1.47	1.75	2.05	2.38	2.73
50.0	834.3	0.21	0.33	0.47	0.64	0.84	1.06	1.31	1.58	1.89	2.21	2.57	2.95
55.0	776.2	0.23	0.35	0.51	0.69	0.90	1.14	1.41	1.70	2.03	2.38	2.76	3.17
60.0	725.7	0.24	0.38	0.54	0.74	0.96	1.22	1.51	1.82	2.17	2.54	2.95	3.39
65.0	681.7	0.26	0.40	0.58	0.79	1.03	1.30	1.60	1.94	2.31	2.71	3.14	3.61
70.0	643.3	0.27	0.42	0.61	0.83	1.09	1.38	1.70	2.05	2.45	2.87	3.33	3.82
75.0	609.7	0.29	0.45	0.65	0.88	1.15	1.45	1.79	2.17	2.58	3.03	3.51	4.03

Pre-tension and Erection Sag/Tension Tables for 175 mm² (30 + 7/2.19) ACSR – Basic Span 90 m

Table A13 – PVC-covered Conductor

Pre-tension kgf (All Temps)	Sag (m) for span lengths (m) at all temperatures											
	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
1800	0.13	0.20	0.28	0.38	0.50	0.63	0.78	0.95	1.13	1.32	1.53	1.76

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for span lengths (m)											
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	1859.2	0.12	0.18	0.26	0.36	0.47	0.59	0.73	0.88	1.05	1.23	1.43	1.64
5.0	1728.2	0.13	0.20	0.28	0.38	0.50	0.63	0.78	0.95	1.13	1.32	1.53	1.76
10.0	1603.4	0.13	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.21	1.43	1.65	1.90
15.0	1485.8	0.15	0.23	0.33	0.45	0.58	0.74	0.91	1.10	1.31	1.54	1.78	2.05
20.0	1376.3	0.16	0.25	0.35	0.48	0.63	0.80	0.98	1.19	1.42	1.66	1.93	2.21
25.0	1275.4	0.17	0.27	0.38	0.52	0.68	0.86	1.06	1.28	1.53	1.79	2.08	2.39
30.0	1183.5	0.18	0.29	0.41	0.56	0.73	0.93	1.14	1.38	1.65	1.93	2.24	2.57
35.0	1100.6	0.20	0.31	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	2.41	2.76
40.0	1026.5	0.21	0.33	0.47	0.65	0.84	1.07	1.32	1.59	1.90	2.23	2.58	2.96
45.0	960.5	0.23	0.35	0.51	0.69	0.90	1.14	1.41	1.70	2.03	2.38	2.76	3.17
50.0	902.0	0.24	0.37	0.54	0.73	0.96	1.21	1.50	1.81	2.16	2.53	2.94	3.37
55.0	850.3	0.25	0.40	0.57	0.78	1.02	1.29	1.59	1.92	2.29	2.69	3.12	3.58
60.0	804.4	0.27	0.42	0.61	0.82	1.08	1.36	1.68	2.03	2.42	2.84	3.30	3.78
65.0	763.6	0.28	0.44	0.64	0.87	1.13	1.43	1.77	2.14	2.55	2.99	3.47	3.99
70.0	727.3	0.30	0.46	0.67	0.91	1.19	1.51	1.86	2.25	2.68	3.14	3.64	4.18
75.0	694.9	0.31	0.49	0.70	0.95	1.25	1.58	1.95	2.36	2.80	3.29	3.81	4.38

Pre-tension and Erection Sag/Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 90 m

Table A14 — Bare Conductor

Pre-tension kgf (All Temps)			Sag (m) for span lengths (m) at all temperatures											
150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
800	950	1050	0.13	0.20	0.29	0.40	0.52	0.66	0.82	0.99	1.18	1.38	1.60	1.84

Erection Temp (°C)	Erection Tension (kgf)			Sag (m) for span lengths (m)											
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	892.7	1028.4	1165.4	0.12	0.18	0.26	0.36	0.47	0.60	0.74	0.89	1.06	1.24	1.44	1.65
5.0	803.6	925.5	1048.6	0.13	0.20	0.29	0.40	0.52	0.66	0.82	0.99	1.18	1.38	1.60	1.84
10.0	722.3	831.7	942.4	0.15	0.23	0.33	0.45	0.58	0.74	0.91	1.10	1.31	1.54	1.78	2.05
15.0	649.7	748.1	847.9	0.16	0.25	0.36	0.50	0.65	0.82	1.01	1.22	1.46	1.71	1.98	2.27
20.0	586.3	675.2	765.6	0.18	0.28	0.40	0.55	0.72	0.91	1.12	1.36	1.61	1.89	2.20	2.52
25.0	531.8	612.6	695.0	0.20	0.31	0.44	0.60	0.79	1.00	1.23	1.49	1.78	2.09	2.42	2.78
30.0	485.5	559.5	635.2	0.22	0.34	0.49	0.66	0.87	1.09	1.35	1.64	1.95	2.28	2.65	3.04
35.0	446.4	514.7	584.6	0.24	0.37	0.53	0.72	0.94	1.19	1.47	1.78	2.12	2.48	2.88	3.31
40.0	413.4	476.8	541.9	0.25	0.40	0.57	0.78	1.02	1.28	1.59	1.92	2.28	2.68	3.11	3.57
45.0	385.3	444.6	505.7	0.27	0.43	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.87	3.33	3.83
50.0	361.4	417.2	474.7	0.29	0.45	0.65	0.89	1.16	1.47	1.81	2.19	2.61	3.06	3.55	4.08
55.0	340.8	393.5	447.9	0.31	0.48	0.69	0.94	1.23	1.56	1.92	2.33	2.77	3.25	3.77	4.32
60.0	322.9	373.0	424.7	0.32	0.51	0.73	0.99	1.30	1.64	2.03	2.45	2.92	3.43	3.97	4.56
65.0	307.3	355.0	404.4	0.34	0.53	0.77	1.04	1.36	1.73	2.13	2.58	3.07	3.60	4.18	4.79
70.0	293.5	339.1	386.4	0.36	0.56	0.80	1.09	1.43	1.81	2.23	2.70	3.21	3.77	4.37	5.02
75.0	281.2	325.0	370.5	0.37	0.58	0.84	1.14	1.49	1.88	2.33	2.82	3.35	3.93	4.56	5.24

Pre-tension and Erection Sag/Tension Tables for 150 mm², 175 mm² and 200 mm² (18 + 1 Construction) ACSR — Basic Span 90 m

Table A15 — PVC-covered Conductor

Pre-tension kgf (All Temps)			Sag (m) for span lengths (m) at all temperatures											
150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
800	950	1050	0.17	0.27	0.39	0.55	0.69	0.87	1.07	1.30	1.54	1.81	2.10	2.41

Erection Temp (°C)	Erection Tension (kgf)			Sag (m) for span lengths (m)											
	150 mm ²	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0
0.0	882.0	1016.2	1152.6	0.16	0.25	0.35	0.48	0.63	0.80	0.98	1.19	1.41	1.66	1.92	2.21
5.0	807.9	930.2	1053.5	0.17	0.27	0.39	0.53	0.69	0.87	1.07	1.30	1.54	1.81	2.10	2.41
10.0	741.4	853.0	964.6	0.19	0.29	0.42	0.57	0.75	0.95	1.17	1.42	1.68	1.98	2.29	2.63
15.0	682.4	784.6	885.9	0.20	0.32	0.46	0.62	0.81	1.03	1.27	1.54	1.83	2.15	2.49	2.86
20.0	630.6	724.6	817.1	0.22	0.34	0.50	0.67	0.88	1.12	1.38	1.67	1.98	2.33	2.70	3.10
25.0	585.4	672.4	757.2	0.24	0.37	0.53	0.73	0.95	1.20	1.48	1.80	2.14	2.51	2.91	3.34
30.0	546.1	627.0	705.3	0.25	0.40	0.57	0.78	1.02	1.29	1.59	1.93	2.29	2.69	3.12	3.58
35.0	511.9	587.5	660.3	0.27	0.42	0.61	0.83	1.09	1.38	1.70	2.05	2.44	2.87	3.33	3.82
40.0	482.1	553.2	621.2	0.29	0.45	0.65	0.88	1.15	1.46	1.80	2.18	2.60	3.05	3.53	4.06
45.0	456.0	523.1	587.0	0.31	0.48	0.69	0.93	1.22	1.54	1.91	2.31	2.75	3.22	3.74	4.29
50.0	433.0	496.7	557.0	0.32	0.50	0.72	0.98	1.29	1.63	2.01	2.43	2.89	3.39	3.94	4.52
55.0	412.7	473.3	530.6	0.34	0.53	0.76	1.03	1.35	1.71	2.11	2.55	3.03	3.56	4.13	4.74
60.0	394.6	452.5	507.1	0.35	0.55	0.79	1.08	1.41	1.79	2.20	2.67	3.17	3.73	4.32	4.96
65.0	378.4	434.0	486.1	0.37	0.57	0.83	1.13	1.47	1.86	2.30	2.78	3.31	3.88	4.51	5.17
70.0	363.9	417.2	467.2	0.38	0.60	0.86	1.17	1.53	1.94	2.39	2.89	3.44	4.04	4.69	5.38
75.0	350.7	402.1	450.2	0.40	0.62	0.89	1.22	1.59	2.01	2.48	3.00	3.57	4.19	4.86	5.58

